

1. Meta Feature Types

1.1. Quality of Non-Bathymetric Data

Definition: An area within which a uniform assessment of the quality of the non-bathymetric data exists.

CamelCase: QualityOfNonBathymetricData

Alias: M_ACCY

Super type:

Feature use type: meta

Primitive: surface

Remarks: No remarks.

Attribute Bindings:

S-10x Attribute	S-57 Acronym	Allowable Encoding Value	Type	Multiplicity
Category of Temporal Variation		1 : Extreme Event 2 : Likely to Change and Significant Shoaling Expected 3 : Likely to Change But Significant Shoaling Not Expected 4 : Likely to Change 5 : Unlikely to Change 6 : Unassessed	EN	1, 1
Orientation Uncertainty			RE	0, 1
Horizontal Distance Uncertainty	(HORACC)		RE	0, 1
Horizontal Position Uncertainty	(POSACC)		C	1, 1
Uncertainty Fixed	(POSACC) (SOUACC) (VERACC)		(S) RE	1, 1
Uncertainty Variable Factor			(S) RE	0, 1
Information	(INFORM)		C	0, 1
Uncertainty Variable Factor			(S) RE	0, 1

1.2. Data Coverage

Definition: A geographical area that describes the coverage and extent of spatial objects.

CamelCase: DataCoverage

Alias: M_COVR M_CSCL

Super type:

Feature use type: meta

Primitive: surface

Remarks: No remarks.

Attribute Bindings:

S-10x Attribute	S-57 Acronym	Allowable Encoding Value	Type	Multiplicity
Maximum Display Scale			IN	1, 1
Minimum Display Scale			IN	1, 1

1.3. Local Direction of Buoyage

Definition: An area within which the navigational system of marks has been established in relation to a specific direction.

CamelCase: LocalDirectionOfBuoyage

Alias: M_NSYS

Super type:

Feature use type: meta

Primitive: surface

Remarks: No remarks.

Attribute Bindings:

S-10x Attribute	S-57 Acronym	Allowable Encoding Value	Type	Multiplicity
Orientation Value	(ORIENT)		(S) RE	1, 1

1.4. Navigational System of Marks

Definition: An area within which the navigational system of marks has been established in relation to a specific direction.

CamelCase: NavigationalSystemOfMarks

Alias: M_NSYS

Super type:

Feature use type: meta

Primitive: surface

Remarks: No remarks.

Attribute Bindings:

S-10x Attribute	S-57 Acronym	Allowable Encoding Value	Type	Multiplicity
Marks Navigational - System Of	(MARSYS)	1 : IALA A 2 : IALA B 9 : No System 10 : Other System 11 : CEVNI 12 : Russian Inland Waterway Regulations 13 : Brazilian National Inland Waterway Regulations - Two Sides 15 : Paraguay-Parana Waterway - Brazilian Complementary Aids	EN	1, 1

1.5. Sounding Datum

Definition: The horizontal plane or tidal datum to which soundings have been reduced. Also called datum for sounding reduction.

CamelCase: SoundingDatum

Alias: M_SDAT

Super type:

Feature use type: meta

Primitive: surface

Remarks: No remarks.

Attribute Bindings:

S-10x Attribute	S-57 Acronym	Allowable Encoding Value	Type	Multiplicity
Vertical Datum	(VERDAT) (Datum Level) (Reference Plane) (Levelling Datum) (Datum for Sounding Reduction) (Datum for Heights)	1 : Mean Low Water Springs 2 : Mean Lower Low Water Springs 3 : Mean Sea Level 4 : Lowest Low Water 5 : Mean Low Water 6 : Lowest Low Water Springs 7 : Approximate Mean Low Water Springs 8 : Indian Spring Low Water 9 : Low Water Springs 10 : Approximate Lowest Astronomical Tide 11 : Nearly Lowest Low Water 12 : Mean Lower Low Water 13 : Low Water 14 : Approximate Mean Low Water 15 : Approximate Mean Lower Low Water 16 : Mean High Water 17 : Mean High Water Springs 18 : High Water 19 : Approximate Mean Sea Level	EN	1, 1

		20 : High Water Springs 21 : Mean Higher High Water 22 : Equinoctial Spring Low Water 23 : Lowest Astronomical Tide 24 : Local Datum 25 : International Great Lakes Datum 1985 26 : Mean Water Level 27 : Lower Low Water Large Tide 28 : Higher High Water Large Tide 29 : Nearly Highest High Water 30 : Highest Astronomical Tide 31 : Local Low Water Reference Level 32 : Local High Water Reference Level 33 : Local Mean Water Reference Level 34 : Equivalent Height of Water (German GIW) 35 : Highest Shipping Height of Water (German HSW) 36 : Reference Low Water Level According to Danube Commission 37 : Highest Shipping Height of Water According to Danube Commission 38 : Dutch River Low Water Reference Level (OLR) 39 : Russian Project Water Level 40 : Russian Normal Backwater Level 41 : Ohio River Datum 43 : Dutch High Water Reference Level 44 : Baltic Sea Chart Datum 2000 45 : Dutch Estuary Low Water Reference Level (OLW) 46 : International Great Lakes Datum 2020 47 : Sea Floor		
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		48 : Sea Surface 49 : Hydrographic Zero		
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1.6. Vertical Datum of Data

Definition: Any level surface (for example Mean Sea Level) taken as a surface of reference to which the elevations within a data set are reduced. Also called datum level, reference level, reference plane, levelling datum, datum for heights.

CamelCase: VerticalDatumOfData

Alias: M_VDAT

Super type:

Feature use type: meta

Primitive: surface

Remarks: No remarks.

Attribute Bindings:

S-10x Attribute	S-57 Acronym	Allowable Encoding Value	Type	Multiplicity
Vertical Datum	(VERDAT) (Datum Level) (Reference Plane) (Levelling Datum) (Datum for Sounding Reduction) (Datum for Heights)	1 : Mean Low Water Springs 2 : Mean Lower Low Water Springs 3 : Mean Sea Level 4 : Lowest Low Water 5 : Mean Low Water 6 : Lowest Low Water Springs 7 : Approximate Mean Low Water Springs 8 : Indian Spring Low Water 9 : Low Water Springs 10 : Approximate Lowest Astronomical Tide 11 : Nearly Lowest Low Water 12 : Mean Lower Low Water 13 : Low Water 14 : Approximate Mean Low Water 15 : Approximate Mean Lower Low Water 16 : Mean High Water 17 : Mean High Water Springs 18 : High Water 19 : Approximate Mean Sea	EN	1, 1

		Level 20 : High Water Springs 21 : Mean Higher High Water 22 : Equinoctial Spring Low Water 23 : Lowest Astronomical Tide 24 : Local Datum 25 : International Great Lakes Datum 1985 26 : Mean Water Level 27 : Lower Low Water Large Tide 28 : Higher High Water Large Tide 29 : Nearly Highest High Water 30 : Highest Astronomical Tide 31 : Local Low Water Reference Level 32 : Local High Water Reference Level 33 : Local Mean Water Reference Level 34 : Equivalent Height of Water (German GIW) 35 : Highest Shipping Height of Water (German HSW) 36 : Reference Low Water Level According to Danube Commission 37 : Highest Shipping Height of Water According to Danube Commission 38 : Dutch River Low Water Reference Level (OLR) 39 : Russian Project Water Level 40 : Russian Normal Backwater Level 41 : Ohio River Datum 43 : Dutch High Water Reference Level 44 : Baltic Sea Chart Datum 2000 45 : Dutch Estuary Low Water Reference Level (OLW) 46 : International Great Lakes Datum 2020		
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		47 : Sea Floor 48 : Sea Surface 49 : Hydrographic Zero		
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2. Geo Feature Types

2.1. Aids to Navigation

Definition: A visual, acoustical, or radio device, external to a ship, designed to assist in determining a safe course or a vessel's position, or to warn of dangers and/or obstructions. Aids to navigation usually include buoys, beacons, fog signals, lights, radio beacons, leading marks, radio position fixing systems and GNSS which are chart-related and assist safe navigation.

CamelCase: AidsToNavigation

Alias:

Super type:

Feature use type: geographic

Primitive: noGeometry

Remarks: No remarks.

Attribute Bindings:

S-10x Attribute	S-57 Acronym	Allowable Encoding Value	Type	Multiplicity
ID Code	(Identification Number) (Identification Code)		TE	0, 1
Information	(INFORM)		C	0, *
File Locator			(S) TE	0, 1
File Reference	(TXTDSC) (NXTDSC)		(S) TE	0, 1
Headline			(S) TE	0, 1
Language			(S) TE	1, 1
Text	(INFORM) (NINFOM)		(S) TE	0, 1
Feature Name			C	0, *
Display Name			(S) BO	0, 1
Language			(S) TE	0, 1
Name	(OBJNAM)		(S) TE	1, 1
Scale Minimum	(SCAMIN)		IN	0, 1

Source Date	(SORDAT)		DA	0, 1
Seasonal Action Required			TE	0, *

Information Bindings:

Type	Association Name	Association Ends					
		Class	Role	Mult	Class	Role	Mult
association	Aton Status				AtonStatusInformation	Statuspart	0, 1

Feature Bindings:

Type	Association Name	Association Ends					
		Class	Role	Mult	Class	Role	Mult
Asso	Aton Aggregations	AidsToNavigation	atonAggregation By	0, *	AtonAggregation	peerAtonAggregation	0, *
Asso	Aton Associations	AidsToNavigation	atonAssociation By	0, *	AtonAssociation	peerAtonAssociation	0, *

2.2. Structure Object

Definition: Something (such as a house, tower, bridge, etc.) that is built by putting parts together and that usually stands on its own.

CamelCase: StructureObject

Alias:

Super type: AidsToNavigation

Feature use type: geographic

Primitive: noGeometry

Remarks: No remarks.

Attribute Bindings:

S-10x Attribute	S-57 Acronym	Allowable Encoding Value	Type	Multiplicity
AtoN Number			UI	1, 1
Aid Availability Category		1 : Category 1 2 : Category 2 3 : Category 3	EN	0, 1
Condition	(CONDTN)	1 : Under Construction 2 : Ruined 3 : Under Reclamation 4 : Wingless 5 : Planned Construction	EN	0, 1
Postal Code	(POSCOD) (Postcode) (ZIP Code)		(S) TE	0, 1

Information Bindings:

Type	Association Name	Association Ends					
		Class	Role	Mult	Class	Role	Mult
association	Aton Positioning Information Association				PositioningInformation	positioningMethod	0, *
association	Aton Fixing Method				AtoNFixingMet	fixingMethod	0, *

ation	Association				hod		
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Feature Bindings:

Type	Association Name	Association Ends					
		Class	Role	Mult	Class	Role	Mult
Asso	Structure Equipment	StructureObject	parent	1, 1	Equipment	child	0, *

2.3. Equipment

Definition: The implements used in an operation or activity.

CamelCase: Equipment

Alias:

Super type: AidsToNavigation

Feature use type: geographic

Primitive: noGeometry

Remarks: No remarks.

Attribute Bindings:

S-10x Attribute	S-57 Acronym	Allowable Encoding Value	Type	Multiplicity
Remote Monitoring System			TE	0, *

Feature Bindings:

Type	Association Name	Association Ends					
		Class	Role	Mult	Class	Role	Mult
Asso	Structure Equipment	Equipment	child	0, *	StructureObject	parent	1, 1

2.4. Electronic Aton

Definition: TBD

CamelCase: ElectronicAton

Alias:

Super type: AidsToNavigation

Feature use type: geographic

Primitive: noGeometry

Remarks: No remarks.

Attribute Bindings:

S-10x Attribute	S-57 Acronym	Allowable Encoding Value	Type	Multiplicity
AtoN Number			UI	0, 1
MMSI Code			TE	1, 1
Status	(STATUS)		EN	0, *

2.5. Generic Beacon

Definition: A fixed artificial navigation mark that can be recognized by its shape, colour, pattern, topmark or light character, or a combination of these. It may carry various additional aids to navigation.

CamelCase: GenericBeacon

Alias:

Super type: StructureObject

Feature use type: geographic

Primitive: noGeometry

Remarks: This term is not commonly used when the navigation mark can be classified as a lighthouse.

Attribute Bindings:

S-10x Attribute	S-57 Acronym	Allowable Encoding Value	Type	Multiplicity
Beacon Shape	(BCNSHP)	1 : Stake, Pole, Perch, Post 2 : Withy 3 : Beacon Tower 4 : Lattice Beacon 5 : Pile Beacon 6 : Cairn 7 : Buoyant Beacon	EN	1, 1
Colour	(COLOUR)	1 : White 2 : Black 3 : Red 4 : Green 5 : Blue 6 : Yellow 7 : Grey 8 : Brown 9 : Amber 10 : Violet 11 : Orange 12 : Magenta 13 : Pink	EN	1, *
Colour Pattern	(COLPAT)	1 : Horizontal Stripes 2 : Vertical Stripes 3 : Diagonal Stripes	EN	0, *

		4 : Squared 5 : Stripes (Direction Unknown) 6 : Border Stripe 7 : Single Colour 8 : Rectangle 9 : Triangle		
Elevation	(ELEVAT)		RE	0, 1
Height	(HEIGHT)		RE	0, 1
Marks Navigational - System Of	(MARSYS)	1 : IALA A 2 : IALA B 9 : No System 10 : Other System 11 : CEVNI 12 : Russian Inland Waterway Regulations 13 : Brazilian National Inland Waterway Regulations - Two Sides 14 : Brazilian National Inland Waterway Regulations - Side Independent 15 : Paraguay-Parana Waterway - Brazilian Complementary Aids	EN	0, 1
Nature of Construction	(NATCON)	1 : Masonry 2 : Concreted 3 : Loose Boulders 4 : Hard Surfaced 5 : Unsurfaced 6 : Wooden 7 : Metal 8 : Glass Reinforced Plastic 9 : Painted 10 : Framework 11 : Latticed 12 : Glass 13 : Fiberglass 14 : Plastic	EN	0, *
Radar Conspicuous	(CONRAD)		BO	0, 1
Status	(STATUS)	1 : Permanent 2 : Occasional 3 : Recommended	EN	0, *

		4 : Not in Use 5 : Periodic/Intermittent 6 : Reserved 7 : Temporary 8 : Private 9 : Mandatory 11 : Extinguished 12 : Illuminated 13 : Historic 14 : Public 15 : Synchronized 16 : Watched 17 : Unwatched 18 : Existence Doubtful 19 : On Request 20 : Drop Away 21 : Rising 22 : Increasing 23 : Decreasing 24 : Strong 25 : Good 26 : Moderately 27 : Poor 28 : Buoyed 29 : Fully Operational 30 : Partially Operational 31 : Drifting 32 : Broken 33 : Offline 34 : Discontinued 35 : Manual Observation 36 : Unknown Status 37 : Confirmed 38 : Candidate 39 : Under Modification 41 : Under Removal / Deletion 42 : Removed / Deleted 43 : Candidate for Modification		
Vertical Length	(VERLEN)		RE	0, 1
Visual Prominence	(CONVIS)	1 : Visually Conspicuous	EN	0, 1

		2 : Not Visually Conspicuous 3 : Prominent		
Vertical Accuracy			RE	0, 1

2.6. Generic Buoy

Definition: A floating object moored to the bottom in a particular (charted) place, as an aid to navigation or for other specific purposes.

CamelCase: GenericBuoy

Alias:

Super type: StructureObject

Feature use type: geographic

Primitive: noGeometry

Remarks: Navigational buoys may be classified according to: (a) their shape, appearance, or construction, such as barrel, can, cask, conical, cylindrical, dan, keg, nun, pillar, spar, spherical, or topmark buoy; (b) their colour, such as black, chequered, green, red buoy; (c) their location, such as bifurcation, fairway, junction, mid-channel, middle-ground, or turning buoy; (d) the various kinds of hazards or dangers to navigation which they mark, such as bar, isolated danger, fish trap, obstruction, spoil ground, telegraph or wreck buoy; (e) their particular purpose or use, such as anchor, anchorage, compass adjustment, dredging, farewell (or landfall), marker, quarantine, station (or watch), or warping buoy.

Attribute Bindings:

S-10x Attribute	S-57 Acronym	Allowable Encoding Value	Type	Multiplicity
Buoy Shape	(BOYSHP)	1 : Conical 2 : Can 3 : Spherical 4 : Pillar 5 : Spar 6 : Barrel 7 : Superbuoy 8 : Ice Buoy	EN	1, 1
Colour	(COLOUR)	1 : White 2 : Black 3 : Red 4 : Green 5 : Blue 6 : Yellow 7 : Grey 8 : Brown 9 : Amber 10 : Violet 11 : Orange 12 : Magenta	EN	1, *

		13 : Pink		
Colour Pattern	(COLPAT)	1 : Horizontal Stripes 2 : Vertical Stripes 3 : Diagonal Stripes 4 : Squared 5 : Stripes (Direction Unknown) 6 : Border Stripe 7 : Single Colour 8 : Rectangle 9 : Triangle	EN	0, *
Marks Navigational - System Of	(MARSYS)	1 : IALA A 2 : IALA B 9 : No System 10 : Other System 11 : CEVNI 12 : Russian Inland Waterway Regulations 13 : Brazilian National Inland Waterway Regulations - Two Sides 14 : Brazilian National Inland Waterway Regulations - Side Independent 15 : Paraguay-Parana Waterway - Brazilian Complementary Aids	EN	0, 1
Nature of Construction	(NATCON)	1 : Masonry 2 : Concreted 3 : Loose Boulders 4 : Hard Surfaced 5 : Unsurfaced 6 : Wooden 7 : Metal 8 : Glass Reinforced Plastic 9 : Painted 10 : Framework 11 : Latticed 12 : Glass 13 : Fiberglass 14 : Plastic	EN	0, *
Radar Conspicuous	(CONRAD)		BO	0, 1
Status	(STATUS)	1 : Permanent	EN	0, *

		2 : Occasional 3 : Recommended 4 : Not in Use 5 : Periodic/Intermittent 6 : Reserved 7 : Temporary 8 : Private 9 : Mandatory 11 : Extinguished 12 : Illuminated 13 : Historic 14 : Public 15 : Synchronized 16 : Watched 17 : Unwatched 18 : Existence Doubtful 19 : On Request 20 : Drop Away 21 : Rising 22 : Increasing 23 : Decreasing 24 : Strong 25 : Good 26 : Moderately 27 : Poor 28 : Buoyed 29 : Fully Operational 30 : Partially Operational 31 : Drifting 32 : Broken 33 : Offline 34 : Discontinued 35 : Manual Observation 36 : Unknown Status 37 : Confirmed 38 : Candidate 39 : Under Modification 41 : Under Removal / Deletion 42 : Removed / Deleted 43 : Candidate for Modification		
Type of Buoy			TE	0, 1

Vertical Length	(VERLEN)		RE	0, 1
Vertical Accuracy			RE	0, 1

Feature Bindings:

Type	Association Name	Association Ends					
		Class	Role	Mult	Class	Role	Mult
Asso	Buoy Topmark	GenericBuoy	buoyPart	1, 1	Topmark	topmarkPart	0, *
Asso	Shackle Connection	GenericBuoy	shackleToBuoyconnectedTo	0, 1	MooringShackle	shackleToBuoyconnected	1, 1
Asso	Bridle Connection	GenericBuoy	bridleholds	1, 1	Bridle	buoyhangs	0, 1
Asso	Buoy Counter Weight	GenericBuoy	counterWeightholds	1, 1	CounterWeight	buoyattached	0, 1

2.7. Generic Light

Definition: -

CamelCase: GenericLight

Alias:

Super type: Equipment

Feature use type: geographic

Primitive: noGeometry

Remarks: -

Attribute Bindings:

S-10x Attribute	S-57 Acronym	Allowable Encoding Value	Type	Multiplicity
Height	(HEIGHT)		RE	0, 1
Status	(STATUS)	1 : Permanent 2 : Occasional 3 : Recommended 4 : Not in Use 5 : Periodic/Intermittent 6 : Reserved 7 : Temporary 8 : Private 9 : Mandatory 11 : Extinguished 12 : Illuminated 13 : Historic 14 : Public 15 : Synchronized 16 : Watched 17 : Unwatched 18 : Existence Doubtful 19 : On Request 20 : Drop Away 21 : Rising 22 : Increasing 23 : Decreasing 24 : Strong 25 : Good	EN	0, *

		26 : Moderately 27 : Poor 28 : Buoyed 29 : Fully Operational 30 : Partially Operational 31 : Drifting 32 : Broken 33 : Offline 34 : Discontinued 35 : Manual Observation 36 : Unknown Status 37 : Confirmed 38 : Candidate 39 : Under Modification 41 : Under Removal / Deletion 42 : Removed / Deleted 43 : Candidate for Modification		
Vertical Datum	(VERDAT) (Datum Level) (Reference Plane) (Levelling Datum) (Datum for Sounding Reduction) (Datum for Heights)	1 : Mean Low Water Springs 2 : Mean Lower Low Water Springs 3 : Mean Sea Level 4 : Lowest Low Water 5 : Mean Low Water 6 : Lowest Low Water Springs 7 : Approximate Mean Low Water Springs 8 : Indian Spring Low Water 9 : Low Water Springs 10 : Approximate Lowest Astronomical Tide 11 : Nearly Lowest Low Water 12 : Mean Lower Low Water 13 : Low Water 14 : Approximate Mean Low Water 15 : Approximate Mean Lower Low Water 16 : Mean High Water 17 : Mean High Water Springs 18 : High Water 19 : Approximate Mean Sea	EN	0, 1

		Level 20 : High Water Springs 21 : Mean Higher High Water 22 : Equinoctial Spring Low Water 23 : Lowest Astronomical Tide 24 : Local Datum 25 : International Great Lakes Datum 1985 26 : Mean Water Level 27 : Lower Low Water Large Tide 28 : Higher High Water Large Tide 29 : Nearly Highest High Water 30 : Highest Astronomical Tide 31 : Local Low Water Reference Level 32 : Local High Water Reference Level 33 : Local Mean Water Reference Level 34 : Equivalent Height of Water (German GIW) 35 : Highest Shipping Height of Water (German HSW) 36 : Reference Low Water Level According to Danube Commission 37 : Highest Shipping Height of Water According to Danube Commission 38 : Dutch River Low Water Reference Level (OLR) 39 : Russian Project Water Level 40 : Russian Normal Backwater Level 41 : Ohio River Datum 43 : Dutch High Water Reference Level 44 : Baltic Sea Chart Datum 2000 45 : Dutch Estuary Low Water Reference Level (OLW) 46 : International Great Lakes Datum 2020		
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		47 : Sea Floor 48 : Sea Surface 49 : Hydrographic Zero		
Vertical Length	(VERLEN)		RE	0, 1
Effective Intensity			RE	0, 1
Peak Intensity			RE	0, 1

2.8. Landmark

Definition: A prominent object at a fixed location on land which can be used in determining a location or a direction.

CamelCase: Landmark

Alias: LNDMRK

Super type: StructureObject

Feature use type: geographic

Primitive: point curve surface

Remarks: No remarks.

Attribute Bindings:

S-10x Attribute	S-57 Acronym	Allowable Encoding Value	Type	Multiplicity
Category of Landmark	(CATLMK)	1 : Cairn 2 : Cemetery 3 : Chimney 4 : Dish Aerial 5 : Flagstaff 6 : Flare Stack 7 : Mast 8 : Windsock 9 : Monument 10 : Column/Pillar 11 : Memorial Plaque 12 : Obelisk 13 : Statue 14 : Cross 15 : Dome 16 : Radar Scanner 17 : Tower 18 : Windmill 19 : Windmotor 20 : Spire/Minaret 21 : Large Rock or Boulder on Land 22 : Triangulation Mark 23 : Boundary Mark 24 : Observation Wheel	EN	1, *

		25 : Torii 26 : Bridge 27 : Dam		
Colour	(COLOUR)	1 : White 2 : Black 3 : Red 4 : Green 5 : Blue 6 : Yellow 7 : Grey 8 : Brown 9 : Amber 10 : Violet 11 : Orange 12 : Magenta 13 : Pink	EN	0, *
Colour Pattern	(COLPAT)	1 : Horizontal Stripes 2 : Vertical Stripes 3 : Diagonal Stripes 4 : Squared 5 : Stripes (Direction Unknown) 6 : Border Stripe 7 : Single Colour 8 : Rectangle 9 : Triangle	EN	0, *
Function	(FUNCTN)	2 : Harbour-Masters Office 3 : Customs Office 4 : Health Office 5 : Hospital 6 : Post Office 7 : Hotel 8 : Railway Station 9 : Police Station 10 : Water-Police Station 11 : Pilot Office 12 : Pilot Lookout 13 : Bank Office 14 : Headquarters for District Control 15 : Transit Shed/Warehouse	EN	0, *

		16 : Factory 17 : Power Station 18 : Administrative 19 : Educational Facility 20 : Church 21 : Chapel 22 : Temple 23 : Pagoda 24 : Shinto Shrine 25 : Buddhist Temple 26 : Mosque 27 : Marabout 28 : Lookout 29 : Communication 30 : Television 31 : Radio 32 : Radar 33 : Light Support 34 : Microwave 35 : Cooling 36 : Observation 37 : Timeball 38 : Clock 39 : Control 40 : Airship Mooring 41 : Stadium 42 : Bus Station 43 : Passenger Terminal Building 44 : Sea Rescue Control 45 : Observatory 46 : Ore Crusher 47 : Boathouse 48 : Pumping Station		
Nature of Construction	(NATCON)	1 : Masonry 2 : Concreted 3 : Loose Boulders 4 : Hard Surfaced 5 : Unsurfaced 6 : Wooden 7 : Metal 8 : Glass Reinforced Plastic	EN	0, *

		9 : Painted 10 : Framework 11 : Latticed 12 : Glass 13 : Fiberglass 14 : Plastic		
Radar Conspicuous	(CONRAD)		BO	0, 1
Status	(STATUS)	1 : Permanent 2 : Occasional 3 : Recommended 4 : Not in Use 5 : Periodic/Intermittent 6 : Reserved 7 : Temporary 8 : Private 9 : Mandatory 11 : Extinguished 12 : Illuminated 13 : Historic 14 : Public 15 : Synchronized 16 : Watched 17 : Unwatched 18 : Existence Doubtful 19 : On Request 20 : Drop Away 21 : Rising 22 : Increasing 23 : Decreasing 24 : Strong 25 : Good 26 : Moderately 27 : Poor 28 : Buoyed 29 : Fully Operational 30 : Partially Operational 31 : Drifting 32 : Broken 33 : Offline 34 : Discontinued 35 : Manual Observation 36 : Unknown Status	EN	0, *

		37 : Confirmed 38 : Candidate 39 : Under Modification 41 : Under Removal / Deletion 42 : Removed / Deleted 43 : Candidate for Modification		
Vertical Datum	(VERDAT) (Datum Level) (Reference Plane) (Levelling Datum) (Datum for Sounding Reduction) (Datum for Heights)	1 : Mean Low Water Springs 2 : Mean Lower Low Water Springs 3 : Mean Sea Level 4 : Lowest Low Water 5 : Mean Low Water 6 : Lowest Low Water Springs 7 : Approximate Mean Low Water Springs 8 : Indian Spring Low Water 9 : Low Water Springs 10 : Approximate Lowest Astronomical Tide 11 : Nearly Lowest Low Water 12 : Mean Lower Low Water 13 : Low Water 14 : Approximate Mean Low Water 15 : Approximate Mean Lower Low Water 16 : Mean High Water 17 : Mean High Water Springs 18 : High Water 19 : Approximate Mean Sea Level 20 : High Water Springs 21 : Mean Higher High Water 22 : Equinoctial Spring Low Water 23 : Lowest Astronomical Tide 24 : Local Datum 25 : International Great Lakes Datum 1985 26 : Mean Water Level 27 : Lower Low Water Large	EN	0, 1

		Tide 28 : Higher High Water Large Tide 29 : Nearly Highest High Water 30 : Highest Astronomical Tide 31 : Local Low Water Reference Level 32 : Local High Water Reference Level 33 : Local Mean Water Reference Level 34 : Equivalent Height of Water (German GIW) 35 : Highest Shipping Height of Water (German HSW) 36 : Reference Low Water Level According to Danube Commission 37 : Highest Shipping Height of Water According to Danube Commission 38 : Dutch River Low Water Reference Level (OLR) 39 : Russian Project Water Level 40 : Russian Normal Backwater Level 41 : Ohio River Datum 43 : Dutch High Water Reference Level 44 : Baltic Sea Chart Datum 2000 45 : Dutch Estuary Low Water Reference Level (OLW) 46 : International Great Lakes Datum 2020 47 : Sea Floor 48 : Sea Surface 49 : Hydrographic Zero		
Visual Prominence	(CONVIS)	1 : Visually Conspicuous 2 : Not Visually Conspicuous 3 : Prominent	EN	1, 1
Elevation	(ELEVAT)		RE	0, 1
Height	(HEIGHT)		RE	0, 1
Manned Structure			BO	0, 1

Vertical Length	(VERLEN)		RE	0, 1
Vertical Accuracy			RE	0, 1

2.9. Lateral Beacon

Definition: A lateral beacon is used to indicate the port or starboard hand side of the route to be followed. They are generally used for well defined channels and are used in conjunction with a conventional direction of buoyage.

CamelCase: LateralBeacon

Alias:

Super type: GenericBeacon

Feature use type: geographic

Primitive: point

Remarks: -

Attribute Bindings:

S-10x Attribute	S-57 Acronym	Allowable Encoding Value	Type	Multiplicity
Category of Lateral Mark	(CATLAM)	1 : Port-Hand Lateral Mark 2 : Starboard-Hand Lateral Mark 3 : Preferred Channel to Starboard Lateral Mark 4 : Preferred Channel to Port Lateral Mark	EN	1, 1

2.10. Lateral Buoy

Definition: A lateral buoy is used to indicate the port or starboard hand side of the route to be followed. They are generally used for well-defined channels and are used in conjunction with a conventional direction of buoyage.

CamelCase: LateralBuoy

Alias:

Super type: GenericBuoy

Feature use type: geographic

Primitive: point

Remarks: -

Attribute Bindings:

S-10x Attribute	S-57 Acronym	Allowable Encoding Value	Type	Multiplicity
Category of Lateral Mark	(CATLAM)	1 : Port-Hand Lateral Mark 2 : Starboard-Hand Lateral Mark 3 : Preferred Channel to Starboard Lateral Mark 4 : Preferred Channel to Port Lateral Mark	EN	1, 1

2.11. Navigation Line

Definition: A straight line extending towards an area of navigational interest and generally generated by two navigational aids or one navigational aid and a bearing.

CamelCase: NavigationLine

Alias: NAVLNE

Super type: AidsToNavigation

Feature use type: geographic

Primitive: curve

Remarks: The extent of the navigation line depends on the visibility of the navigational aid(s). The recommended track is that portion of a 'navigation line' that a ship should use for navigation.

Attribute Bindings:

S-10x Attribute	S-57 Acronym	Allowable Encoding Value	Type	Multiplicity
Category of Navigation Line	(CATNAV)	1 : Clearing Line 2 : Transit Line 3 : Leading Line Bearing a Recommended Track	EN	1, 1
Status	(STATUS)	1 : Permanent 2 : Occasional 3 : Recommended 4 : Not in Use 5 : Periodic/Intermittent 6 : Reserved 7 : Temporary 8 : Private 9 : Mandatory 11 : Extinguished 12 : Illuminated 13 : Historic 14 : Public 15 : Synchronized 16 : Watched 17 : Unwatched 18 : Existence Doubtful 19 : On Request 20 : Drop Away 21 : Rising	EN	0, *

		22 : Increasing 23 : Decreasing 24 : Strong 25 : Good 26 : Moderately 27 : Poor 28 : Buoyed 29 : Fully Operational 30 : Partially Operational 31 : Drifting 32 : Broken 33 : Offline 34 : Discontinued 35 : Manual Observation 36 : Unknown Status 37 : Confirmed 38 : Candidate 39 : Under Modification 41 : Under Removal / Deletion 42 : Removed / Deleted 43 : Candidate for Modification		
Orientation Value	(ORIENT)		(S) RE	1, 1

Feature Bindings:

Type	Association Name	Association Ends					
		Class	Role	Mult	Class	Role	Mult
Asso	Range System	NavigationLine	navigationLine	1, *	Recommended Track	navigableTrack	0, *

2.12. Recommended Track

Definition: A route which has been specially examined to ensure so far as possible that it is free of dangers and along which ships are advised to navigate.

CamelCase: RecommendedTrack

Alias: RECTRC

Super type: AidsToNavigation

Feature use type: geographic

Primitive: curve

Remarks: Recommended tracks include all channels recommended for hydrographic reasons to lead safely between shoal depths. The use of such tracks is generally left to the discretion of the mariner and will depend on the vessel's draught, the state of the tide, adequacy of navigational aids and so on (IHO Chart Specifications, M-4). The recommended track is that portion of a 'navigation line' that a ship should use for navigation. In the case of a two-way recommended track only one value of orientation is encoded; the other value can be deduced (that is, the value encoded + 180 degrees).

Attribute Bindings:

S-10x Attribute	S-57 Acronym	Allowable Encoding Value	Type	Multiplicity
Based On Fixed Marks	(CATTRK=1)		BO	1, 1
Depth Range Minimum Value	(DRVAL1)		RE	0, 1
Maximal Permitted Draught	(lg_drt)		RE	0, 1
Status	(STATUS)	1 : Permanent 2 : Occasional 3 : Recommended 4 : Not in Use 5 : Periodic/Intermittent 6 : Reserved 7 : Temporary 8 : Private 9 : Mandatory 11 : Extinguished 12 : Illuminated 13 : Historic 14 : Public 15 : Synchronized 16 : Watched 17 : Unwatched 18 : Existence Doubtful	EN	0, *

		19 : On Request 20 : Drop Away 21 : Rising 22 : Increasing 23 : Decreasing 24 : Strong 25 : Good 26 : Moderately 27 : Poor 28 : Buoyed 29 : Fully Operational 30 : Partially Operational 31 : Drifting 32 : Broken 33 : Offline 34 : Discontinued 35 : Manual Observation 36 : Unknown Status 37 : Confirmed 38 : Candidate 39 : Under Modification 41 : Under Removal / Deletion 42 : Removed / Deleted 43 : Candidate for Modification		
Vertical Datum	(VERDAT) (Datum Level) (Reference Plane) (Levelling Datum) (Datum for Sounding Reduction) (Datum for Heights)	1 : Mean Low Water Springs 2 : Mean Lower Low Water Springs 3 : Mean Sea Level 4 : Lowest Low Water 5 : Mean Low Water 6 : Lowest Low Water Springs 7 : Approximate Mean Low Water Springs 8 : Indian Spring Low Water 9 : Low Water Springs 10 : Approximate Lowest Astronomical Tide 11 : Nearly Lowest Low Water 12 : Mean Lower Low Water 13 : Low Water 14 : Approximate Mean Low	EN	0, 1

		Water 15 : Approximate Mean Lower Low Water 16 : Mean High Water 17 : Mean High Water Springs 18 : High Water 19 : Approximate Mean Sea Level 20 : High Water Springs 21 : Mean Higher High Water 22 : Equinoctial Spring Low Water 23 : Lowest Astronomical Tide 24 : Local Datum 25 : International Great Lakes Datum 1985 26 : Mean Water Level 27 : Lower Low Water Large Tide 28 : Higher High Water Large Tide 29 : Nearly Highest High Water 30 : Highest Astronomical Tide 31 : Local Low Water Reference Level 32 : Local High Water Reference Level 33 : Local Mean Water Reference Level 34 : Equivalent Height of Water (German GIW) 35 : Highest Shipping Height of Water (German HSW) 36 : Reference Low Water Level According to Danube Commission 37 : Highest Shipping Height of Water According to Danube Commission 38 : Dutch River Low Water Reference Level (OLR) 39 : Russian Project Water Level 40 : Russian Normal Backwater Level 41 : Ohio River Datum		
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		43 : Dutch High Water Reference Level 44 : Baltic Sea Chart Datum 2000 45 : Dutch Estuary Low Water Reference Level (OLW) 46 : International Great Lakes Datum 2020 47 : Sea Floor 48 : Sea Surface 49 : Hydrographic Zero		
Orientation			C	1, 1
Orientation Uncertainty			(S) RE	0, 1
Orientation Value	(ORIENT)		(S) RE	1, 1
Vertical Uncertainty	(VERACC) (SOUACC)		C	0, 1
Traffic Flow	(TRAFFIC)	1 : Inbound 2 : Outbound 3 : One-Way 4 : Two-Way	EN	1, 1

Feature Bindings:

Type	Association Name	Association Ends					
		Class	Role	Mult	Class	Role	Mult
Asso	Range System	Recommended Track	navigableTrack	0, *	NavigationLine	navigationLine	1, *

2.13. Light Sectored

Definition: A light presenting different appearances (in particular, different colours) over various parts of the horizon of interest to maritime navigation.

CamelCase: LightSectored

Alias: LIGHTS Sector Light

Super type: GenericLight

Feature use type: geographic

Primitive: point

Remarks: No remarks.

Attribute Bindings:

S-10x Attribute	S-57 Acronym	Allowable Encoding Value	Type	Multiplicity
Colour	(COLOUR)	1 : White 3 : Red 4 : Green 5 : Blue 6 : Yellow 9 : Amber 10 : Violet 11 : Orange 14 : Green A 15 : Green B 16 : White Temporary 17 : Red Temporary 18 : Yellow Temporary 19 : Green Preferred 20 : Green Temporary	EN	1, *
Category of Light	(CATLIT)	1 : Directional Function 4 : Leading Light 5 : Aero Light 6 : Air Obstruction Light 8 : Flood Light 9 : Strip Light 10 : Subsidiary Light 11 : Spotlight 12 : Front	EN	0, *

		13 : Rear 14 : Lower 15 : Upper 17 : Emergency 18 : Bearing Light 19 : Horizontally Disposed 20 : Vertically Disposed		
Exhibition Condition of Light	(EXCLIT)	1 : Light Shown Without Change of Character 2 : Daytime Light 3 : Fog Light 4 : Night Light	EN	0, 1
Marks Navigational - System Of	(MARSYS)	1 : IALA A 2 : IALA B 9 : No System 10 : Other System 11 : CEVNI 12 : Russian Inland Waterway Regulations 13 : Brazilian National Inland Waterway Regulations - Two Sides 14 : Brazilian National Inland Waterway Regulations - Side Independent 15 : Paraguay-Parana Waterway - Brazilian Complementary Aids	EN	0, 1
Signal Generation	(SIGGEN)	1 : Automatically 2 : By Wave Action 3 : By Hand 4 : By Wind 5 : Radio Activated 6 : Call Activated	EN	0, 1
Obscured Sector			C	0, *
Signal Status		1 : Lit/Sound 2 : Eclipsed/Silent	(S) EN	1, 1

2.14. Light All Around

Definition: An all around light is a light that is visible over the whole horizon of interest to marine navigation and having no change in the characteristics of the light.

CamelCase: LightAllAround

Alias: LIGHTS

Super type: GenericLight

Feature use type: geographic

Primitive: point

Remarks: No remarks.

Attribute Bindings:

S-10x Attribute	S-57 Acronym	Allowable Encoding Value	Type	Multiplicity
Colour	(COLOUR)	1 : White 3 : Red 4 : Green 5 : Blue 6 : Yellow 9 : Amber 10 : Violet 11 : Orange 14 : Green A 15 : Green B 16 : White Temporary 17 : Red Temporary 18 : Yellow Temporary 19 : Green Preferred 20 : Green Temporary	EN	1, *
Category of Light	(CATLIT)	1 : Directional Function 4 : Leading Light 5 : Aero Light 6 : Air Obstruction Light 8 : Flood Light 9 : Strip Light 10 : Subsidiary Light 11 : Spotlight 12 : Front	EN	0, *

		13 : Rear 14 : Lower 15 : Upper 17 : Emergency 18 : Bearing Light 19 : Horizontally Disposed 20 : Vertically Disposed		
Exhibition Condition of Light	(EXCLIT)	1 : Light Shown Without Change of Character 2 : Daytime Light 3 : Fog Light 4 : Night Light	EN	0, 1
Light Visibility	(LITVIS)	1 : High Intensity 2 : Low Intensity 3 : Faint 4 : Intensified 5 : Unintensified 6 : Visibility Deliberately Restricted 7 : Obscured 8 : Partially Obscured 9 : Visible in Line of Range	EN	0, 1
Major Light			BO	0, 1
Marks Navigational - System Of	(MARSYS)	1 : IALA A 2 : IALA B 9 : No System 10 : Other System 11 : CEVNI 12 : Russian Inland Waterway Regulations 13 : Brazilian National Inland Waterway Regulations - Two Sides 14 : Brazilian National Inland Waterway Regulations - Side Independent 15 : Paraguay-Parana Waterway - Brazilian Complementary Aids	EN	0, 1
Signal Generation	(SIGGEN)	1 : Automatically 2 : By Wave Action 3 : By Hand 4 : By Wind 5 : Radio Activated	EN	0, 1

		6 : Call Activated		
Value of Nominal Range	(VALNMR)		RE	0, 1
Multiplicity of Features			C	0, 1
Multiplicity Known			(S) BO	1, 1
Flare Bearing			IN	0, 1

2.15. Light Air Obstruction

Definition: An air obstruction light is a light marking an obstacle which constitutes a danger to air navigation.

CamelCase: LightAirObstruction

Alias: LIGHTS

Super type: GenericLight

Feature use type: geographic

Primitive: point

Remarks: No remarks.

Attribute Bindings:

S-10x Attribute	S-57 Acronym	Allowable Encoding Value	Type	Multiplicity
Exhibition Condition of Light	(EXCLIT)	1 : Light Shown Without Change of Character 2 : Daytime Light 3 : Fog Light 4 : Night Light	EN	0, *
Light Visibility	(LITVIS)	1 : High Intensity 2 : Low Intensity 3 : Faint 4 : Intensified 5 : Unintensified 6 : Visibility Deliberately Restricted 7 : Obscured 8 : Partially Obscured 9 : Visible in Line of Range	EN	0, *
Value of Nominal Range	(VALNMR)		RE	0, 1
Multiplicity of Features			C	0, 1
Multiplicity Known			(S) BO	1, 1
Flare Bearing			IN	0, 1

2.16. Light Fog Detector

Definition: A fog detector light is a light used to automatically determine conditions of visibility which warrant the turning on or off of a sound signal.

CamelCase: LightFogDetector

Alias:

Super type: GenericLight

Feature use type: geographic

Primitive: point

Remarks: No remarks.

Attribute Bindings:

S-10x Attribute	S-57 Acronym	Allowable Encoding Value	Type	Multiplicity
Signal Generation	(SIGGEN)	1 : Automatically 2 : By Wave Action 3 : By Hand 4 : By Wind 5 : Radio Activated 6 : Call Activated	EN	0, 1
Signal Status		1 : Lit/Sound 2 : Eclipsed/Silent	(S) EN	1, 1

2.17. Radar Reflector

Definition: A device capable of, or intended for, reflecting radar signals.

CamelCase: RadarReflector

Alias: RADRFL

Super type: Equipment

Feature use type: geographic

Primitive: point

Remarks: A radar reflector is usually a tetrahedron or pentagonal corner reflector to facilitate reflection towards the sender (International Maritime Dictionary, 2nd Ed.).

Attribute Bindings:

S-10x Attribute	S-57 Acronym	Allowable Encoding Value	Type	Multiplicity
Height	(HEIGHT)		RE	0, 1
Status	(STATUS)	1 : Permanent 2 : Occasional 3 : Recommended 4 : Not in Use 5 : Periodic/Intermittent 6 : Reserved 7 : Temporary 8 : Private 9 : Mandatory 11 : Extinguished 12 : Illuminated 13 : Historic 14 : Public 15 : Synchronized 16 : Watched 17 : Unwatched 18 : Existence Doubtful 19 : On Request 20 : Drop Away 21 : Rising 22 : Increasing 23 : Decreasing 24 : Strong	EN	0, *

		25 : Good 26 : Moderately 27 : Poor 28 : Buoyed 29 : Fully Operational 30 : Partially Operational 31 : Drifting 32 : Broken 33 : Offline 34 : Discontinued 35 : Manual Observation 36 : Unknown Status 37 : Confirmed 38 : Candidate 39 : Under Modification 41 : Under Removal / Deletion 42 : Removed / Deleted 43 : Candidate for Modification		
Vertical Datum	(VERDAT) (Datum Level) (Reference Plane) (Levelling Datum) (Datum for Sounding Reduction) (Datum for Heights)	1 : Mean Low Water Springs 2 : Mean Lower Low Water Springs 3 : Mean Sea Level 4 : Lowest Low Water 5 : Mean Low Water 6 : Lowest Low Water Springs 7 : Approximate Mean Low Water Springs 8 : Indian Spring Low Water 9 : Low Water Springs 10 : Approximate Lowest Astronomical Tide 11 : Nearly Lowest Low Water 12 : Mean Lower Low Water 13 : Low Water 14 : Approximate Mean Low Water 15 : Approximate Mean Lower Low Water 16 : Mean High Water 17 : Mean High Water Springs 18 : High Water	EN	0, 1

		19 : Approximate Mean Sea Level 20 : High Water Springs 21 : Mean Higher High Water 22 : Equinoctial Spring Low Water 23 : Lowest Astronomical Tide 24 : Local Datum 25 : International Great Lakes Datum 1985 26 : Mean Water Level 27 : Lower Low Water Large Tide 28 : Higher High Water Large Tide 29 : Nearly Highest High Water 30 : Highest Astronomical Tide 31 : Local Low Water Reference Level 32 : Local High Water Reference Level 33 : Local Mean Water Reference Level 34 : Equivalent Height of Water (German GIW) 35 : Highest Shipping Height of Water (German HSW) 36 : Reference Low Water Level According to Danube Commission 37 : Highest Shipping Height of Water According to Danube Commission 38 : Dutch River Low Water Reference Level (OLR) 39 : Russian Project Water Level 40 : Russian Normal Backwater Level 41 : Ohio River Datum 43 : Dutch High Water Reference Level 44 : Baltic Sea Chart Datum 2000 45 : Dutch Estuary Low Water Reference Level (OLW) 46 : International Great		
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		Lakes Datum 2020 47 : Sea Floor 48 : Sea Surface 49 : Hydrographic Zero		
Vertical Accuracy			RE	0, 1

2.18. Fog Signal

Definition: A warning signal transmitted by a vessel, or aid to navigation, during periods of low visibility. Also, the device producing such a signal.

CamelCase: FogSignal

Alias: FOGSIG

Super type: Equipment

Feature use type: geographic

Primitive: point

Remarks: No remarks.

Attribute Bindings:

S-10x Attribute	S-57 Acronym	Allowable Encoding Value	Type	Multiplicity
Category of Fog Signal	(CATFOG)	1 : Explosive 2 : Diaphone 3 : Siren 4 : Nautophone 5 : Reed 6 : Tyfon 7 : Bell 8 : Whistle 9 : Gong 10 : Horn	EN	1, 1
Signal Frequency	(SIGFRQ)		IN	0, 1
Signal Generation	(SIGGEN)	1 : Automatically 2 : By Wave Action 3 : By Hand 4 : By Wind 5 : Radio Activated 6 : Call Activated	EN	0, 1
Signal Group	(SIGGRP)		TE	0, 1
Signal Output			RE	0, 1
Signal Period	(SIGPER)		RE	0, 1
Status	(STATUS)	1 : Permanent 2 : Occasional	EN	0, *

		3 : Recommended 4 : Not in Use 5 : Periodic/Intermittent 6 : Reserved 7 : Temporary 8 : Private 9 : Mandatory 11 : Extinguished 12 : Illuminated 13 : Historic 14 : Public 15 : Synchronized 16 : Watched 17 : Unwatched 18 : Existence Doubtful 19 : On Request 20 : Drop Away 21 : Rising 22 : Increasing 23 : Decreasing 24 : Strong 25 : Good 26 : Moderately 27 : Poor 28 : Buoyed 29 : Fully Operational 30 : Partially Operational 31 : Drifting 32 : Broken 33 : Offline 34 : Discontinued 35 : Manual Observation 36 : Unknown Status 37 : Confirmed 38 : Candidate 39 : Under Modification 41 : Under Removal / Deletion 42 : Removed / Deleted 43 : Candidate for Modification		
Value of Maximum Range	(VALMXR)		RE	0, 1

Signal Status		1 : Lit/Sound 2 : Eclipsed/Silent	(S) EN	1, 1
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2.19. Environment Observation Equipment

Definition: A sensor used to observe the environment.

CamelCase: EnvironmentObservationEquipment

Alias:

Super type: Equipment

Feature use type: geographic

Primitive: point

Remarks: No remarks.

Attribute Bindings:

S-10x Attribute	S-57 Acronym	Allowable Encoding Value	Type	Multiplicity
Height	(HEIGHT)		RE	0, 1
Status	(STATUS)	1 : Permanent 2 : Occasional 3 : Recommended 4 : Not in Use 5 : Periodic/Intermittent 6 : Reserved 7 : Temporary 8 : Private 9 : Mandatory 11 : Extinguished 12 : Illuminated 13 : Historic 14 : Public 15 : Synchronized 16 : Watched 17 : Unwatched 18 : Existence Doubtful 19 : On Request 20 : Drop Away 21 : Rising 22 : Increasing 23 : Decreasing 24 : Strong 25 : Good	EN	0, *

		26 : Moderately 27 : Poor 28 : Buoyed 29 : Fully Operational 30 : Partially Operational 31 : Drifting 32 : Broken 33 : Offline 34 : Discontinued 35 : Manual Observation 36 : Unknown Status 37 : Confirmed 38 : Candidate 39 : Under Modification 41 : Under Removal / Deletion 42 : Removed / Deleted 43 : Candidate for Modification		
Type of Environmental Observation Equipment			TE	1, *

2.20. Radio Station

Definition: A place equipped to transmit radio waves. Such a station may be either stationary or mobile, and may also be provided with a radio receiver.

CamelCase: RadioStation

Alias: RDOSTA W/T Station

Super type: Equipment

Feature use type: geographic

Primitive: point

Remarks: The transmission of a radio station may serve to provide mariners with a line of position (IHO Chart Specifications, M-4). The feature 'radio station' is used to encode the point of transmission of the signal.

Attribute Bindings:

S-10x Attribute	S-57 Acronym	Allowable Encoding Value	Type	Multiplicity
Category of Radio Station	(CATROS)	1 : Circular (Non-Directional) Marine or Aero-Marine Radiobeacon 2 : Directional Radiobeacon 3 : Rotating Pattern Radiobeacon 4 : Consol Beacon 5 : Radio Direction-Finding Station 6 : Coast Radio Station Providing QTG Service 7 : Aeronautical Radiobeacon 8 : Decca 9 : Loran C 10 : Differential GNSS 11 : Toran 12 : Omega 13 : Syledis 14 : Chaika 19 : Radio Telephone Station 20 : AIS Base Station	EN	1, 1
Estimated Range of Transmission	(ESTRNG)		RE	0, 1
Status	(STATUS)	1 : Permanent 2 : Occasional	EN	0, 1

		3 : Recommended 4 : Not in Use 5 : Periodic/Intermittent 6 : Reserved 7 : Temporary 8 : Private 9 : Mandatory 11 : Extinguished 12 : Illuminated 13 : Historic 14 : Public 15 : Synchronized 16 : Watched 17 : Unwatched 18 : Existence Doubtful 19 : On Request 20 : Drop Away 21 : Rising 22 : Increasing 23 : Decreasing 24 : Strong 25 : Good 26 : Moderately 27 : Poor 28 : Buoyed 29 : Fully Operational 30 : Partially Operational 31 : Drifting 32 : Broken 33 : Offline 34 : Discontinued 35 : Manual Observation 36 : Unknown Status 37 : Confirmed 38 : Candidate 39 : Under Modification 41 : Under Removal / Deletion 42 : Removed / Deleted 43 : Candidate for Modification		
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Feature Bindings:

Type	Association Name	Association Ends					
		Class	Role	Mult	Class	Role	Mult
Asso	Physical AIS	RadioStation	physicalAISbroa dcasts	0, *	PhysicalAISaid ToNavigation	physicalAISbroa dcastBy	0, 1
Asso	Synthetic AIS	RadioStation	syntheticAISbro adcasts	0, *	SyntheticAISai dToNavigation	syntheticAISbro adcastBy	0, 1
Asso	Virtual AIS	RadioStation	virtualAISbroadc asts	0, *	VirtualAISaidT oNavigation	virtualAISbroadc astBy	0, 1

2.21. Daymark

Definition: (1) The identifying characteristics of an aid to navigation which serve to facilitate its recognition against a daylight viewing background. On those structures that do not by themselves present an adequate viewing area to be seen at the required distance, the aid is made more visible by affixing a daymark to the structure. A daymark so affixed has a distinctive colour and shape depending on the purpose of the aid. (2) An unlighted navigational mark.

CamelCase: Daymark

Alias: DAYMAR

Super type: Equipment

Feature use type: geographic

Primitive: point

Remarks: No remarks.

Attribute Bindings:

S-10x Attribute	S-57 Acronym	Allowable Encoding Value	Type	Multiplicity
Category of Special Purpose Mark	(CATSPM)	1 : Firing Danger Mark 2 : Target Mark 3 : Marker Ship Mark 4 : Degaussing Range Mark 5 : Barge Mark 6 : Cable Mark 7 : Spoil Ground Mark 8 : Outfall Mark 9 : ODAS 10 : Recording Mark 11 : Seaplane Anchorage Mark 12 : Recreation Zone Mark 13 : Private Mark 14 : Mooring Mark 15 : LANBY 16 : Leading Mark 17 : Measured Distance Mark 18 : Notice Mark 19 : TSS Mark 20 : Anchoring Prohibited Mark 21 : Berthing Prohibited	EN	0, 1

		Mark 22 : Overtaking Prohibited Mark 23 : Two-Way Traffic Prohibited Mark 24 : Reduced Wake Mark 25 : Speed Limit Mark 26 : Stop Mark 27 : General Warning Mark 28 : Sound Ship's Siren Mark 29 : Restricted Vertical Clearance Mark 30 : Maximum Vessel's Draught Mark 31 : Restricted Horizontal Clearance Mark 32 : Strong Current Warning Mark 33 : Berthing Permitted Mark 34 : Overhead Power Cable Mark 35 : Channel Edge Gradient Mark 36 : Telephone Mark 37 : Ferry Crossing Mark 39 : Pipeline Mark 40 : Anchorage Mark 41 : Clearing Mark 42 : Control Mark 43 : Diving Mark 44 : Refuge Beacon 45 : Foul Ground Mark 46 : Yachting Mark 47 : Heliport Mark 48 : GNSS Mark 49 : Seaplane Landing Mark 50 : Entry Prohibited Mark 51 : Work in Progress Mark 52 : Mark With Unknown Purpose 53 : Wellhead Mark 54 : Channel Separation Mark 55 : Marine Farm Mark 56 : Artificial Reef Mark		
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		57 : Ice Mark 58 : Nature Reserve Mark 59 : Fish Aggregating Device 60 : Wreck Mark 61 : Customs Mark 62 : Causeway Mark 63 : Wave Recorder 64 : Jetski Prohibited 65 : Facility Protection Mark 66 : Oil Pipeline Protection Mark 67 : Marine Cable Protection Mark		
Colour	(COLOUR)	1 : White 2 : Black 3 : Red 4 : Green 5 : Blue 6 : Yellow 7 : Grey 8 : Brown 9 : Amber 10 : Violet 11 : Orange 12 : Magenta 13 : Pink	EN	1, *
Colour Pattern	(COLPAT)	1 : Horizontal Stripes 2 : Vertical Stripes 3 : Diagonal Stripes 4 : Squared 5 : Stripes (Direction Unknown) 6 : Border Stripe 7 : Single Colour 8 : Rectangle 9 : Triangle	EN	0, *
Elevation	(ELEVAT)		RE	0, 1
Height	(HEIGHT)		RE	0, 1
Nature of Construction	(NATCON)	1 : Masonry 2 : Concreted 3 : Loose Boulders	EN	0, *

		4 : Hard Surfaced 5 : Unsurfaced 6 : Wooden 7 : Metal 8 : Glass Reinforced Plastic 9 : Painted 10 : Framework 11 : Latticed 12 : Glass 13 : Fiberglass 14 : Plastic		
Orientation Value	(ORIENT)		RE	0, 1
Status	(STATUS)	1 : Permanent 2 : Occasional 3 : Recommended 4 : Not in Use 5 : Periodic/Intermittent 6 : Reserved 7 : Temporary 8 : Private 9 : Mandatory 11 : Extinguished 12 : Illuminated 13 : Historic 14 : Public 15 : Synchronized 16 : Watched 17 : Unwatched 18 : Existence Doubtful 19 : On Request 20 : Drop Away 21 : Rising 22 : Increasing 23 : Decreasing 24 : Strong 25 : Good 26 : Moderately 27 : Poor 28 : Buoyed 29 : Fully Operational 30 : Partially Operational 31 : Drifting	EN	0, *

		32 : Broken 33 : Offline 34 : Discontinued 35 : Manual Observation 36 : Unknown Status 37 : Confirmed 38 : Candidate 39 : Under Modification 41 : Under Removal / Deletion 42 : Removed / Deleted 43 : Candidate for Modification		
Topmark/Daymark Shape	(TOPSHP)	1 : Cone (Point Up) 2 : Cone (Point Down) 3 : Sphere 4 : 2 Spheres 5 : Cylinder 6 : Board 7 : X-Shaped 8 : Upright Cross 9 : Cube (Point Up) 10 : 2 Cones (Point to Point) 11 : 2 Cones (Base to Base) 12 : Rhombus 13 : 2 Cones (Points Upward) 14 : 2 Cones (Points Downward) 15 : Besom (Point Up) 16 : Besom (Point Down) 17 : Flag 18 : Sphere Over a Rhombus 19 : Square 20 : Rectangle (Horizontal) 21 : Rectangle (Vertical) 22 : Trapezium (Up) 23 : Trapezium (Down) 24 : Triangle (Point Up) 25 : Triangle (Point Down) 26 : Circle 27 : Two Upright Crosses (One Over the Other)	EN	1, 1

		28 : T-Shape 29 : Triangle Pointing Up Over a Circle 30 : Upright Cross Over a Circle 31 : Rhombus Over a Circle 32 : Circle Over a Triangle Pointing Up 33 : Other Shape (See Shape Information) 34 : Tubular		
Vertical Datum	(VERDAT) (Datum Level) (Reference Plane) (Levelling Datum) (Datum for Sounding Reduction) (Datum for Heights)	1 : Mean Low Water Springs 2 : Mean Lower Low Water Springs 3 : Mean Sea Level 4 : Lowest Low Water 5 : Mean Low Water 6 : Lowest Low Water Springs 7 : Approximate Mean Low Water Springs 8 : Indian Spring Low Water 9 : Low Water Springs 10 : Approximate Lowest Astronomical Tide 11 : Nearly Lowest Low Water 12 : Mean Lower Low Water 13 : Low Water 14 : Approximate Mean Low Water 15 : Approximate Mean Lower Low Water 16 : Mean High Water 17 : Mean High Water Springs 18 : High Water 19 : Approximate Mean Sea Level 20 : High Water Springs 21 : Mean Higher High Water 22 : Equinoctial Spring Low Water 23 : Lowest Astronomical Tide 24 : Local Datum 25 : International Great	EN	0, 1

		Lakes Datum 1985 26 : Mean Water Level 27 : Lower Low Water Large Tide 28 : Higher High Water Large Tide 29 : Nearly Highest High Water 30 : Highest Astronomical Tide 31 : Local Low Water Reference Level 32 : Local High Water Reference Level 33 : Local Mean Water Reference Level 34 : Equivalent Height of Water (German GIW) 35 : Highest Shipping Height of Water (German HSW) 36 : Reference Low Water Level According to Danube Commission 37 : Highest Shipping Height of Water According to Danube Commission 38 : Dutch River Low Water Reference Level (OLR) 39 : Russian Project Water Level 40 : Russian Normal Backwater Level 41 : Ohio River Datum 43 : Dutch High Water Reference Level 44 : Baltic Sea Chart Datum 2000 45 : Dutch Estuary Low Water Reference Level (OLW) 46 : International Great Lakes Datum 2020 47 : Sea Floor 48 : Sea Surface 49 : Hydrographic Zero		
Vertical Length	(VERLEN)		RE	0, 1
Shape Information			C	0, 1
IsSlatted			BO	1, 1

2.22. Retroreflector

Definition: A means of distinguishing unlighted marks at night. Retro-reflective material is secured to the mark in a particular pattern to reflect back light.

CamelCase: Retroreflector

Alias: RETRFL

Super type: Equipment

Feature use type: geographic

Primitive: point

Remarks: The body carrying the retro-reflector is a separate feature.

Attribute Bindings:

S-10x Attribute	S-57 Acronym	Allowable Encoding Value	Type	Multiplicity
Colour	(COLOUR)	1 : White 3 : Red 4 : Green 5 : Blue 6 : Yellow 7 : Grey 8 : Brown 9 : Amber 10 : Violet 11 : Orange 12 : Magenta 13 : Pink	EN	0, *
Colour Pattern	(COLPAT)	1 : Horizontal Stripes 2 : Vertical Stripes 3 : Diagonal Stripes 4 : Squared 5 : Stripes (Direction Unknown) 6 : Border Stripe 7 : Single Colour 8 : Rectangle 9 : Triangle	EN	0, *
Marks Navigational - System Of	(MARSYS)	1 : IALA A 2 : IALA B	EN	0, 1

		9 : No System 10 : Other System 11 : CEVNI 12 : Russian Inland Waterway Regulations 13 : Brazilian National Inland Waterway Regulations - Two Sides 14 : Brazilian National Inland Waterway Regulations - Side Independent 15 : Paraguay-Parana Waterway - Brazilian Complementary Aids		
Status	(STATUS)	1 : Permanent 2 : Occasional 3 : Recommended 4 : Not in Use 5 : Periodic/Intermittent 6 : Reserved 7 : Temporary 8 : Private 9 : Mandatory 11 : Extinguished 12 : Illuminated 13 : Historic 14 : Public 15 : Synchronized 16 : Watched 17 : Unwatched 18 : Existence Doubtful 19 : On Request 20 : Drop Away 21 : Rising 22 : Increasing 23 : Decreasing 24 : Strong 25 : Good 26 : Moderately 27 : Poor 28 : Buoyed 29 : Fully Operational 30 : Partially Operational 31 : Drifting 32 : Broken	EN	0, *

		33 : Offline 34 : Discontinued 35 : Manual Observation 36 : Unknown Status 37 : Confirmed 38 : Candidate 39 : Under Modification 41 : Under Removal / Deletion 42 : Removed / Deleted 43 : Candidate for Modification		
Vertical Datum	(VERDAT) (Datum Level) (Reference Plane) (Levelling Datum) (Datum for Sounding Reduction) (Datum for Heights)	1 : Mean Low Water Springs 2 : Mean Lower Low Water Springs 3 : Mean Sea Level 4 : Lowest Low Water 5 : Mean Low Water 6 : Lowest Low Water Springs 7 : Approximate Mean Low Water Springs 8 : Indian Spring Low Water 9 : Low Water Springs 10 : Approximate Lowest Astronomical Tide 11 : Nearly Lowest Low Water 12 : Mean Lower Low Water 13 : Low Water 14 : Approximate Mean Low Water 15 : Approximate Mean Lower Low Water 16 : Mean High Water 17 : Mean High Water Springs 18 : High Water 19 : Approximate Mean Sea Level 20 : High Water Springs 21 : Mean Higher High Water 22 : Equinoctial Spring Low Water 23 : Lowest Astronomical Tide	EN	0, 1

		24 : Local Datum 25 : International Great Lakes Datum 1985 26 : Mean Water Level 27 : Lower Low Water Large Tide 28 : Higher High Water Large Tide 29 : Nearly Highest High Water 30 : Highest Astronomical Tide 31 : Local Low Water Reference Level 32 : Local High Water Reference Level 33 : Local Mean Water Reference Level 34 : Equivalent Height of Water (German GLW) 35 : Highest Shipping Height of Water (German HSW) 36 : Reference Low Water Level According to Danube Commission 37 : Highest Shipping Height of Water According to Danube Commission 38 : Dutch River Low Water Reference Level (OLR) 39 : Russian Project Water Level 40 : Russian Normal Backwater Level 41 : Ohio River Datum 43 : Dutch High Water Reference Level 44 : Baltic Sea Chart Datum 2000 45 : Dutch Estuary Low Water Reference Level (OLW) 46 : International Great Lakes Datum 2020 47 : Sea Floor 48 : Sea Surface 49 : Hydrographic Zero		
Height	(HEIGHT)		RE	0, 1
Vertical Accuracy			RE	0, 1

2.23. Radar Transponder Beacon

Definition: A transponder beacon transmitting a coded signal on radar frequency, permitting an interrogating craft to determine the bearing and range of the transponder.

CamelCase: RadarTransponderBeacon

Alias: RTPBCN Radar Beacon RACON

Super type: Equipment

Feature use type: geographic

Primitive: point

Remarks: The feature 'radar transponder beacon' is only used to encode the technical equipment independent of the structure on which it is located (for example a beacon, light-vessel or tower).

Attribute Bindings:

S-10x Attribute	S-57 Acronym	Allowable Encoding Value	Type	Multiplicity
Category of Radar Transponder Beacon	(CATRTB)	1 : Ramark, Radar Beacon Transmitting Continuously 2 : Racon, Radar Transponder Beacon 3 : Leading Racon/Radar Transponder Beacon	EN	1, 1
Radar Wave Length			C	0, 1
Radar Band			(S) TE	1, 1
Wave Length Value	(RadarWaveLength)		(S) RE	1, 1
Signal Group	(SIGGRP)		TE	0, 1
Status	(STATUS)	1 : Permanent 2 : Occasional 3 : Recommended 4 : Not in Use 5 : Periodic/Intermittent 6 : Reserved 7 : Temporary 8 : Private 9 : Mandatory 11 : Extinguished 12 : Illuminated 13 : Historic	EN	0, *

		14 : Public 15 : Synchronized 16 : Watched 17 : Unwatched 18 : Existence Doubtful 19 : On Request 20 : Drop Away 21 : Rising 22 : Increasing 23 : Decreasing 24 : Strong 25 : Good 26 : Moderately 27 : Poor 28 : Buoyed 29 : Fully Operational 30 : Partially Operational 31 : Drifting 32 : Broken 33 : Offline 34 : Discontinued 35 : Manual Observation 36 : Unknown Status 37 : Confirmed 38 : Candidate 39 : Under Modification 41 : Under Removal / Deletion 42 : Removed / Deleted 43 : Candidate for Modification		
Value of Nominal Range	(VALNMR)		RE	0, 1
Manufacturer			TE	0, 1
Signal Status		1 : Lit/Sound 2 : Eclipsed/Silent	(S) EN	1, 1

2.24. Virtual AIS Aid to Navigation

Definition: An Automatic Identification System (AIS) message 21 transmitted from an AIS station to simulate on navigation systems an Aid to Navigation which does not physically exist.

CamelCase: VirtualAIS AidToNavigation

Alias:

Super type: ElectronicAton

Feature use type: geographic

Primitive: point

Remarks: No remarks.

Attribute Bindings:

S-10x Attribute	S-57 Acronym	Allowable Encoding Value	Type	Multiplicity
Virtual AIS Aid to Navigation Type		1 : North Cardinal 2 : East Cardinal 3 : South Cardinal 4 : West Cardinal 5 : Port Lateral 6 : Starboard Lateral 7 : Preferred Channel to Port 8 : Preferred Channel to Starboard 9 : Isolated Danger 10 : Safe Water 11 : Special Purpose 12 : New Danger Marking	EN	1, 1

Feature Bindings:

Type	Association Name	Association Ends					
		Class	Role	Mult	Class	Role	Mult
Asso	Virtual AIS	VirtualAIS AidToNavigation	virtualAISbroadcastBy	0, 1	RadioStation	virtualAISbroadcasts	0, *

2.25. Physical AIS Aid to Navigation

Definition: An Automatic Identification System (AIS) message 21 transmitted from a physical Aid to Navigation, or transmitted from an AIS station for an Aid to Navigation which physically exists.

CamelCase: PhysicalAIS AidToNavigation

Alias:

Super type: ElectronicAton

Feature use type: geographic

Primitive: point

Remarks: No remarks.

Attribute Bindings:

S-10x Attribute	S-57 Acronym	Allowable Encoding Value	Type	Multiplicity
Category Of Physical AIS Aid To Navigation		1 : Physical AIS Type 1 2 : Physical AIS Type 2 3 : Physical AIS Type 3	EN	1, 1

Feature Bindings:

Type	Association Name	Association Ends					
		Class	Role	Mult	Class	Role	Mult
Asso	Physical AIS	PhysicalAIS AidToNavigation	physicalAISbroadcastBy	0, 1	RadioStation	physicalAISbroadcasts	0, *

2.26. Synthetic AIS Aid To Navigation

Definition: -

CamelCase: SyntheticAISaidToNavigation

Alias:

Super type: ElectronicAton

Feature use type: geographic

Primitive: point

Remarks: -

Attribute Bindings:

S-10x Attribute	S-57 Acronym	Allowable Encoding Value	Type	Multiplicity
Category Of Synthetic AIS Aid To Navigation		1 : predicted 2 : monitored	EN	1, 1
Virtual AIS Aid to Navigation Type		1 : North Cardinal 2 : East Cardinal 3 : South Cardinal 4 : West Cardinal 5 : Port Lateral 6 : Starboard Lateral 7 : Preferred Channel to Port 8 : Preferred Channel to Starboard 9 : Isolated Danger 10 : Safe Water 11 : Special Purpose 12 : New Danger Marking	EN	1, 1

Feature Bindings:

Type	Association Name	Association Ends					
		Class	Role	Mult	Class	Role	Mult
Asso	Synthetic AIS	SyntheticAISaidToNavigation	syntheticAISbroadcastBy	0, 1	RadioStation	syntheticAISbroadcasts	0, *

2.27. Power Source

Definition: -

CamelCase: PowerSource

Alias:

Super type: Equipment

Feature use type: geographic

Primitive: point

Remarks: -

Attribute Bindings:

S-10x Attribute	S-57 Acronym	Allowable Encoding Value	Type	Multiplicity
Category Of Power Source		1 : battery 2 : generator 3 : solar panel 4 : electrical service	EN	1, 1
Status	(STATUS)	1 : Permanent 2 : Occasional 3 : Recommended 4 : Not in Use 5 : Periodic/Intermittent 6 : Reserved 7 : Temporary 8 : Private 9 : Mandatory 11 : Extinguished 12 : Illuminated 13 : Historic 14 : Public 15 : Synchronized 16 : Watched 17 : Unwatched 18 : Existence Doubtful 19 : On Request 20 : Drop Away 21 : Rising 22 : Increasing	EN	0, *

		23 : Decreasing 24 : Strong 25 : Good 26 : Moderately 27 : Poor 28 : Buoyed 29 : Fully Operational 30 : Partially Operational 31 : Drifting 32 : Broken 33 : Offline 34 : Discontinued 35 : Manual Observation 36 : Unknown Status 37 : Confirmed 38 : Candidate 39 : Under Modification 41 : Under Removal / Deletion 42 : Removed / Deleted 43 : Candidate for Modification		
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2.28. Isolated Danger Beacon

Definition: An isolated danger beacon is a beacon erected on an isolated danger of limited extent, which has navigable water all around it.

CamelCase: IsolatedDangerBeacon

Alias:

Super type: GenericBeacon

Feature use type: geographic

Primitive: point

Remarks: -

2.29. Cardinal Beacon

Definition: A cardinal beacon is used in conjunction with the compass to indicate where the mariner may find the best navigable water. It is placed in one of the four quadrants (North, East, South and West), bounded by inter-cardinal bearings from the point marked.

CamelCase: CardinalBeacon

Alias:

Super type: GenericBeacon

Feature use type: geographic

Primitive: point

Remarks: -

Attribute Bindings:

S-10x Attribute	S-57 Acronym	Allowable Encoding Value	Type	Multiplicity
Category of Cardinal Mark	(CATCAM)	1 : North Cardinal Mark 2 : East Cardinal Mark 3 : South Cardinal Mark 4 : West Cardinal Mark	EN	1, 1

2.30. Isolated Danger Buoy

Definition: An isolated danger buoy is a buoy moored on or above an isolated danger of limited extent, which has navigable water all around it.

CamelCase: IsolatedDangerBuoy

Alias:

Super type: GenericBuoy

Feature use type: geographic

Primitive: point

Remarks: -

2.31. Cardinal Buoy

Definition: A cardinal buoy is used in conjunction with the compass to indicate where the mariner may find the best navigable water. It is placed in one of the four quadrants (North, East, South and West), bounded by inter-cardinal bearings from the point marked.

CamelCase: CardinalBuoy

Alias:

Super type: GenericBuoy

Feature use type: geographic

Primitive: point

Remarks: -

Attribute Bindings:

S-10x Attribute	S-57 Acronym	Allowable Encoding Value	Type	Multiplicity
Category of Cardinal Mark	(CATCAM)	1 : North Cardinal Mark 2 : East Cardinal Mark 3 : South Cardinal Mark 4 : West Cardinal Mark	EN	1, 1

2.32. Installation Buoy

Definition: A buoy is a floating object moored to the bottom in a particular place, as an aid to navigation or for other specific purposes. (IHO Dictionary, S-32, 5th Edition, 565). An installation buoy is a buoy used for loading tankers with gas or oil. (IHO Chart Specifications, M-4)

CamelCase: InstallationBuoy

Alias:

Super type: GenericBuoy

Feature use type: geographic

Primitive: point

Remarks: -

Attribute Bindings:

S-10x Attribute	S-57 Acronym	Allowable Encoding Value	Type	Multiplicity
Category of Installation Buoy		1 : Catenary Anchor Leg Mooring 2 : Single Buoy Mooring	EN	1, 1

2.33. Mooring Buoy

Definition: The equipment or structure used to secure a vessel. (IHO Registry)

CamelCase: MooringBuoy

Alias:

Super type: GenericBuoy

Feature use type: geographic

Primitive: point

Remarks: -

2.34. Emergency Wreck Marking Buoy

Definition: An emergency wreck marking buoy is a buoy moored on or above a new wreck, designed to provide a prominent (both visual and radio) and easily identifiable temporary (24-72 hours) first response. (IHO Registry)

CamelCase: EmergencyWreckMarkingBuoy

Alias:

Super type: GenericBuoy

Feature use type: geographic

Primitive: point

Remarks: -

2.35. Lighthouse

Definition: A distinctive structure on or off a coast exhibiting a major light designed to serve as an aid to navigation.

CamelCase: Lighthouse

Alias:

Super type: Landmark

Feature use type: geographic

Primitive: point surface

Remarks: No remarks.

2.36. Light Float

Definition: A boat-like structure used instead of a light buoy in waters where strong streams or currents are experienced, or when a greater elevation than that of a light buoy is necessary.

CamelCase: LightFloat

Alias: LITFLT

Super type: StructureObject

Feature use type: geographic

Primitive: point

Remarks: The light of a light float is a separate feature, handled as with buoys, beacons, etc.

Attribute Bindings:

S-10x Attribute	S-57 Acronym	Allowable Encoding Value	Type	Multiplicity
Colour	(COLOUR)	1 : White 2 : Black 3 : Red 4 : Green 5 : Blue 6 : Yellow 7 : Grey 8 : Brown 9 : Amber 10 : Violet 11 : Orange 12 : Magenta 13 : Pink	EN	1, *
Colour Pattern	(COLPAT)	1 : Horizontal Stripes 2 : Vertical Stripes 3 : Diagonal Stripes 4 : Squared 5 : Stripes (Direction Unknown) 6 : Border Stripe 7 : Single Colour 8 : Rectangle 9 : Triangle	EN	0, *
Horizontal Length	(HORLEN)		RE	0, 1

Horizontal Width	(HORWID)		RE	0, 1
Manned Structure			BO	0, 1
Nature of Construction	(NATCON)	1 : Masonry 2 : Concreted 3 : Loose Boulders 4 : Hard Surfaced 5 : Unsurfaced 6 : Wooden 7 : Metal 8 : Glass Reinforced Plastic 9 : Painted 10 : Framework 11 : Latticed 12 : Glass 13 : Fiberglass 14 : Plastic	EN	0, *
Radar Conspicuous	(CONRAD)		BO	0, 1
Status	(STATUS)	1 : Permanent 2 : Occasional 3 : Recommended 4 : Not in Use 5 : Periodic/Intermittent 6 : Reserved 7 : Temporary 8 : Private 9 : Mandatory 11 : Extinguished 12 : Illuminated 13 : Historic 14 : Public 15 : Synchronized 16 : Watched 17 : Unwatched 18 : Existence Doubtful 19 : On Request 20 : Drop Away 21 : Rising 22 : Increasing 23 : Decreasing 24 : Strong	EN	0, *

		25 : Good 26 : Moderately 27 : Poor 28 : Buoyed 29 : Fully Operational 30 : Partially Operational 31 : Drifting 32 : Broken 33 : Offline 34 : Discontinued 35 : Manual Observation 36 : Unknown Status 37 : Confirmed 38 : Candidate 39 : Under Modification 41 : Under Removal / Deletion 42 : Removed / Deleted 43 : Candidate for Modification		
Vertical Length	(VERLEN)		RE	0, 1
Visual Prominence	(CONVIS)	1 : Visually Conspicuous 2 : Not Visually Conspicuous 3 : Prominent	EN	0, 1
Vertical Accuracy			RE	0, 1
Horizontal Accuracy			RE	0, 1

2.37. Light Vessel

Definition: A distinctively marked vessel anchored or moored at a charted point, to serve as an aid to navigation. By night, it displays a characteristic light(s) and is usually equipped with other devices, such as fog signal, submarine sound signal, and radio-beacon, to assist navigation.

CamelCase: LightVessel

Alias: LITVES Lightship

Super type: StructureObject

Feature use type: geographic

Primitive: point

Remarks: The light(s), fog signal etc. of a light vessel are separate features, handled as with buoys, beacons, etc.

Attribute Bindings:

S-10x Attribute	S-57 Acronym	Allowable Encoding Value	Type	Multiplicity
Colour	(COLOUR)	1 : White 2 : Black 3 : Red 4 : Green 5 : Blue 6 : Yellow 7 : Grey 8 : Brown 9 : Amber 10 : Violet 11 : Orange 12 : Magenta 13 : Pink	EN	1, *
Colour Pattern	(COLPAT)	1 : Horizontal Stripes 2 : Vertical Stripes 3 : Diagonal Stripes 4 : Squared 5 : Stripes (Direction Unknown) 6 : Border Stripe 7 : Single Colour 8 : Rectangle 9 : Triangle	EN	0, *

Horizontal Length	(HORLEN)		RE	0, 1
Horizontal Width	(HORWID)		RE	0, 1
Manned Structure			BO	0, 1
Nature of Construction	(NATCON)	1 : Masonry 2 : Concreted 3 : Loose Boulders 4 : Hard Surfaced 5 : Unsurfaced 6 : Wooden 7 : Metal 8 : Glass Reinforced Plastic 9 : Painted 10 : Framework 11 : Latticed 12 : Glass 13 : Fiberglass 14 : Plastic	EN	0, *
Radar Conspicuous	(CONRAD)		BO	0, 1
Status	(STATUS)	1 : Permanent 2 : Occasional 3 : Recommended 4 : Not in Use 5 : Periodic/Intermittent 6 : Reserved 7 : Temporary 8 : Private 9 : Mandatory 11 : Extinguished 12 : Illuminated 13 : Historic 14 : Public 15 : Synchronized 16 : Watched 17 : Unwatched 18 : Existence Doubtful 19 : On Request 20 : Drop Away 21 : Rising 22 : Increasing 23 : Decreasing	EN	0, *

		24 : Strong 25 : Good 26 : Moderately 27 : Poor 28 : Buoyed 29 : Fully Operational 30 : Partially Operational 31 : Drifting 32 : Broken 33 : Offline 34 : Discontinued 35 : Manual Observation 36 : Unknown Status 37 : Confirmed 38 : Candidate 39 : Under Modification 41 : Under Removal / Deletion 42 : Removed / Deleted 43 : Candidate for Modification		
Vertical Length	(VERLEN)		RE	0, 1
Visual Prominence	(CONVIS)	1 : Visually Conspicuous 2 : Not Visually Conspicuous 3 : Prominent	EN	0, 1
Vertical Accuracy			RE	0, 1
Horizontal Accuracy			RE	0, 1

2.38. Offshore Platform

Definition: A permanent offshore structure, either fixed or floating.

CamelCase: OffshorePlatform

Alias: OFSPLF

Super type: StructureObject

Feature use type: geographic

Primitive: point surface

Remarks: No remarks.

Attribute Bindings:

S-10x Attribute	S-57 Acronym	Allowable Encoding Value	Type	Multiplicity
Category of Offshore Platform	(CATOFP)	1 : Oil Rig 2 : Production Platform 3 : Observation/Research Platform 4 : Articulated Loading Platform 5 : Single Anchor Leg Mooring 6 : Mooring Tower 7 : Artificial Island 8 : Floating Production, Storage and Off-Loading Vessel 9 : Accommodation Platform 10 : Navigation, Communication and Control Buoy 11 : Floating Oil Tank	EN	0, *
Colour	(COLOUR)	1 : White 2 : Black 3 : Red 4 : Green 5 : Blue 6 : Yellow 7 : Grey 8 : Brown 9 : Amber	EN	0, *

		10 : Violet 11 : Orange 12 : Magenta 13 : Pink		
Colour Pattern	(COLPAT)	1 : Horizontal Stripes 2 : Vertical Stripes 3 : Diagonal Stripes 4 : Squared 5 : Stripes (Direction Unknown) 6 : Border Stripe 7 : Single Colour 8 : Rectangle 9 : Triangle	EN	0, *
Height	(HEIGHT)		RE	0, 1
Manned Structure			BO	0, 1
Nature of Construction	(NATCON)	1 : Masonry 2 : Concreted 3 : Loose Boulders 4 : Hard Surfaced 5 : Unsurfaced 6 : Wooden 7 : Metal 8 : Glass Reinforced Plastic 9 : Painted 10 : Framework 11 : Latticed 12 : Glass 13 : Fiberglass 14 : Plastic	EN	0, *
Product	(PRODCT)	1 : Oil 2 : Gas 3 : Water 4 : Stone 5 : Coal 6 : Ore 7 : Chemicals 8 : Drinking Water 9 : Milk 10 : Bauxite 11 : Coke	EN	0, *

		12 : Iron Ingots 13 : Salt 14 : Sand 15 : Timber 16 : Sawdust/Wood Chips 17 : Scrap Metal 18 : Liquefied Natural Gas 19 : Liquefied Petroleum Gas 20 : Wine 21 : Cement 22 : Grain 23 : Electricity 24 : Ice 25 : Clay		
Radar Conspicuous	(CONRAD)		BO	0, 1
Status	(STATUS)	1 : Permanent 2 : Occasional 3 : Recommended 4 : Not in Use 5 : Periodic/Intermittent 6 : Reserved 7 : Temporary 8 : Private 9 : Mandatory 11 : Extinguished 12 : Illuminated 13 : Historic 14 : Public 15 : Synchronized 16 : Watched 17 : Unwatched 18 : Existence Doubtful 19 : On Request 20 : Drop Away 21 : Rising 22 : Increasing 23 : Decreasing 24 : Strong 25 : Good 26 : Moderately 27 : Poor	EN	0, *

		28 : Buoyed 29 : Fully Operational 30 : Partially Operational 31 : Drifting 32 : Broken 33 : Offline 34 : Discontinued 35 : Manual Observation 36 : Unknown Status 37 : Confirmed 38 : Candidate 39 : Under Modification 41 : Under Removal / Deletion 42 : Removed / Deleted 43 : Candidate for Modification		
Vertical Datum	(VERDAT) (Datum Level) (Reference Plane) (Levelling Datum) (Datum for Sounding Reduction) (Datum for Heights)	1 : Mean Low Water Springs 2 : Mean Lower Low Water Springs 3 : Mean Sea Level 4 : Lowest Low Water 5 : Mean Low Water 6 : Lowest Low Water Springs 7 : Approximate Mean Low Water Springs 8 : Indian Spring Low Water 9 : Low Water Springs 10 : Approximate Lowest Astronomical Tide 11 : Nearly Lowest Low Water 12 : Mean Lower Low Water 13 : Low Water 14 : Approximate Mean Low Water 15 : Approximate Mean Lower Low Water 16 : Mean High Water 17 : Mean High Water Springs 18 : High Water 19 : Approximate Mean Sea Level 20 : High Water Springs	EN	0, 1

		21 : Mean Higher High Water 22 : Equinoctial Spring Low Water 23 : Lowest Astronomical Tide 24 : Local Datum 25 : International Great Lakes Datum 1985 26 : Mean Water Level 27 : Lower Low Water Large Tide 28 : Higher High Water Large Tide 29 : Nearly Highest High Water 30 : Highest Astronomical Tide 31 : Local Low Water Reference Level 32 : Local High Water Reference Level 33 : Local Mean Water Reference Level 34 : Equivalent Height of Water (German GIW) 35 : Highest Shipping Height of Water (German HSW) 36 : Reference Low Water Level According to Danube Commission 37 : Highest Shipping Height of Water According to Danube Commission 38 : Dutch River Low Water Reference Level (OLR) 39 : Russian Project Water Level 40 : Russian Normal Backwater Level 41 : Ohio River Datum 43 : Dutch High Water Reference Level 44 : Baltic Sea Chart Datum 2000 45 : Dutch Estuary Low Water Reference Level (OLW) 46 : International Great Lakes Datum 2020 47 : Sea Floor 48 : Sea Surface		
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		49 : Hydrographic Zero		
Vertical Length	(VERLEN)		RE	0, 1
Visual Prominence	(CONVIS)	1 : Visually Conspicuous 2 : Not Visually Conspicuous 3 : Prominent	EN	0, 1
Vertical Accuracy			RE	0, 1

2.39. Silo/Tank

Definition: A large storage structure used for storing loose materials, liquids and/or gases.

CamelCase: SiloTank

Alias: SILTNK

Super type: StructureObject

Feature use type: geographic

Primitive: point surface

Remarks: No remarks.

Attribute Bindings:

S-10x Attribute	S-57 Acronym	Allowable Encoding Value	Type	Multiplicity
Building Shape	(BUISHP)	5 : High-Rise Building 6 : Pyramid 7 : Cylindrical 8 : Spherical 9 : Cubic	EN	0, 1
Category of Silo/Tank	(CATSIL)	1 : Silo in General 2 : Tank in General 3 : Grain Elevator 4 : Water Tower	EN	0, 1
Colour	(COLOUR)	1 : White 2 : Black 3 : Red 4 : Green 5 : Blue 6 : Yellow 7 : Grey 8 : Brown 9 : Amber 10 : Violet 11 : Orange 12 : Magenta 13 : Pink	EN	0, *
Colour Pattern	(COLPAT)	1 : Horizontal Stripes 2 : Vertical Stripes	EN	0, *

		3 : Diagonal Stripes 4 : Squared 5 : Stripes (Direction Unknown) 6 : Border Stripe 7 : Single Colour 8 : Rectangle 9 : Triangle		
Elevation	(ELEVAT)		RE	0, 1
Height	(HEIGHT)		RE	0, 1
Nature of Construction	(NATCON)	1 : Masonry 2 : Concreted 3 : Loose Boulders 4 : Hard Surfaced 5 : Unsurfaced 6 : Wooden 7 : Metal 8 : Glass Reinforced Plastic 9 : Painted 10 : Framework 11 : Latticed 12 : Glass 13 : Fiberglass 14 : Plastic	EN	0, *
Radar Conspicuous	(CONRAD)		BO	0, 1
Status	(STATUS)	1 : Permanent 2 : Occasional 3 : Recommended 4 : Not in Use 5 : Periodic/Intermittent 6 : Reserved 7 : Temporary 8 : Private 9 : Mandatory 11 : Extinguished 12 : Illuminated 13 : Historic 14 : Public 15 : Synchronized 16 : Watched 17 : Unwatched	EN	0, *

		18 : Existence Doubtful 19 : On Request 20 : Drop Away 21 : Rising 22 : Increasing 23 : Decreasing 24 : Strong 25 : Good 26 : Moderately 27 : Poor 28 : Buoyed 29 : Fully Operational 30 : Partially Operational 31 : Drifting 32 : Broken 33 : Offline 34 : Discontinued 35 : Manual Observation 36 : Unknown Status 37 : Confirmed 38 : Candidate 39 : Under Modification 41 : Under Removal / Deletion 42 : Removed / Deleted 43 : Candidate for Modification		
Vertical Datum	(VERDAT) (Datum Level) (Reference Plane) (Levelling Datum) (Datum for Sounding Reduction) (Datum for Heights)	1 : Mean Low Water Springs 2 : Mean Lower Low Water Springs 3 : Mean Sea Level 4 : Lowest Low Water 5 : Mean Low Water 6 : Lowest Low Water Springs 7 : Approximate Mean Low Water Springs 8 : Indian Spring Low Water 9 : Low Water Springs 10 : Approximate Lowest Astronomical Tide 11 : Nearly Lowest Low Water 12 : Mean Lower Low Water 13 : Low Water	EN	0, 1

		14 : Approximate Mean Low Water 15 : Approximate Mean Lower Low Water 16 : Mean High Water 17 : Mean High Water Springs 18 : High Water 19 : Approximate Mean Sea Level 20 : High Water Springs 21 : Mean Higher High Water 22 : Equinoctial Spring Low Water 23 : Lowest Astronomical Tide 24 : Local Datum 25 : International Great Lakes Datum 1985 26 : Mean Water Level 27 : Lower Low Water Large Tide 28 : Higher High Water Large Tide 29 : Nearly Highest High Water 30 : Highest Astronomical Tide 31 : Local Low Water Reference Level 32 : Local High Water Reference Level 33 : Local Mean Water Reference Level 34 : Equivalent Height of Water (German GIW) 35 : Highest Shipping Height of Water (German HSW) 36 : Reference Low Water Level According to Danube Commission 37 : Highest Shipping Height of Water According to Danube Commission 38 : Dutch River Low Water Reference Level (OLR) 39 : Russian Project Water Level 40 : Russian Normal Backwater Level		
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		41 : Ohio River Datum 43 : Dutch High Water Reference Level 44 : Baltic Sea Chart Datum 2000 45 : Dutch Estuary Low Water Reference Level (OLW) 46 : International Great Lakes Datum 2020 47 : Sea Floor 48 : Sea Surface 49 : Hydrographic Zero		
Vertical Length	(VERLEN)		RE	0, 1
Visual Prominence	(CONVIS)	1 : Visually Conspicuous 2 : Not Visually Conspicuous 3 : Prominent	EN	0, 1
Vertical Accuracy			RE	0, 1

2.40. Pile

Definition: A long heavy timber or section of steel, wood, concrete, etc., forced into the earth or sea floor to serve as a support, as for a pier, or to resist lateral pressure; or as a free standing pole within a marine environment.

CamelCase: Pile

Alias: PILPNT

Super type: StructureObject

Feature use type: geographic

Primitive: point

Remarks: No remarks.

Attribute Bindings:

S-10x Attribute	S-57 Acronym	Allowable Encoding Value	Type	Multiplicity
Category of Pile	(CATPLE)	1 : Stake 3 : Post 4 : Tripodal 5 : Piling 6 : Area of Piles 7 : Pipe	EN	0, 1
Colour	(COLOUR)	1 : White 2 : Black 3 : Red 4 : Green 5 : Blue 6 : Yellow 7 : Grey 8 : Brown 9 : Amber 10 : Violet 11 : Orange 12 : Magenta 13 : Pink	EN	0, *
Colour Pattern	(COLPAT)	1 : Horizontal Stripes 2 : Vertical Stripes 3 : Diagonal Stripes 4 : Squared 5 : Stripes (Direction	EN	0, *

		Unknown) 6 : Border Stripe 7 : Single Colour 8 : Rectangle 9 : Triangle		
Height	(HEIGHT)		RE	0, 1
Vertical Datum	(VERDAT) (Datum Level) (Reference Plane) (Levelling Datum) (Datum for Sounding Reduction) (Datum for Heights)	1 : Mean Low Water Springs 2 : Mean Lower Low Water Springs 3 : Mean Sea Level 4 : Lowest Low Water 5 : Mean Low Water 6 : Lowest Low Water Springs 7 : Approximate Mean Low Water Springs 8 : Indian Spring Low Water 9 : Low Water Springs 10 : Approximate Lowest Astronomical Tide 11 : Nearly Lowest Low Water 12 : Mean Lower Low Water 13 : Low Water 14 : Approximate Mean Low Water 15 : Approximate Mean Lower Low Water 16 : Mean High Water 17 : Mean High Water Springs 18 : High Water 19 : Approximate Mean Sea Level 20 : High Water Springs 21 : Mean Higher High Water 22 : Equinoctial Spring Low Water 23 : Lowest Astronomical Tide 24 : Local Datum 25 : International Great Lakes Datum 1985 26 : Mean Water Level 27 : Lower Low Water Large Tide	EN	0, 1

		28 : Higher High Water Large Tide 29 : Nearly Highest High Water 30 : Highest Astronomical Tide 31 : Local Low Water Reference Level 32 : Local High Water Reference Level 33 : Local Mean Water Reference Level 34 : Equivalent Height of Water (German GIW) 35 : Highest Shipping Height of Water (German HSW) 36 : Reference Low Water Level According to Danube Commission 37 : Highest Shipping Height of Water According to Danube Commission 38 : Dutch River Low Water Reference Level (OLR) 39 : Russian Project Water Level 40 : Russian Normal Backwater Level 41 : Ohio River Datum 43 : Dutch High Water Reference Level 44 : Baltic Sea Chart Datum 2000 45 : Dutch Estuary Low Water Reference Level (OLW) 46 : International Great Lakes Datum 2020 47 : Sea Floor 48 : Sea Surface 49 : Hydrographic Zero		
Vertical Length	(VERLEN)		RE	0, 1
Visual Prominence	(CONVIS)	1 : Visually Conspicuous 2 : Not Visually Conspicuous 3 : Prominent	EN	0, 1
Vertical Accuracy			RE	0, 1

2.41. Building

Definition: A free-standing self-supporting construction that is roofed, usually walled, and is intended for human occupancy (for example: a place of work or recreation) and/or habitation.

CamelCase: Building

Alias: BUISGL

Super type: StructureObject

Feature use type: geographic

Primitive: point

Remarks: This feature is used to encode single buildings, including those with a particular function or service of major interest.

2.42. Bridge

Definition: (1) An elevated structure extending across or over the weather deck of a vessel, or part of such a structure. The term is sometimes modified to indicate the intended use, such as navigating bridge or signal bridge. (2) A structure erected over a depression or an obstacle such as a body of water, railroad, etc., to provide a roadway for vehicles or pedestrians.

CamelCase: Bridge

Alias: BRIDGE

Super type: StructureObject

Feature use type: geographic

Primitive: noGeometry

Remarks: A bridge may consist of portions which cover the land and the water.

2.43. Sinker Anchor

Definition: A heavy weight (of concrete, cast-iron, etc..) that rests on the sea bed and to which a mooring line can be attached. (IALA Dictionary, 8-5-025)

CamelCase: SinkerAnchor

Alias:

Super type: AidsToNavigation

Feature use type: geographic

Primitive: point

Remarks: -

Attribute Bindings:

S-10x Attribute	S-57 Acronym	Allowable Encoding Value	Type	Multiplicity
Nature of Construction	(NATCON)	1 : Masonry 2 : Concreted 3 : Loose Boulders 4 : Hard Surfaced 5 : Unsurfaced 6 : Wooden 7 : Metal 8 : Glass Reinforced Plastic 9 : Painted 10 : Framework 11 : Latticed 12 : Glass 13 : Fiberglass 14 : Plastic	EN	0, 1
Sinker Dimensions			C	0, 1
Height Length Units		1 : Metres 2 : Feet 3 : Kilometres 4 : Hectometres 5 : Statute Miles 6 : Nautical Miles	(S) EN	1, 1
Sinker Type			TE	0, 1

Feature Bindings:

Type	Association Name	Association Ends					
		Class	Role	Mult	Class	Role	Mult
Asso	ShackleToAnchorConnection	SinkerAnchor	shackleToAnchorconnectedTo	0, 1	MooringShackle	shackleToAnchorconnected	1, 1

2.44. Mooring Shackle

Definition: A shackle at the lower end of a mooring chain, for attachment to an anchor or sinker. (IALA Dictionary, 8-5-150)

CamelCase: MooringShackle

Alias:

Super type: AidsToNavigation

Feature use type: geographic

Primitive: point

Remarks: -

Attribute Bindings:

S-10x Attribute	S-57 Acronym	Allowable Encoding Value	Type	Multiplicity
Nature of Construction	(NATCON)	1 : Masonry 2 : Concreted 3 : Loose Boulders 4 : Hard Surfaced 5 : Unsurfaced 6 : Wooden 7 : Metal 8 : Glass Reinforced Plastic 9 : Painted 10 : Framework 11 : Latticed 12 : Glass 13 : Fiberglass 14 : Plastic	EN	0, 1
Shackle Type		1 : forelock shackles 2 : clenching shackles 3 : bolt shackles 4 : screw pin shackles 5 : kenter shackle 6 : quick release link	EN	0, 1
Weight			RE	0, 1

Feature Bindings:

Type	Association Name	Association Ends					
		Class	Role	Mult	Class	Role	Mult
Asso	Shackle Connection	MooringShackle	shackleToBuoyconnected	1, 1	GenericBuoy	shackleToBuoyconnectedTo	0, 1
Asso	Shackle To Bridle Connection	MooringShackle	shackleToBridleconnected	1, 1	Bridle	shackleToBridleconnectedTo	0, 1
Asso	Bridle Cable Connection	MooringShackle	shackleToCableconnected	1, 1	CableSubmarine	bridleattached	0, *
Asso	Shackle To Swivel Connection	MooringShackle	shackleToSwivelconnected	1, 1	Swivel	shackleToSwivelconnectedTo	0, *
Asso	ShackleToAnchorConnection	MooringShackle	shackleToAnchorconnected	1, 1	SinkerAnchor	shackleToAnchorconnectedTo	0, 1

2.45. Cable Submarine

Definition: An assembly of wires or fibres, or a wire rope or chain, which has been laid underwater or buried beneath the sea floor.

CamelCase: CableSubmarine

Alias: CBLSUB Submarine Cable

Super type: AidsToNavigation

Feature use type: geographic

Primitive: point

Remarks: No remarks.

Attribute Bindings:

S-10x Attribute	S-57 Acronym	Allowable Encoding Value	Type	Multiplicity
Cable Dimensions			C	0, 1
Cable Length			(S) RE	1, 1
Status	(STATUS)	1 : Permanent 2 : Occasional 3 : Recommended 4 : Not in Use 5 : Periodic/Intermittent 6 : Reserved 7 : Temporary 8 : Private 9 : Mandatory 11 : Extinguished 12 : Illuminated 13 : Historic 14 : Public 15 : Synchronized 16 : Watched 17 : Unwatched 18 : Existence Doubtful 19 : On Request 20 : Drop Away 21 : Rising 22 : Increasing	EN	0, *

		23 : Decreasing 24 : Strong 25 : Good 26 : Moderately 27 : Poor 28 : Buoyed 29 : Fully Operational 30 : Partially Operational 31 : Drifting 32 : Broken 33 : Offline 34 : Discontinued 35 : Manual Observation 36 : Unknown Status 37 : Confirmed 38 : Candidate 39 : Under Modification 41 : Under Removal / Deletion 42 : Removed / Deleted 43 : Candidate for Modification		
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Feature Bindings:

Type	Association Name	Association Ends					
		Class	Role	Mult	Class	Role	Mult
Asso	Bridle Cable Connection	CableSubmarine	bridleattached	0, *	Bridle	cableholds	1, 1
Asso	Swivel Cable Connection	CableSubmarine	swivelattached	0, *	Swivel	cableholds	1, 1
Asso	Shackle Connection From Cable	CableSubmarine	bridleattached	0, *	MooringShackle	shackleToCable connected	1, 1

2.46. Swivel

Definition: A chain link that provides for rotary motion between the lengths of chain that it connects. (IALA Dictionary, 8-5-165)

CamelCase: Swivel

Alias:

Super type: AidsToNavigation

Feature use type: geographic

Primitive: point

Remarks: -

Attribute Bindings:

S-10x Attribute	S-57 Acronym	Allowable Encoding Value	Type	Multiplicity
Nature of Construction	(NATCON)	1 : Masonry 2 : Concreted 3 : Loose Boulders 4 : Hard Surfaced 5 : Unsurfaced 6 : Wooden 7 : Metal 8 : Glass Reinforced Plastic 9 : Painted 10 : Framework 11 : Latticed 12 : Glass 13 : Fiberglass 14 : Plastic	EN	0, 1
Weight			RE	0, 1
Swivel Type			TE	0, 1

Feature Bindings:

Type	Association Name	Association Ends					
		Class	Role	Mult	Class	Role	Mult

Asso	Swivel Connection	Swivel	bridlehangs	0, 1	Bridle	swivelholds	1, 1
Asso	Swivel Cable Connection	Swivel	cableholds	1, 1	CableSubmarine	swivelattached	0, *
Asso	Shackle To Swivel Connection	Swivel	shackleToSwivel connectedTo	0, *	MooringShackle	shackleToSwivel connected	1, 1

2.47. Bridle

Definition: Two lengths of chain connected by a central ring and used for lifting wide loads. (IALA Dictionary,8-3-195)

CamelCase: Bridle

Alias:

Super type: AidsToNavigation

Feature use type: geographic

Primitive: point

Remarks: -

Attribute Bindings:

S-10x Attribute	S-57 Acronym	Allowable Encoding Value	Type	Multiplicity
BridleLink Type			TE	0, 1
Legs Details			TE	0, 1

Feature Bindings:

Type	Association Name	Association Ends					
		Class	Role	Mult	Class	Role	Mult
Asso	Bridle Connection	Bridle	buoyhangs	0, 1	GenericBuoy	bridleholds	1, 1
Asso	Swivel Connection	Bridle	swivelholds	1, 1	Swivel	bridlehangs	0, 1
Asso	Shackle To Bridle Connection	Bridle	shackleToBridle connectedTo	0, 1	MooringShackle	shackleToBridle connected	1, 1
Asso	Bridle Cable Connection	Bridle	cableholds	1, 1	CableSubmarine	bridleattached	0, *

2.48. Counter Weight

Definition: -

CamelCase: CounterWeight

Alias:

Super type: AidsToNavigation

Feature use type: geographic

Primitive: point

Remarks: -

Attribute Bindings:

S-10x Attribute	S-57 Acronym	Allowable Encoding Value	Type	Multiplicity
Nature of Construction	(NATCON)	1 : Masonry 2 : Concreted 3 : Loose Boulders 4 : Hard Surfaced 5 : Unsurfaced 6 : Wooden 7 : Metal 8 : Glass Reinforced Plastic 9 : Painted 10 : Framework 11 : Latticed 12 : Glass 13 : Fiberglass 14 : Plastic	EN	0, 1
Weight			RE	1, 1
Counter Weight Type			TE	0, 1

Feature Bindings:

Type	Association Name	Association Ends					
		Class	Role	Mult	Class	Role	Mult
Asso	Buoy Counter	CounterWeight	buoyattached	0, 1	GenericBuoy	counterWeighth	1, 1

	Weight					olds	
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2.49. Topmark

Definition: A characteristic shape secured at the top of a buoy or beacon to aid in its identification. (IHO Dictionary, S-32, 5th Edition, 5548)

CamelCase: Topmark

Alias:

Super type: AidsToNavigation

Feature use type: geographic

Primitive: point

Remarks: -

Attribute Bindings:

S-10x Attribute	S-57 Acronym	Allowable Encoding Value	Type	Multiplicity
Colour	(COLOUR)	1 : White 2 : Black 3 : Red 4 : Green 5 : Blue 6 : Yellow 7 : Grey 8 : Brown 9 : Amber 10 : Violet 11 : Orange 12 : Magenta 13 : Pink	EN	0, *
Colour Pattern	(COLPAT)	1 : Horizontal Stripes 2 : Vertical Stripes 3 : Diagonal Stripes 4 : Squared 5 : Stripes (Direction Unknown) 6 : Border Stripe 7 : Single Colour 8 : Rectangle 9 : Triangle	EN	0, *
Status	(STATUS)	1 : Permanent	EN	0, *

		2 : Occasional 3 : Recommended 4 : Not in Use 5 : Periodic/Intermittent 6 : Reserved 7 : Temporary 8 : Private 9 : Mandatory 11 : Extinguished 12 : Illuminated 13 : Historic 14 : Public 15 : Synchronized 16 : Watched 17 : Unwatched 18 : Existence Doubtful 19 : On Request 20 : Drop Away 21 : Rising 22 : Increasing 23 : Decreasing 24 : Strong 25 : Good 26 : Moderately 27 : Poor 28 : Buoyed 29 : Fully Operational 30 : Partially Operational 31 : Drifting 32 : Broken 33 : Offline 34 : Discontinued 35 : Manual Observation 36 : Unknown Status 37 : Confirmed 38 : Candidate 39 : Under Modification 41 : Under Removal / Deletion 42 : Removed / Deleted 43 : Candidate for Modification		
Topmark/Daymark Shape	(TOPSHP)	1 : Cone (Point Up)	EN	1, 1

		2 : Cone (Point Down) 3 : Sphere 4 : 2 Spheres 5 : Cylinder 6 : Board 7 : X-Shaped 8 : Upright Cross 9 : Cube (Point Up) 10 : 2 Cones (Point to Point) 11 : 2 Cones (Base to Base) 12 : Rhombus 13 : 2 Cones (Points Upward) 14 : 2 Cones (Points Downward) 15 : Besom (Point Up) 16 : Besom (Point Down) 17 : Flag 18 : Sphere Over a Rhombus 19 : Square 20 : Rectangle (Horizontal) 21 : Rectangle (Vertical) 22 : Trapezium (Up) 23 : Trapezium (Down) 24 : Triangle (Point Up) 25 : Triangle (Point Down) 26 : Circle 27 : Two Upright Crosses (One Over the Other) 28 : T-Shape 29 : Triangle Pointing Up Over a Circle 30 : Upright Cross Over a Circle 31 : Rhombus Over a Circle 32 : Circle Over a Triangle Pointing Up 33 : Other Shape (See Shape Information) 34 : Tubular		
Vertical Length	(VERLEN)		RE	0, 1

Feature Bindings:

Type	Association Name	Association Ends					
		Class	Role	Mult	Class	Role	Mult
Asso	Buoy Topmark	Topmark	topmarkPart	0, *	GenericBuoy	buoyPart	1, 1

2.50. Safe Water Beacon

Definition: A safe water beacon is used to indicate that there is navigable water around the mark.

CamelCase: SafeWaterBeacon

Alias:

Super type: GenericBeacon

Feature use type: geographic

Primitive: point

Remarks: -

2.51. Special Purpose General Beacon

Definition: A special purpose beacon is primarily used to indicate an area or feature, the nature of which is apparent from reference to a chart, Sailing Directions or Notices to Mariners.

CamelCase: SpecialPurposeGeneralBeacon

Alias:

Super type: GenericBeacon

Feature use type: geographic

Primitive: point

Remarks: -

Attribute Bindings:

S-10x Attribute	S-57 Acronym	Allowable Encoding Value	Type	Multiplicity
Category of Special Purpose Mark	(CATSPM)	1 : Firing Danger Mark 2 : Target Mark 3 : Marker Ship Mark 4 : Degaussing Range Mark 5 : Barge Mark 6 : Cable Mark 7 : Spoil Ground Mark 8 : Outfall Mark 9 : ODAS 10 : Recording Mark 11 : Seaplane Anchorage Mark 12 : Recreation Zone Mark 13 : Private Mark 14 : Mooring Mark 15 : LANBY 16 : Leading Mark 17 : Measured Distance Mark 18 : Notice Mark 19 : TSS Mark 20 : Anchoring Prohibited Mark 21 : Berthing Prohibited Mark 22 : Overtaking Prohibited	EN	1, *

		Mark 23 : Two-Way Traffic Prohibited Mark 24 : Reduced Wake Mark 25 : Speed Limit Mark 26 : Stop Mark 27 : General Warning Mark 28 : Sound Ship's Siren Mark 29 : Restricted Vertical Clearance Mark 30 : Maximum Vessel's Draught Mark 31 : Restricted Horizontal Clearance Mark 32 : Strong Current Warning Mark 33 : Berthing Permitted Mark 34 : Overhead Power Cable Mark 35 : Channel Edge Gradient Mark 36 : Telephone Mark 37 : Ferry Crossing Mark 39 : Pipeline Mark 40 : Anchorage Mark 41 : Clearing Mark 42 : Control Mark 43 : Diving Mark 44 : Refuge Beacon 45 : Foul Ground Mark 46 : Yachting Mark 47 : Heliport Mark 48 : GNSS Mark 49 : Seaplane Landing Mark 50 : Entry Prohibited Mark 51 : Work in Progress Mark 52 : Mark With Unknown Purpose 53 : Wellhead Mark 54 : Channel Separation Mark 55 : Marine Farm Mark 56 : Artificial Reef Mark 57 : Ice Mark 58 : Nature Reserve Mark		
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		59 : Fish Aggregating Device 60 : Wreck Mark 61 : Customs Mark 62 : Causeway Mark 63 : Wave Recorder 64 : Jetski Prohibited 65 : Facility Protection Mark 66 : Oil Pipeline Protection Mark 67 : Marine Cable Protection Mark		
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2.52. Safe Water Buoy

Definition: A safe water buoy is used to indicate that there is navigable water around the mark.

CamelCase: SafeWaterBuoy

Alias:

Super type: GenericBuoy

Feature use type: geographic

Primitive: point

Remarks: -

2.53. Special Purpose General Buoy

Definition: A special purpose buoy is primarily used to indicate an area or feature, the nature of which is apparent from reference to a chart, Sailing Directions or Notices to Mariners.

CamelCase: SpecialPurposeGeneralBuoy

Alias:

Super type: GenericBuoy

Feature use type: geographic

Primitive: point

Remarks: -

Attribute Bindings:

S-10x Attribute	S-57 Acronym	Allowable Encoding Value	Type	Multiplicity
Category of Special Purpose Mark	(CATSPM)	1 : Firing Danger Mark 2 : Target Mark 3 : Marker Ship Mark 4 : Degaussing Range Mark 5 : Barge Mark 6 : Cable Mark 7 : Spoil Ground Mark 8 : Outfall Mark 9 : ODAS 10 : Recording Mark 11 : Seaplane Anchorage Mark 12 : Recreation Zone Mark 13 : Private Mark 14 : Mooring Mark 15 : LANBY 16 : Leading Mark 17 : Measured Distance Mark 18 : Notice Mark 19 : TSS Mark 20 : Anchoring Prohibited Mark 21 : Berthing Prohibited Mark 22 : Overtaking Prohibited	EN	1, *

		Mark 23 : Two-Way Traffic Prohibited Mark 24 : Reduced Wake Mark 25 : Speed Limit Mark 26 : Stop Mark 27 : General Warning Mark 28 : Sound Ship's Siren Mark 29 : Restricted Vertical Clearance Mark 30 : Maximum Vessel's Draught Mark 31 : Restricted Horizontal Clearance Mark 32 : Strong Current Warning Mark 33 : Berthing Permitted Mark 34 : Overhead Power Cable Mark 35 : Channel Edge Gradient Mark 36 : Telephone Mark 37 : Ferry Crossing Mark 39 : Pipeline Mark 40 : Anchorage Mark 41 : Clearing Mark 42 : Control Mark 43 : Diving Mark 44 : Refuge Beacon 45 : Foul Ground Mark 46 : Yachting Mark 47 : Heliport Mark 48 : GNSS Mark 49 : Seaplane Landing Mark 50 : Entry Prohibited Mark 51 : Work in Progress Mark 52 : Mark With Unknown Purpose 53 : Wellhead Mark 54 : Channel Separation Mark 55 : Marine Farm Mark 56 : Artificial Reef Mark 57 : Ice Mark 58 : Nature Reserve Mark		
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		59 : Fish Aggregating Device 60 : Wreck Mark 61 : Customs Mark 62 : Causeway Mark 63 : Wave Recorder 64 : Jetski Prohibited 65 : Facility Protection Mark 66 : Oil Pipeline Protection Mark 67 : Marine Cable Protection Mark		
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2.54. Dangerous Feature

Definition: -

CamelCase: DangerousFeature

Alias:

Super type:

Feature use type: geographic

Primitive: point

Remarks: -

Attribute Bindings:

S-10x Attribute	S-57 Acronym	Allowable Encoding Value	Type	Multiplicity
Text	(INFORM) (NINFOM)		(S) TE	0, 1

Feature Bindings:

Type	Association Name	Association Ends					
		Class	Role	Mult	Class	Role	Mult
Asso	Dangerous Feature Association	DangerousFeature	danger	0, *	AtonAssociation	markingAton	1, *

2.55. Aton Aggregation

Definition: Used to identify an aggregation of two or more objects. This aggregation may be named content of categoryOfAggregation should be put in information attribute when converting to S-57.

CamelCase: AtonAggregation

Alias:

Super type:

Feature use type: geographic

Primitive: noGeometry

Remarks: -

Attribute Bindings:

S-10x Attribute	S-57 Acronym	Allowable Encoding Value	Type	Multiplicity
Category Of Aggregation		1 : leading line 3 : measured distance 2 : range system	CL	1, 1

Feature Bindings:

Type	Association Name	Association Ends					
		Class	Role	Mult	Class	Role	Mult
Asso	Aton Aggregations	AtonAggregation	peerAtonAggregation	0, *	AidsToNavigation	atonAggregation By	0, *

2.56. Aton Association

Definition: Used to identify an association between two or more objects. The association may be named content of categoryOfAssociation should be put in information attribute when converting to S-57

CamelCase: AtonAssociation

Alias:

Super type:

Feature use type: geographic

Primitive: noGeometry

Remarks: -

Attribute Bindings:

S-10x Attribute	S-57 Acronym	Allowable Encoding Value	Type	Multiplicity
Category Of Association		1 : channel markings 2 : danger markings	CL	1, 1

Feature Bindings:

Type	Association Name	Association Ends					
		Class	Role	Mult	Class	Role	Mult
Asso	Dangerous Feature Association	AtonAssociation	markingAton	1, *	DangerousFeature	danger	0, *
Asso	Aton Associations	AtonAssociation	peerAtonAssociation	0, *	AidsToNavigation	atonAssociationBy	0, *

3. Carto Feature Types

4. Information Types

4.1. AtoN Fixing Method

Definition: Method used for fixing the position of an aid to navigation.

CamelCase: AtoNFixingMethod

Alias:

Super type:

Remarks: -

Attribute Bindings:

S-10x Attribute	S-57 Acronym	Allowable Encoding Value	Type	Multiplicity
Reference Point			TE	0, 1
Horizontal Datum	(HORDAT)	1 : WGS 72 2 : WGS 84 3 : European 1950 4 : Potsdam Datum 5 : Adindan 6 : Afgooye 7 : Ain el Abd 1970 8 : Anna 1 Astro 1965 9 : Antigua Island Astro 1943 10 : Arc 1950 11 : Arc 1960 12 : Ascension Island 1958 13 : Astro Beacon 'E' 1945 14 : Astro DOS 71/4 15 : Astro Tern Island (FRIG) 1961 16 : Astronomical Station 1952 17 : Australian Geodetic 1966 18 : Australian Geodetic 1984 19 : Ayabelle Lighthouse 20 : Bellevue (IGN)	EN	0, 1

		21 : Bermuda 1957 22 : Bissau 23 : Bogota Observatory 24 : Bukit Rimpah 25 : Camp Area Astro 26 : Campo Inchauspe 1969 27 : Canton Astro 1966 28 : Cape Datum 29 : Cape Canaveral 30 : Carthage 31 : Chatam Island Astro 1971 32 : Chua Astro 33 : Corrego Alegre 34 : Dabola 35 : Djakarta (Batavia) 36 : DOS 1968 37 : Easter Island 1967 38 : European 1979 39 : Fort Thomas 1955 40 : Gan 1970 41 : Geodetic Datum 1949 42 : Graciosa Base SW 1948 43 : Guam 1963 44 : Gunung Segara 45 : GUX 1 Astro 46 : Herat North 47 : Hjorsey 1955 48 : Hong Kong 1963 49 : Hu-Tzu-Shan 50 : Indian 51 : Indian 1954 52 : Indian 1975 53 : Ireland 1965 54 : ISTS 061 Astro 1968 55 : ISTS 073 Astro 1969 56 : Johnston Island 1961 57 : Kandawala 58 : Kerguelen Island 1949 59 : Kertau 1968 60 : Kusaie Astro 1951 61 : L. C. 5 Astro 1961		
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		62 : Leigon 63 : Liberia 1964 64 : Luzon 65 : Mahe 1971 66 : Massawa 67 : Merchich 68 : Midway Astro 1961 69 : Minna 70 : Montserrat Island Astro 1958 71 : M'poraloko 72 : Nahrwan 73 : Naparima, BWI 74 : North American 1927 75 : North American 1983 76 : Observatorio Meteorologico 1939 77 : Old Egyptian 1907 78 : Old Hawaiian 79 : Oman 80 : Ordnance Survey of Great Britain 1936 81 : Pico de las Nieves 82 : Pitcairn Astro 1967 83 : Point 58 84 : Pointe Noire 1948 85 : Porto Santo 1936 86 : Provisional South American 1956 87 : Provisional South Chilean 1963 88 : Puerto Rico 89 : Qatar National 90 : Qornoq 91 : Reunion 92 : Rome 1940 93 : Santo (DOS) 1965 94 : Sao Braz 95 : Sapper Hill 1943 96 : Schwarzeck 97 : Selvagem Grande 1938 98 : South American 1969 99 : South Asia 100 : Tananarive Observatory 1925		
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		101 : Timbalai 1948 102 : Tokyo 103 : Tristan Astro 1968 104 : Viti Levu 1916 105 : Wake-Eniwetok 1960 106 : Wake Island Astro 1952 107 : Yacare 108 : Zanderij 109 : American Samoa 1962 110 : Deception Island 111 : Indian 1960 112 : Indonesian 1974 113 : North Sahara 1959 114 : Pulkovo 1942 116 : S-JTSK 117 : Voirol 1950 118 : Average Terrestrial System 1977 119 : Compensation Geodesique du Quebec 1977 120 : Finnish (KKJ) 121 : Ordnance Survey of Ireland 122 : Revised Kertau 123 : Revised Nahrwan 124 : GGRS 76 (Greece) 125 : Nouvelle Triangulation de France 126 : RT 90 (Sweden) 127 : Geocentric Datum of Australia 128 : BJZ54 (A954 Beijing Coordinates) 129 : Modified BJZ54 130 : GDZ80 131 : Local Datum		
Source Date	(SORDAT)		DA	1, 1
Positioning Procedure			TE	1, 1

4.2. Aton Status Information

Definition: -

CamelCase: AtonStatusInformation

Alias:

Super type:

Remarks: -

Attribute Bindings:

S-10x Attribute	S-57 Acronym	Allowable Encoding Value	Type	Multiplicity
Change Details			C	1, 1
Change Types		1 : Advanced notice of changes 2 : Discrepancy 3 : Proposed changes 4 : Temporary changes	EN	0, 1

4.3. Positioning Information

Definition: Information about how a position was obtained. (proposed by CCG)

CamelCase: PositioningInformation

Alias:

Super type:

Remarks: -

Attribute Bindings:

S-10x Attribute	S-57 Acronym	Allowable Encoding Value	Type	Multiplicity
Positioning Device			TE	1, 1
NMEAString			(S) TE	1, 1

4.4. Spatial Quality

Definition: The indication of the quality of the locational information for features in a dataset.

CamelCase: SpatialQuality

Alias:

Super type:

Remarks: No remarks.

Attribute Bindings:

S-10x Attribute	S-57 Acronym	Allowable Encoding Value	Type	Multiplicity
Quality of Horizontal Measurement	(QUAPOS)	1 : Surveyed 2 : Unsurveyed 3 : Inadequately Surveyed 4 : Approximate 5 : Position Doubtful 6 : Unreliable 7 : Reported (Not Surveyed) 8 : Reported (Not Confirmed) 9 : Estimated 10 : Precisely Known 11 : Calculated	EN	0, 1
Uncertainty Variable Factor			(S) RE	0, 1

5. Simple Attributes

5.1. Category Of Association

Definition: named associations between two or more aids to navigation and/or navigationally relevant features

CamelCase: CategoryOfAssociation

Alias:

Value type: S100_CodeList

Remarks: -

Listed Values:

Code	Label	Definition
1	channel markings	-
2	danger markings	-

5.2. Seasonal Action Required

Definition: -

CamelCase: SeasonalActionRequired

Alias:

Value type: text

Remarks: -

5.3. Counter Weight Type

Definition: -

CamelCase: counterWeightType

Alias:

Value type: text

Remarks: -

5.4. Swivel Type

Definition: -

CamelCase: swivelType

Alias:

Value type: text

Remarks: -

5.5. Legs Details

Definition: -

CamelCase: legsDetails

Alias:

Value type: text

Remarks: -

5.6. BridleLink Type

Definition: -

CamelCase: bridleLinkType

Alias:

Value type: text

Remarks: -

5.7. Sinkers Type

Definition: -

CamelCase: sinkerType

Alias:

Value type: text

Remarks: -

5.8. Weight

Definition: -

CamelCase: weight

Alias:

Value type: real

Remarks: -

5.9. Cable Length

Definition: Total length of a cable.

CamelCase: cableLength

Alias:

Value type: real

Remarks: -

5.10. Diameter

Definition: -

CamelCase: diameter

Alias:

Value type: real

Remarks: -

5.11. Horizontal Accuracy

Definition: -

CamelCase: horizontalAccuracy

Alias:

Value type: real

Remarks: -

5.12. Manufacturer

Definition: -

CamelCase: manufacturer

Alias:

Value type: text

Remarks: -

5.13. IsSlatted

Definition: -

CamelCase: isSlatted

Alias:

Value type: boolean

Remarks: -

5.14. Candela

Definition: -

CamelCase: candela

Alias:

Value type: real

Remarks: -

5.15. Effective Intensity

Definition: The luminous intensity of a fictitious juxtaposed steady-burning point light source that would appear to exhibit a luminosity equal to that of the rhythmic point light source it describes. The apparent reduction in intensity of the rhythmic light is subjective and is due to the nature of the response of the eye of the observer.

CamelCase: effectiveIntensity

Alias:

Value type: real

Remarks: -

5.16. Peak Intensity

Definition: The maximum luminous intensity of a light during its flash cycle.

CamelCase: peakIntensity

Alias:

Value type: real

Remarks: -

5.17. Vertical Accuracy

Definition: -

CamelCase: verticalAccuracy

Alias:

Value type: real

Remarks: -

5.18. Quality of Horizontal Measurement

Definition: The degree of reliability attributed to a position.

CamelCase: qualityOfHorizontalMeasurement

Alias: QUAPOS

Value type: enumeration

Remarks: No remarks.

Listed Values:

Code	Label	Definition
1	Surveyed	The position(s) was(were) determined by the operation of making measurements for determining the relative position of points on, above or beneath the earth's surface. Survey implies a regular, controlled survey of any date.
2	Unsurveyed	Survey data is does not exist or is very poor.
3	Inadequately Surveyed	Not surveyed to modern standards; or due to its age, scale, or positional or vertical uncertainties is not suitable to the type of navigation expected in the area.
4	Approximate	A position that is considered to be less than third-order accuracy, but is generally considered to be within 30.5 metres of its correct geographic location. Also may apply to an object whose position does not remain fixed.
5	Position Doubtful	Of uncertain position. The expression is used principally on charts to indicate that a wreck, shoal, etc., has been reported in various positions and not definitely determined in any.
6	Unreliable	A feature's position has been obtained from questionable or unreliable data.
7	Reported (Not Surveyed)	An object whose position has been reported and its position confirmed by some means other than a formal survey such as an independent report of the same object.
8	Reported (Not Confirmed)	An object whose position has been reported and its position has not been confirmed.
9	Estimated	The most probable position of an object determined from incomplete data or data of questionable accuracy.
10	Precisely Known	A position that is of a known value, such as the position of an anchor berth or other defined object.
11	Calculated	A position that is computed from data.

5.19. Positioning Device

Definition: -

CamelCase: positioningDevice

Alias:

Value type: text

Remarks: -

5.20. Change Types

Definition: -

CamelCase: ChangeTypes

Alias:

Value type: enumeration

Remarks: -

Listed Values:

Code	Label	Definition
1	Advanced notice of changes	-
2	Discrepancy	-
3	Proposed changes	-
4	Temporary changes	-

5.21. Height Length Units

Definition: Units of measure of waterway distances. (IHO Registry)

CamelCase: heightLengthUnits

Alias:

Value type: enumeration

Remarks: -

Listed Values:

Code	Label	Definition
1	Metres	-
2	Feet	-
3	Kilometres	-
4	Hectometres	-
5	Statute Miles	-
6	Nautical Miles	-

5.22. Positioning Procedure

Definition: -

CamelCase: positioningProcedure

Alias:

Value type: text

Remarks: -

5.23. Horizontal Datum

Definition: Horizontal reference surface or the reference coordinate system used for geodetic control in the calculation of coordinates of points on the earth.

CamelCase: horizontalDatum

Alias: HORDAT

Value type: enumeration

Remarks: All necessary information for conversion of geographic coordinates from most of the Geodetic Datums in the above list to WGS-84 is contained in the 'User's Handbook on Datum Transformations involving WGS-84', prepared by the US Defense Mapping Agency and which is available from the IHB as IHO Publication S-60 (English and French Versions), along with an associated standard datum transformation software on floppy disk called 'MADTRAN'.

Listed Values:

Code	Label	Definition
1	WGS 72	A standard for use in cartography, geodesy, and satellite navigation including GPS. This standard includes the definition of the coordinate system's fundamental and derived constants, the ellipsoidal (normal) Earth Gravitational Model (EGM), a description of the associated World Magnetic Model (WMM), and a current list of local datum transformations. The WGS 72 is based on selected satellite, surface gravity and astrogeodetic data available through 1972.
2	WGS 84	A standard for use in cartography, geodesy, and satellite navigation including GPS. This standard includes the definition of the coordinate system's fundamental and derived constants, the ellipsoidal (normal) Earth Gravitational Model (EGM), a description of the associated World Magnetic Model (WMM), and a current list of local datum transformations. WGS 84 is the reference coordinate system used by the Global Positioning System.
3	European 1950	A geodetic datum first defined in 1950 suitable for use in Europe - west: Andorra; Cyprus; Denmark - onshore and offshore; Faroe Islands - onshore; France - offshore; Germany - offshore North Sea; Gibraltar; Greece - offshore; Israel - offshore; Italy including San Marino and Vatican City State; Ireland offshore; Malta; Netherlands - offshore; North Sea; Norway including Svalbard - onshore and offshore; Portugal - mainland - offshore; Spain - onshore; Turkey - onshore and offshore; United Kingdom UKCS offshore east of 6W including Channel Islands (Guernsey and Jersey). Egypt - Western Desert; Iraq - onshore; Jordan. European Datum 1950 references the International 1924 ellipsoid and the Greenwich prime meridian. European Datum 1950 origin is Fundamental point: Potsdam (Helmert Tower). Latitude: 5222'51.4456"N, longitude: 1303'58.9283"E (of Greenwich). European Datum 1950 is a geodetic datum for Topographic mapping, geodetic survey.
4	Potsdam Datum	A geodetic datum first defined in 1990 suitable for use in Germany - Thuringen. Potsdam Datum/83 references the Bessel 1841 ellipsoid and the Greenwich prime meridian. Potsdam Datum/83 origin is Fundamental point: Rauenberg. Latitude: 5227'12.021"N, longitude: 1322'04.928"E (of Greenwich). This station was destroyed in 1910 and the station at Potsdam substituted as the fundamental point. Potsdam Datum/83 is a geodetic datum for Geodetic survey,

		cadastre, topographic mapping, engineering survey. It was defined by information from BKG via EuroGeographics. http://crs.bkg.bund.de PD/83 is the realisation of DHDN in Thuringen. It is the resultant of applying a transformation derived at 13 points on the border between East and West Germany to Pulkovo 1942/83 points in Thuringen.
5	Adindan	A geodetic datum first defined in 1958 suitable for use in Eritrea; Ethiopia; South Sudan; Sudan. Adindan references the Clarke 1880 (RGS) ellipsoid and the Greenwich prime meridian. Adindan origin is Fundamental point: Station 15; Adindan. Latitude: 2210'07.110"N, longitude: 3129'21.608"E (of Greenwich). Adindan is a geodetic datum for Topographic mapping. It was defined by information from US Coast and Geodetic Survey via Geophysical Research vol 67 #11, October 1962. The 12th parallel traverse of 1966-70 (Point 58 datum, code 6620) is connected to the Blue Nile 1958 network in western Sudan. This has given rise to misconceptions that the Blue Nile network is used in west Africa.
6	Afgooye	A geodetic datum first defined in and suitable for use in Somalia - onshore. Afgooye references the Krassowsky 1940 ellipsoid and the Greenwich prime meridian. Afgooye is a geodetic datum for Topographic mapping.
7	Ain el Abd 1970	A geodetic datum first defined in 1970 and suitable for use in Bahrain, Kuwait and Saudi Arabia - onshore. Ain el Abd 1970 references the International 1924 ellipsoid and the Greenwich prime meridian. Ain el Abd 1970 origin is Fundamental point: Ain El Abd. Latitude: 2814'06.171"N, longitude: 4816'20.906"E (of Greenwich). Ain el Abd 1970 is a geodetic datum for Topographic mapping.
8	Anna 1 Astro 1965	A geodetic datum first defined in 1965 suitable for use in Cocos (Keeling) Islands - onshore. Cocos Islands 1965 references the Australian National Spheroid ellipsoid and the Greenwich prime meridian. Cocos Islands 1965 origin is Fundamental point: Anna 1. Cocos Islands 1965 is a geodetic datum for Military and topographic mapping It was defined by information from DMA / NIMA / NGA TR8350.2 (3rd edition, Amendment 1, 3 January 2000).
9	Antigua Island Astro 1943	A geodetic datum first defined in 1943 suitable for use in Antigua island - onshore. Antigua 1943 references the Clarke 1880 (RGS) ellipsoid and the Greenwich prime meridian. Antigua 1943 origin is Fundamental point: station A14. Antigua 1943 is a geodetic datum for Topographic mapping. It was defined by information from Ordnance Survey of Great Britain.
10	Arc 1950	A geodetic datum first defined in 1950 suitable for use in Botswana; Malawi; Zambia; Zimbabwe. Arc 1950 references the Clarke 1880 (Arc) ellipsoid and the Greenwich prime meridian. Arc 1950 origin is Fundamental point: Buffelsfontein. Latitude: 3359'32.000"S, longitude: 2530'44.622"E (of Greenwich). Arc 1950 is a geodetic datum for Topographic mapping, geodetic survey.
11	Arc 1960	A geodetic datum first defined in 1960 suitable for use in Kenya; Tanzania; Uganda. Arc 1960 references the Clarke 1880 (RGS) ellipsoid and the Greenwich prime meridian. Arc 1960 origin is Fundamental point: Buffelsfontein. Latitude: 3359'32.000"S, longitude: 2530'44.622"E (of Greenwich). Arc 1960 is a geodetic datum for Topographic mapping, geodetic survey.
12	Ascension Island 1958	A geodetic datum first defined in 1958 suitable for use in St Helena, Ascension and Tristan da Cunha - Ascension Island - onshore. Ascension Island 1958 references the International 1924 ellipsoid and the Greenwich prime meridian. Ascension Island 1958 is a geodetic datum for Military and topographic mapping. It was defined by information from DMA / NIMA / NGA TR8350.2 (3rd edition,

		Amendment 1, 3 January 2000).
13	Astro Beacon 'E' 1945	Astro beacon 'E' 1945
14	Astro DOS 71/4	A geodetic datum first defined in 1971 suitable for use in St Helena, Ascension and Tristan da Cunha - St Helena Island - onshore. Astro DOS 71 references the International 1924 ellipsoid and the Greenwich prime meridian. Astro DOS 71 origin is Fundamental point: DOS 71/4, Ladder Hill Fort, latitude: 1555'30"S, longitude: 543'25"W (of Greenwich). Astro DOS 71 is a geodetic datum for Geodetic control, military and topographic mapping. It was defined by information from DMA / NIMA / NGA TR8350.2 (3rd edition, Amendment 1, 3 January 2000) and St. Helena Government, Environment and Natural Resources Directorate (ENRD).
15	Astro Tern Island (FRIG) 1961	A geodetic datum first defined in 1961 suitable for use in United States (USA) - Hawaii - Tern Island and Sorel Atoll. Tern Island 1961 references the International 1924 ellipsoid and the Greenwich prime meridian. Tern Island 1961 origin is Fundamental point: station FRIG on tern island, station B4 on Sorel Atoll. Tern Island 1961 is a geodetic datum for Military and topographic mapping. It was defined by information from DMA / NIMA / NGA TR8350.2 (original 1987 first edition and 3rd edition, Amendment 1, 3 January 2000). Two independent astronomic determinations considered to be consistent through adoption of common transformation to WGS 84 (see tfm code 15795).
16	Astronomical Station 1952	Astronomical station 1952.
17	Australian Geodetic 1966	A geodetic datum first defined in 1966 suitable for use in Australia - onshore and offshore. Papua New Guinea - onshore. Australian Geodetic Datum 1966 references the Australian National Spheroid ellipsoid and the Greenwich prime meridian. Australian Geodetic Datum 1966 origin is Fundamental point: Johnson Memorial Cairn. Latitude: 2556'54.5515"S, longitude: 13312'30.0771"E (of Greenwich). Australian Geodetic Datum 1966 is a geodetic datum for Topographic mapping. It was defined by information from Australian Map Grid Technical Manual. National Mapping Council of Australia Technical Publication 7; 1972.
18	Australian Geodetic 1984	A geodetic datum first defined in 1984 suitable for use in Australia - Queensland, South Australia, Western Australia, federal areas offshore west of 129E. Australian Geodetic Datum 1984 references the Australian National Spheroid ellipsoid and the Greenwich prime meridian. Australian Geodetic Datum 1984 origin is Fundamental point: Johnson Memorial Cairn. Latitude: 2556'54.5515"S, longitude: 13312'30.0771"E (of Greenwich). Australian Geodetic Datum 1984 is a geodetic datum for Topographic mapping. It was defined by information from "GDA technical manual v2_2", Intergovernmental Committee on Surveying and Mapping. www.anzlic.org.au/icsm/gdtm/ Uses all data from 1966 adjustment with additional observations, improved software and a geoid model.
19	Ayabelle Lighthouse	A geodetic datum suitable for use in Djibouti - onshore and offshore. Ayabelle Lighthouse references the Clarke 1880 (RGS) ellipsoid and the Greenwich prime meridian. Ayabelle Lighthouse origin is Fundamental point: Ayabelle Lighthouse. Ayabelle Lighthouse is a geodetic datum for Military and topographic mapping. It was defined by information from DMA / NIMA / NGA TR8350.2 (3rd edition, Amendment 1, 3 January 2000).
20	Bellevue (IGN)	A geodetic datum first defined in 1960 suitable for use in Vanuatu - southern islands - Aneityum, Efate, Erromango and Tanna. Bellevue references the International 1924 ellipsoid and the Greenwich prime meridian. Bellevue is a geodetic datum for Military and topographic mapping. It was defined by information from DMA / NIMA / NGA

		TR8350.2 (3rd edition, Amendment 1, 3 January 2000). Datum covers all the major islands of Vanuatu in two different adjustment blocks, but practical usage is as given in the area of use.
21	Bermuda 1957	A geodetic datum first defined in 1957 suitable for use in Bermuda - onshore. Bermuda 1957 references the Clarke 1866 ellipsoid and the Greenwich prime meridian. Bermuda 1957 origin is Fundamental point: Fort George base. Latitude 3222'44.36"N, longitude 6440'58.11"W (of Greenwich). Bermuda 1957 is a geodetic datum for Topographic mapping. It was defined by information from Various oil industry sources.
22	Bissau	A geodetic datum first defined in and is suitable for use in Guinea-Bissau - onshore. Bissau references the International 1924 ellipsoid and the Greenwich prime meridian. Bissau origin is Bissau is a geodetic datum for Topographic mapping. It was defined by information from NIMA TR8350.2 ftp://164.214.2.65/pub/gig/tr8350.2/changes.pdf .
23	Bogota Observatory	A geodetic datum first defined in 1975 suitable for use in Colombia - mainland and offshore Caribbean. Bogota 1975 references the International 1924 ellipsoid and the Greenwich prime meridian. Bogota 1975 origin is Fundamental point: Bogota observatory. Latitude: 435'56.570"N, longitude: 7404'51.300"W (of Greenwich). Bogota 1975 is a geodetic datum for Topographic mapping. It was defined by information from Instituto Geografico Agustin Codazzi (IGAC) special publication no. 1, 4th edition (1975) "Geodesia: Resultados Definitivos de Parte de las Redes Geodesicas Establecidas en el Pais". Replaces 1951 adjustment. Replaced by MAGNA-SIRGAS (datum code 6685).
24	Bukit Rimpah	A geodetic datum suitable for use in Indonesia - Banga and Belitung Islands. Bukit Rimpah references the Bessel 1841 ellipsoid and the Greenwich prime meridian. Bukit Rimpah origin is 200'40.16"S, 10551'39.76"E (of Greenwich). Bukit Rimpah is a geodetic datum for Topographic mapping.
25	Camp Area Astro	A geodetic datum suitable for use in Antarctica - McMurdo Sound, Camp McMurdo area. Camp Area Astro references the International 1924 ellipsoid and the Greenwich prime meridian. Camp Area Astro is a geodetic datum for Geodetic and topographic survey. It was defined by information from DMA / NIMA / NGA TR8350.2 (3rd edition, Amendment 1, 3 January 2000).
26	Campo Inchauspe 1969	A geodetic datum suitable for use in Argentina - mainland onshore and Atlantic offshore Tierra del Fuego. Campo Inchauspe references the International 1924 ellipsoid and the Greenwich prime meridian. Campo Inchauspe origin is Fundamental point: Campo Inchauspe. Latitude: 3558'16.56"S, longitude: 6210'12.03"W (of Greenwich). Campo Inchauspe is a geodetic datum for Topographic mapping. It was defined by information from NIMA http://earth-info.nima.mil/
27	Canton Astro 1966	A geodetic datum first defined in 1966 suitable for use in Kiribati - Phoenix Islands: Kanton, Orona, McKean Atoll, Birnie Atoll, Phoenix Seamounts. Phoenix Islands 1966 references the International 1924 ellipsoid and the Greenwich prime meridian. Phoenix Islands 1966 is a geodetic datum for Military and topographic mapping. It was defined by information from DMA / NIMA / NGA TR8350.2 (3rd edition, Amendment 1, 3 January 2000).
28	Cape Datum	A geodetic datum suitable for use in Botswana; Lesotho; South Africa - mainland; Swaziland. Cape references the Clarke 1880 (Arc) ellipsoid and the Greenwich prime meridian. Cape origin is Fundamental point: Buffelsfontein. Latitude: 3359'32.000"S, longitude: 2530'44.622"E (of Greenwich). Cape is a geodetic datum for Geodetic survey, cadastre, topographic mapping, engineering

		survey. It was defined by information from Private Communication, Directorate of Surveys and Land Information, Cape Town.
29	Cape Canaveral	A geodetic datum first defined in 1963 suitable for use in North America - onshore - Bahamas and USA - Florida (east). Cape Canaveral references the Clarke 1866 ellipsoid and the Greenwich prime meridian. Cape Canaveral origin is Fundamental point: Central 1950. Latitude: 28 29'32.36555"N, longitude 80 34'38.77362"W (of Greenwich). Cape Canaveral is a geodetic datum for US space and military operations. It was defined by information from US NGS and DMA / NIMA / NGA TR8350.2 (3rd edition, Amendment 1, 3 January 2000).
30	Carthage	A geodetic datum first defined in 1925 suitable for use in Tunisia - onshore and offshore. Carthage references the Clarke 1880 (IGN) ellipsoid and the Greenwich prime meridian. Carthage origin is Fundamental point: Carthage. Latitude: 40.9464506g = 3651'06.50"N, longitude: 8.8724368g E of Paris = 1019'20.72"E (of Greenwich). Carthage is a geodetic datum for Topographic mapping. Fundamental point astronomic coordinates determined in 1878.
31	Chatam Island Astro 1971	A geodetic datum first defined in 1971 suitable for use in New Zealand - Chatham Islands group - onshore. Chatham Islands Datum 1971 references the International 1924 ellipsoid and the Greenwich prime meridian. Chatham Islands Datum 1971 is a geodetic datum for Geodetic survey, topographic mapping, engineering survey. It was defined by information from Office of Surveyor General (OSG) Technical Report 14, June 2001. Replaced by Chatham Islands Datum 1979 (code 6673).
32	Chua Astro	A geodetic datum suitable for use in Brazil - south of 18S and west of 54W, plus Distrito Federal. Paraguay - north. Chua references the International 1924 ellipsoid and the Greenwich prime meridian. Chua origin is Fundamental point: Chua. Latitude: 1945'41.160"S, longitude: 4806'07.560"W (of Greenwich). Chua is a geodetic datum for Geodetic survey. It was defined by information from NIMA http://earth-info.nima.mil/ . The Chua origin and associated network is in Brazil with a connecting traverse through northern Paraguay. It was used in Brazil only as input into the Corrego Alegre adjustment and for government work in Distrito Federal.
33	Corrego Alegre	A geodetic datum first defined in 1972 suitable for use in Brazil - onshore - west of 54W and south of 18S; also south of 15S between 54W and 42W; also east of 42W. Corrego Alegre 1970-72 references the International 1924 ellipsoid and the Greenwich prime meridian. Corrego Alegre 1970-72 origin is Fundamental point: Corrego Alegre. Latitude: 1950'14.91"S, longitude: 4857'41.98"W (of Greenwich). Corrego Alegre 1970-72 is a geodetic datum for Topographic mapping, geodetic survey. Superseded by SAD69. It was defined by information from IBGE. Replaces 1961 adjustment (datum code 1074). NIMA gives coordinates of origin as latitude: 1950'15.14"S, longitude: 4857'42.75"W; these may refer to 1961 adjustment.
34	Dabola	A geodetic datum first defined in 1981 suitable for use in Guinea - onshore. Dabola 1981 references the Clarke 1880 (IGN) ellipsoid and the Greenwich prime meridian. Dabola 1981 is a geodetic datum for Topographic mapping. It was defined by information from IGN Paris.
35	Djakarta (Batavia)	A geodetic datum suitable for use in Indonesia - onshore Java and Bali. Batavia (Jakarta) references the Bessel 1841 ellipsoid and the Jakarta prime meridian. Batavia (Jakarta) origin is Fundamental point: Longitude at Batavia astronomical station. Latitude: 607'39.522"S, longitude: 000'00.0"E (of Jakarta). Latitude and azimuth at Genuk. Batavia (Jakarta) is a geodetic datum for Topographic mapping.

36	DOS 1968	DOS 1968.
37	Easter Island 1967	A geodetic datum first defined in 1967 suitable for use in Chile - Easter Island onshore. Easter Island 1967 references the International 1924 ellipsoid and the Greenwich prime meridian. Easter Island 1967 is a geodetic datum for Military and topographic mapping, +/- 25 meters in each component. It was defined by information from DMA / NIMA / NGA TR8350.2 (3rd edition, Amendment 1, 3 January 2000).
38	European 1979	A geodetic datum first defined in 1979 suitable for use in Europe - west. European Datum 1979 references the International 1924 ellipsoid and the Greenwich prime meridian. European Datum 1979 origin is Fundamental point: Potsdam (Helmert Tower). Latitude: 5222'51.4456"N, longitude: 1303'58.9283"E (of Greenwich). European Datum 1979 is a geodetic datum for Scientific network. Replaced by 1987 adjustment.
39	Fort Thomas 1955	Fort Thomas 1955 datum.
40	Gan 1970	A geodetic datum first defined in 1970 suitable for use in Maldives - onshore. Gan 1970 references the International 1924 ellipsoid and the Greenwich prime meridian. Gan 1970 is a geodetic datum for Topographic mapping. It was defined by information from Various industry sources. In some references incorrectly named "Gandajika 1970".
41	Geodetic Datum 1949	A geodetic datum first defined in 1949 suitable for use in New Zealand - North Island, South Island, Stewart Island - onshore and nearshore. New Zealand Geodetic Datum 1949 references the International 1924 ellipsoid and the Greenwich prime meridian. New Zealand Geodetic Datum 1949 origin is Fundamental point: Papatahi. Latitude: 4119' 8.900"S, longitude: 17502'51.000"E (of Greenwich). New Zealand Geodetic Datum 1949 is a geodetic datum for Geodetic survey, cadastre, topographic mapping, engineering survey. It was defined by information from Land Information New Zealand. http://www.linz.govt.nz/rcs/linz/pub/web/root/core/SurveySystem/GeodeticInfo/GeodeticDatums/nzgd2000factsheet/index.jsp . Replaced by New Zealand Geodetic Datum 2000 (code 6167) from March 2000.
42	Graciosa Base SW 1948	Graciosa Base SW 1948 datum.
43	Guam 1963	A geodetic datum first defined in 1963 suitable for use in Guam - onshore. Guam 1963 references the Clarke 1866 ellipsoid and the Greenwich prime meridian. Guam 1963 origin is Fundamental point: Tagcha. Latitude: 1322'38.49"N, longitude: 14445'51.56"E (of Greenwich). Guam 1963 is a geodetic datum for Topographic mapping. It was defined by information from US National Geospatial Intelligence Agency (NGA). http://earth-info.nga.mil/ Replaced by NAD83(HARN)
44	Gunung Segara	A geodetic datum suitable for use in Indonesia - Kalimantan - onshore east coastal area including Mahakam delta coastal and offshore shelf areas. Gunung Segara references the Bessel 1841 ellipsoid and the Greenwich prime meridian. Gunung Segara origin is Station P5 (Gunung Segara). Latitude 032'12.83"S, longitude 11708'48.47"E (of Greenwich). Gunung Segara is a geodetic datum for Topographic mapping. It was defined by information from TotalFinaElf.
45	GUX 1 Astro	GUX 1 Astro datum.
46	Herat North	A geodetic datum suitable for use in Afghanistan. Herat North references the International 1924 ellipsoid and the Greenwich prime meridian. Herat North origin is Fundamental point: Herat North. Latitude: 3423'09.08"N, longitude: 6410'58.94"E (of Greenwich).

		Herat North is a geodetic datum for Topographic mapping. It was defined by information from NIMA http://earth-info.nima.mil/ .
47	Hjorsey 1955	A geodetic datum first defined in 1955 suitable for use in Iceland - onshore. Hjorsey 1955 references the International 1924 ellipsoid and the Greenwich prime meridian. Hjorsey 1955 origin is Fundamental point: Latitude: 6431'29.26"N, longitude: 2222'05.84"W (of Greenwich). Hjorsey 1955 is a geodetic datum for 1/50,000 scale topographic mapping. It was defined by information from Landmaelingar Islands (National Survey of Iceland).
48	Hong Kong 1963	A geodetic datum first defined in 1963 suitable for use in China - Hong Kong - onshore and offshore. Hong Kong 1963 references the Clarke 1858 ellipsoid and the Greenwich prime meridian. Hong Kong 1963 origin is Fundamental point: Trig "Zero", 38.4 feet south along the transit circle of the Kowloon Observatory. Latitude 2218'12.82"N, longitude 11410'18.75"E (of Greenwich). Hong Kong 1963 is a geodetic datum for Topographic mapping and hydrographic charting. It was defined by information from Survey and Mapping Office, Lands Department. http://www.info.gov.hk/landsd/ . Replaced by Hong Kong 1963(67) for military purposes only in 1967. Replaced by Hong Kong 1980.
49	Hu-Tzu-Shan	A geodetic datum first defined in 1950 suitable for use in Taiwan, Republic of China - onshore - Taiwan Island, Penghu (Pescadores) Islands. Hu Tzu Shan 1950 references the International 1924 ellipsoid and the Greenwich prime meridian. Hu Tzu Shan 1950 origin is Fundamental point: Hu Tzu Shan. Latitude: 2358'32.34"N, longitude: 12058'25.975"E (of Greenwich). Hu Tzu Shan 1950 is a geodetic datum for Topographic mapping. It was defined by information from NIMA US NGA, http://earth-info.nga.mil/GandG/index.html
50	Indian	Indian datum.
51	Indian 1954	A geodetic datum first defined in 1954 suitable for use in Myanmar (Burma) - onshore; Thailand - onshore. Indian 1954 references the Everest 1830 (1937 Adjustment) ellipsoid and the Greenwich prime meridian. Indian 1954 origin is Extension of Kalianpur 1937 over Myanmar and Thailand. Indian 1954 is a geodetic datum for Topographic mapping.
52	Indian 1975	A geodetic datum first defined in 1975 suitable for use in Thailand - onshore plus offshore Gulf of Thailand. Indian 1975 references the Everest 1830 (1937 Adjustment) ellipsoid and the Greenwich prime meridian. Indian 1975 origin is Fundamental point: Khau Sakaerang. Indian 1975 is a geodetic datum for Topographic mapping.
53	Ireland 1965	A geodetic datum first defined in 1975 suitable for use in Ireland - onshore. United Kingdom (UK) - Northern Ireland (Ulster) - onshore. Ireland 1965 references the Airy Modified 1849 ellipsoid and the Greenwich prime meridian. Ireland 1965 origin is Adjusted to best mean fit 9 stations of the OSNI 1952 primary adjustment in Northern Ireland plus the 1965 values of 3 stations in the Republic of Ireland. Ireland 1965 is a geodetic datum for Geodetic survey, topographic mapping and engineering survey. It was defined by information from "The Irish Grid - A Description of the Co-ordinate Reference System" published by Ordnance Survey of Ireland, Dublin and Ordnance Survey of Northern Ireland, Belfast. Differences from the 1965 adjustment (datum code 6299) are: average difference in Eastings 0.092m; average difference in Northings 0.108m; maximum vector difference 0.548m.
54	ISTS 061 Astro 1968	A geodetic datum first defined in 1968 suitable for use in South Georgia and the South Sandwich Islands - South Georgia onshore. ISTS 061 Astro 1968 references the International 1924 ellipsoid and

		the Greenwich prime meridian. ISTS 061 Astro 1968 origin is Fundamental point: ISTS 061. ISTS 061 Astro 1968 is a geodetic datum for Military and topographic mapping It was defined by information from DMA / NIMA / NGA TR8350.2 (3rd edition, Amendment 1, 3 January 2000).
55	ISTS 073 Astro 1969	A geodetic datum first defined in 1969 suitable for use in British Indian Ocean Territory - Chagos Archipelago - Diego Garcia. ISTS 073 Astro 1969 references the International 1924 ellipsoid and the Greenwich prime meridian. ISTS 073 Astro 1969 origin is Fundamental point: ISTS 073. ISTS 073 Astro 1969 is a geodetic datum for Military and topographic mapping. It was defined by information from DMA / NIMA / NGA TR8350.2 (3rd edition, Amendment 1, 3 January 2000).
56	Johnston Island 1961	A geodetic datum first defined in 1961 suitable for use in United States Minor Outlying Islands - Johnston Island. Johnston Island 1961 references the International 1924 ellipsoid and the Greenwich prime meridian. Johnston Island 1961 is a geodetic datum for Military and topographic mapping. It was defined by information from DMA / NIMA / NGA TR8350.2 (3rd edition, Amendment 1, 3 January 2000).
57	Kandawala	A geodetic datum first defined in 1930 suitable for use in Sri Lanka - onshore. Kandawala references the Everest 1830 (1937 Adjustment) ellipsoid and the Greenwich prime meridian. Kandawala origin is Fundamental point: Kandawala. Latitude: 714'06.838"N, longitude: 7952'36.670"E. Kandawala is a geodetic datum for Topographic mapping. It was defined by information from Abeyratne, Featherstone and Tantrigoda in Survey Review vol. 42 no. 317 (July 2010).
58	Kerguelen Island 1949	A geodetic datum first defined in 1949 suitable for use in French Southern Territories - Kerguelen onshore. References the International 1924 ellipsoid and the Greenwich prime meridian. Origin is K0 1949. Is a geodetic datum for Geodetic survey, cadastre, topographic mapping, engineering survey. It was defined by information from IGN Paris.
59	Kertau 1968	A geodetic datum first defined in 1968 suitable for use in Malaysia - West Malaysia onshore and offshore east coast; Singapore - onshore and offshore. Kertau 1968 references the Everest 1830 Modified ellipsoid and the Greenwich prime meridian. Kertau 1968 origin is Fundamental point: Kertau. Latitude: 327'50.710"N, longitude: 10237'24.550"E (of Greenwich). Kertau 1968 is a geodetic datum for Geodetic survey, cadastre. It was defined by information from Defence Geographic Centre. Replaces MRT48 and earlier adjustments. Adopts metric conversion of 39.370113 inches per metre. Not used for 1969 metrication of RSO grid - see Kertau (RSO) (code 6751).
60	Kusaie Astro 1951	A geodetic datum first defined in 1951 suitable for use in Federated States of Micronesia - Kosrae (Kusaie). Kusaie 1951 references the International 1924 ellipsoid and the Greenwich prime meridian. Kusaie 1951 is a geodetic datum for Military and topographic mapping. It was defined by information from DMA / NIMA / NGA TR8350.2 (3rd edition, Amendment 1, 3 January 2000).
61	L. C. 5 Astro 1961	L. C. 5 Astro 1961 datum.
62	Leigon	A geodetic datum suitable for use in Ghana - onshore and offshore. Leigon references the Clarke 1880 (RGS) ellipsoid and the Greenwich prime meridian. Leigon origin is Fundamental point: GCS Station 121, Leigon. Latitude: 538'52.27"N, longitude: 011'46.08"W (of Greenwich). Leigon is a geodetic datum for Topographic mapping. It was defined by information from Ordnance Survey International. Replaced Accra datum (code 6168) from 1978. Coordinates at Leigon

		fundamental point defined as Accra datum values for that point.
63	Liberia 1964	A geodetic datum first defined in 1964 suitable for use in Liberia - onshore. Liberia 1964 references the Clarke 1880 (RGS) ellipsoid and the Greenwich prime meridian. Liberia 1964 origin is Fundamental point: Robertsfield. Latitude: 613'53.02"N, longitude: 1021'35.44"W (of Greenwich). Liberia 1964 is a geodetic datum for Topographic mapping. It was defined by information from NIMA http://earth-info.nima.mil/ .
64	Luzon	A geodetic datum first defined in 1911 suitable for use in Philippines - onshore. Luzon references the Clarke 1866 ellipsoid and the Greenwich prime meridian. Luzon origin is Fundamental point: Balacan. Latitude: 1333'41.000"N, longitude: 12152'03.000"E (of Greenwich). Luzon is a geodetic datum for Topographic mapping. It was defined by information from Coast and Geodetic Survey Replaced by Philippine Reference system of 1992 (datum code 6683).
65	Mahe 1971	A geodetic datum first defined in 1971 suitable for use in Seychelles - Mahe Island. Mahe 1971 references the Clarke 1880 (RGS) ellipsoid and the Greenwich prime meridian. Mahe 1971 origin is Fundamental point: Station SITE. Latitude: 440'14.644"S, longitude: 5528'44.488"E (of Greenwich). Mahe 1971 is a geodetic datum for US military survey. It was defined by information from Clifford Mugnier's September 2007 PE&RS "Grids and Datums" article on Seychelles (www.asprs.org/resources/grids/). South East Island 1943 (datum code 1138) used for topographic mapping, cadastral and hydrographic survey.
66	Massawa	A geodetic datum suitable for use in Eritrea - onshore and offshore. Massawa references the Bessel 1841 ellipsoid and the Greenwich prime meridian. Massawa is a geodetic datum for Topographic mapping.
67	Merchich	A geodetic datum first defined in 1922 suitable for use in Morocco - onshore. Merchich references the Clarke 1880 (IGN) ellipsoid and the Greenwich prime meridian. Merchich origin is Fundamental point: Merchich. Latitude: 3326'59.672"N, longitude: 733'27.295"W (of Greenwich). Merchich is a geodetic datum for Topographic mapping.
68	Midway Astro 1961	A geodetic datum first defined in 1961 suitable for use in United States Minor Outlying Islands - Midway Islands - Sand Island and Eastern Island. Midway 1961 references the International 1924 ellipsoid and the Greenwich prime meridian. Midway 1961 is a geodetic datum for Military and topographic mapping. It was defined by information from DMA / NIMA / NGA TR8350.2 (3rd edition, Amendment 1, 3 January 2000).
69	Minna	A geodetic datum suitable for use in Nigeria - onshore and offshore. Minna references the Clarke 1880 (RGS) ellipsoid and the Greenwich prime meridian. Minna origin is Fundamental point: Minna base station L40. Latitude: 938'08.87"N, longitude: 630'58.76"E (of Greenwich). Minna is a geodetic datum for Topographic mapping. It was defined by information from NIMA http://earth-info.nima.mil/ .
70	Montserrat Island Astro 1958	A geodetic datum first defined in 1958 suitable for use in Montserrat - onshore. Montserrat 1958 references the Clarke 1880 (RGS) ellipsoid and the Greenwich prime meridian. Montserrat 1958 origin is Fundamental point: station M36. Montserrat 1958 is a geodetic datum for Topographic mapping. It was defined by information from Ordnance Survey of Great Britain.
71	M'poraloko	A geodetic datum suitable for use in Gabon - onshore and offshore. M'poraloko references the Clarke 1880 (IGN) ellipsoid and the

		Greenwich prime meridian. M'poraloko is a geodetic datum for Topographic mapping.
72	Nahrwan	A geodetic datum first defined in 1934 suitable for use in Iraq - onshore; Iran - onshore northern Gulf coast and west bordering southeast Iraq. Nahrwan 1934 references the Clarke 1880 (RGS) ellipsoid and the Greenwich prime meridian. Nahrwan 1934 origin is Fundamental point: Nahrwan south base. Latitude: 3319'10.87"N, longitude: 4443'25.54"E (of Greenwich). Nahrwan 1934 is a geodetic datum for Oil exploration and production. It was defined by information from Various industry sources. This adjustment later discovered to have a significant orientation error. In Iran replaced by FD58. In Iraq, replaced by Karbala 1979.
73	Naparima, BWI	A geodetic datum first defined in 1972 suitable for use in Trinidad and Tobago - Tobago - onshore. Naparima 1972 references the International 1924 ellipsoid and the Greenwich prime meridian. Naparima 1972 origin is Fundamental point: Naparima. Latitude: 1016'44.860"N, longitude: 6127'34.620"W (of Greenwich). Naparima 1972 is a geodetic datum for Topographic mapping. It was defined by information from Ordnance Survey International. Naparima 1972 is an extension of the Naparima 1955 network of Trinidad to include Tobago.
74	North American 1927	A geodetic datum first defined in 1927 suitable for use in North and central America; Antigua and Barbuda; Bahamas; Belize; British Virgin Islands. Usage shall be onshore only except that onshore and offshore shall apply to Canada east coast (New Brunswick; Newfoundland and Labrador; Prince Edward Island; Quebec). Cuba. Mexico (Gulf of Mexico and Caribbean coasts only). USA Alaska. USA Gulf of Mexico (Alabama; Florida; Louisiana; Mississippi; Texas). USA East Coast. Bahamas onshore plus offshore over internal continental shelf only. North American Datum 1927 references the Clarke 1866 ellipsoid and the Greenwich prime meridian. North American Datum 1927 origin is Fundamental point: Meade's Ranch. Latitude: 3913'26.686"N, longitude: 9832'30.506"W (of Greenwich). North American Datum 1927 is a geodetic datum for Topographic mapping. In United States (USA) and Canada, replaced by North American Datum 1983 (NAD83) (code 6269) ; in Mexico, replaced by Mexican Datum of 1993 (code 1042).
75	North American 1983	A geodetic datum first defined in 1986 suitable for use in North America - onshore and offshore: Canada; Puerto Rico; United States (USA); US Virgin Islands; British Virgin Islands. North American Datum 1983 references the GRS 1980 ellipsoid and the Greenwich prime meridian. North American Datum 1983 origin is Origin at geocentre. North American Datum 1983 is a geodetic datum for Topographic mapping. Although the 1986 adjustment included connections to Greenland and Mexico, it has not been adopted there. In Canada and US, replaced NAD27.
76	Observatorio Meteorologico 1939	A geodetic datum first defined in 1939 suitable for use in Portugal - western Azores onshore - Flores, Corvo. Azores Occidental Islands 1939 references the International 1924 ellipsoid and the Greenwich prime meridian. Azores Occidental Islands 1939 origin is Fundamental point: Observatorio Meteorologico Flores. Azores Occidental Islands 1939 is a geodetic datum for Topographic mapping. It was defined by information from Instituto Geografico e Cadastral Lisbon via EuroGeographics; http://crs.bkg.bund.de/crs-eu/ .
77	Old Egyptian 1907	A geodetic datum first defined in 1907 suitable for use in Egypt - onshore and offshore. Egypt 1907 references the Helmert 1906 ellipsoid and the Greenwich prime meridian. Egypt 1907 origin is Fundamental point: Station F1 (Venus). Latitude: 3001'42.86"N, longitude: 3116'33.60"E (of Greenwich). Egypt 1907 is a geodetic

		datum for Geodetic survey, cadastre, topographic mapping, engineering survey.
78	Old Hawaiian	A geodetic datum suitable for use in United States (USA) - Hawaii - main islands onshore. Old Hawaiian references the Clarke 1866 ellipsoid and the Greenwich prime meridian. Old Hawaiian origin is Fundamental point: Oahu West Base Astro. Latitude: 2118'13.89"N, longitude 15750'55.79"W (of Greenwich). Old Hawaiian is a geodetic datum for Topographic mapping. It was defined by information from http://www.ngs.noaa.gov/ (NADCON readme file). Hawaiian Islands were never on NAD27 but rather on Old Hawaiian Datum. NADCON conversion program provides transformation from Old Hawaiian Datum to NAD83 (original 1986 realization) but making the transformation appear to user as if from NAD27.
79	Oman	A geodetic datum first defined in 2013 suitable for use in Oman - onshore and offshore. Oman National Geodetic Datum 2014 references the GRS 1980 ellipsoid and the Greenwich prime meridian. Oman National Geodetic Datum 2014 origin is 20 stations of the Oman primary network tied to ITRF2008 at epoch 2013.15. Oman National Geodetic Datum 2014 is a geodetic datum for Geodetic Survey. It was defined by information from National Survey Authority, Sultanate of Oman. Replaces WGS 84 (G874).
80	Ordnance Survey of Great Britain 1936	A geodetic datum first defined in 1936 suitable for use in United Kingdom (UK) - offshore to boundary of UKCS within 4946'N to 6101'N and 733'W to 333'E; onshore Great Britain (England, Wales and Scotland). Isle of Man onshore. OSGB 1936 references the Airy 1830 ellipsoid and the Greenwich prime meridian. OSGB 1936 origin is Prior to 2002, fundamental point: Herstmonceux, Latitude: 5051'55.271"N, longitude: 020'45.882"E (of Greenwich). From April 2002 the datum is defined through the application of the OSTN transformation from ETRS89. OSGB 1936 is a geodetic datum for Topographic mapping. It was defined by information from Ordnance Survey of Great Britain. The average accuracy of OSTN compared to the old triangulation network (down to 3rd order) is 0.1m. With the introduction of OSTN15, the area for OSGB 1936 has effectively been extended from Britain to cover the adjacent UK Continental Shelf.
81	Pico de las Nieves	A geodetic datum suitable for use in Spain - Canary Islands onshore. Pico de las Nieves 1984 references the International 1924 ellipsoid and the Greenwich prime meridian. Pico de las Nieves 1984 is a geodetic datum for Military and topographic mapping. It was defined by information from DMA / NIMA / NGA TR8350.2 (3rd edition, Amendment 1, 3 January 2000). Replaces Pico de las Nieves 1968 (PN68). Replaced by REGCAN95.
82	Pitcairn Astro 1967	A geodetic datum first defined in 1967 suitable for use in Pitcairn - Pitcairn Island. Pitcairn 1967 references the International 1924 ellipsoid and the Greenwich prime meridian. Pitcairn 1967 origin is Fundamental point: Pitcairn Astro. Latitude: 2504'06.87"S, longitude: 13006'47.83"W (of Greenwich). Pitcairn 1967 is a geodetic datum for Military and topographic mapping. It was defined by information from DMA / NIMA / NGA TR8350.2 (3rd edition, Amendment 1, 3 January 2000). Replaced by Pitcairn 2006.
83	Point 58	A geodetic datum first defined in 1969 suitable for use in Senegal - central, Mali - southwest, Burkina Faso - central, Niger - southwest, Nigeria - north, Chad - central. All in proximity to the parallel of latitude of 12N. Point 58 references the Clarke 1880 (RGS) ellipsoid and the Greenwich prime meridian. Point 58 origin is Fundamental point: Point 58. Latitude: 1252'44.045"N, longitude: 358'37.040"E (of Greenwich). Point 58 is a geodetic datum for Geodetic survey. It was defined by information from IGN Paris. Used as the basis for computation of the 12th Parallel traverse conducted 1966-70 from

		Senegal to Chad and connecting to the Blue Nile 1958 (Adindan) triangulation in Sudan.
84	Pointe Noire 1948	Pointe Noire 1948 datum.
85	Porto Santo 1936	A geodetic datum first defined in 1936 suitable for use in Portugal - Madeira, Porto Santo and Desertas islands - onshore. Porto Santo 1936 references the International 1924 ellipsoid and the Greenwich prime meridian. Porto Santo 1936 origin is SE Base on Porto Santo island. Porto Santo 1936 is a geodetic datum for Topographic mapping. It was defined by information from Instituto Geografico e Cadastral Lisbon http://www.igeo.pt Replaced by 1995 adjustment (datum code 6663). For Selvagens see Selvagem Grande (code 6616).
86	Provisional South American 1956	A geodetic datum first defined in 1956 suitable for use in Aruba - onshore; Bolivia; Bonaire - onshore; Brazil - offshore - Amazon Cone shelf; Chile - onshore north of 4330'S; Curacao - onshore; Ecuador - mainland onshore; Guyana - onshore; Peru - onshore; Venezuela - onshore. Provisional South American Datum 1956 references the International 1924 ellipsoid and the Greenwich prime meridian. Provisional South American Datum 1956 origin is Fundamental point: La Canoa. Latitude: 834'17.170"N, longitude: 6351'34.880"W (of Greenwich). Provisional South American Datum 1956 is a geodetic datum for Topographic mapping. Same origin as La Canoa datum.
87	Provisional South Chilean 1963	A geodetic datum first defined in 1963 suitable for use in Argentina and Chile - Tierra del Fuego, onshore. Hito XVIII 1963 references the International 1924 ellipsoid and the Greenwich prime meridian. Hito XVIII 1963 origin is Chile-Argentina boundary survey. Hito XVIII 1963 is a geodetic datum for Geodetic survey. It was defined by information from Various oil company records. Used in Tierra del Fuego.
88	Puerto Rico	A geodetic datum first defined in 1901 suitable for use in Puerto Rico, US Virgin Islands and British Virgin Islands - onshore. Puerto Rico references the Clarke 1866 ellipsoid and the Greenwich prime meridian. Puerto Rico origin is Fundamental point: Cardona Island Lighthouse. Latitude: 1757'31.40"N, longitude: 6638'07.53"W (of Greenwich). Puerto Rico is a geodetic datum for Topographic mapping. It was defined by information from Ordnance Survey of Great Britain and http://www.ngs.noaa.gov/ (NADCON readme file). NADCON conversion program provides transformation from Puerto Rico Datum to NAD83 (original 1986 realization) but making the transformation appear to user as if from NAD27.
89	Qatar National	A geodetic datum first defined in 1995 suitable for use in Qatar - onshore. Qatar National Datum 1995 references the International 1924 ellipsoid and the Greenwich prime meridian. Qatar National Datum 1995 origin is defined by transformation from WGS 84 - see coordinate operation code 1840. Qatar National Datum 1995 is a geodetic datum for Topographic mapping. It was defined by information from Qatar Centre for Geographic Information.
90	Qornoq	A geodetic datum first defined in 1927 suitable for use in Greenland - west coast onshore. Qornoq 1927 references the International 1924 ellipsoid and the Greenwich prime meridian. Qornoq 1927 origin is Fundamental point: Station 7008. Latitude: 6431'06.27"N, longitude: 5112'24.86"W (of Greenwich). Qornoq 1927 is a geodetic datum for Topographic mapping. It was defined by information from Kort & Matrikelstyrelsen, Copenhagen. Origin coordinates from NIMA http://earth-info.nima.mil/ .
91	Reunion	A geodetic datum first defined in 1947 suitable for use in Reunion - onshore. Reunion 1947 references the International 1924 ellipsoid and the Greenwich prime meridian. Reunion 1947 origin is Fundamental point: Piton des Neiges (Borne). Latitude:

		2105'13.119"S, longitude: 5529'09.193"E (of Greenwich). Reunion 1947 is a geodetic datum for Geodetic survey, cadastre, topographic mapping, engineering survey. It was defined by information from IGN Paris. Replaced by RGR92 (datum code 6627).
92	Rome 1940	A geodetic datum first defined in and is suitable for use in Italy - onshore and offshore; San Marino, Vatican City State. Monte Mario (Rome) references the International 1924 ellipsoid and the Rome prime meridian. Monte Mario (Rome) origin is Fundamental point: Monte Mario. Latitude: 4155'25.51"N, longitude: 000' 00.00"E (of Rome). Monte Mario (Rome) is a geodetic datum for Topographic mapping. Replaced Genova datum, Bessel 1841 ellipsoid, from 1940.
93	Santo (DOS) 1965	A geodetic datum first defined in 1965 suitable for use in Vanuatu - northern islands - Aese, Ambrym, Aoba, Epi, Espiritu Santo, Maewo, Malo, Malkula, Paama, Pentecost, Shepherd and Tutuba. Santo 1965 references the International 1924 ellipsoid and the Greenwich prime meridian. Santo 1965 is a geodetic datum for Military and topographic mapping. It was defined by information from DMA / NIMA / NGA TR8350.2 (3rd edition, Amendment 1, 3 January 2000). Datum covers all the major islands of Vanuatu in two different adjustment blocks, but practical usage is as given in the area of use.
94	Sao Braz	A geodetic datum first defined in 1995 suitable for use in Portugal - eastern Azores onshore - Sao Miguel, Santa Maria, Formigas. Azores Oriental Islands 1995 references the International 1924 ellipsoid and the Greenwich prime meridian. Azores Oriental Islands 1995 origin is Fundamental point: Forte de So Bras. Origin and orientation constrained to those of the 1940 adjustment. Azores Oriental Islands 1995 is a geodetic datum for Topographic mapping. It was defined by information from Instituto Geografico e Cadastral Lisbon; http://www.igeo.pt/ Classical and GPS observations. Replaces 1940 adjustment (datum code 6184).
95	Sapper Hill 1943	A geodetic datum first defined in 1943 suitable for use in Falkland Islands (Malvinas) - onshore. Sapper Hill 1943 references the International 1924 ellipsoid and the Greenwich prime meridian. Sapper Hill 1943 is a geodetic datum for Topographic mapping.
96	Schwarzeck	A geodetic datum suitable for use in Namibia - onshore and offshore. Schwarzeck references the Bessel Namibia (GLM) ellipsoid and the Greenwich prime meridian. Schwarzeck origin is Fundamental point: Schwarzeck. Latitude: 2245'35.820"S, longitude: 1840'34.549"E (of Greenwich). Fixed during German South West Africa-British Bechuanaland boundary survey of 1898-1903. Schwarzeck is a geodetic datum for Topographic mapping. It was defined by information from Private Communication, Directorate of Surveys and Land Information, Cape Town.
97	Selvagem Grande 1938	A geodetic datum suitable for use in Portugal - Selvagens islands (Madeira province) - onshore. Selvagem Grande references the International 1924 ellipsoid and the Greenwich prime meridian. Selvagem Grande is a geodetic datum for Topographic mapping. It was defined by information from Instituto Geografico e Cadastral Lisbon http://www.igeo.pt .
98	South American 1969	A geodetic datum first defined in 1969 suitable for use in Brazil - onshore and offshore. In rest of South America - onshore north of approximately 45S and Tierra del Fuego. South American Datum 1969 references the GRS 1967 Modified ellipsoid and the Greenwich prime meridian. South American Datum 1969 origin is Fundamental point: Chua. Geodetic latitude: 1945'41.6527"S; geodetic longitude: 4806'04.0639"W (of Greenwich). (Astronomic coordinates: Latitude 1945'41.34"S +/- 0.05", longitude 4806'07.80"W +/- 0.08"). South American Datum 1969 is a geodetic datum for Topographic mapping. It was defined by information from DMA 1974. SAD69 uses GRS

		1967 ellipsoid but with 1/f to exactly 2 decimal places. In Brazil only, replaced by SAD69(96) (datum code 1075).
99	South Asia	South Asia datum.
100	Tananarive Observatory 1925	A geodetic datum first defined in 1925 suitable for use in Madagascar - onshore and nearshore. Tananarive 1925 references the International 1924 ellipsoid and the Greenwich prime meridian. Tananarive 1925 origin is Fundamental point: Tananarive observatory. Latitude: 1855'02.10"S, longitude: 4733'06.75"E (of Greenwich). Tananarive 1925 is a geodetic datum for Topographic mapping. It was defined by information from IGN Paris.
101	Timbalai 1948	A geodetic datum first defined in 1948 suitable for use in Brunei - onshore and offshore; Malaysia - East Malaysia (Sabah; Sarawak) - onshore and offshore. Timbalai 1948 references the Everest 1830 (1967 Definition) ellipsoid and the Greenwich prime meridian. Timbalai 1948 origin is Fundamental point: Station P85 at Timbalai. Latitude: 517' 3.548"N, longitude: 11510'56.409"E (of Greenwich). Timbalai 1948 is a geodetic datum for Topographic mapping. It was defined by information from Defence Geographic Centre. In 1968, the original adjustment was densified in Sarawak and extended to Sabah.
102	Tokyo	A geodetic datum first defined in 1918 suitable for use in Japan - onshore; North Korea - onshore; South Korea - onshore. Tokyo references the Bessel 1841 ellipsoid and the Greenwich prime meridian. Tokyo origin is Fundamental point: Nikon-Keido-Genten. Latitude: 3539'17.5148"N, longitude: 13944'40.5020"E (of Greenwich). Longitude derived in 1918. Tokyo is a geodetic datum for Geodetic survey, cadastre, topographic mapping, engineering survey. It was defined by information from Geographic Survey Institute; Japan; Bulletin 40 (March 1994). Also http://vldb.gsi.go.jp/sokuchi/datum/tokyodatum.html . In Japan, replaces Tokyo 1892 (code 1048) and replaced by Japanese Geodetic Datum 2000 (code 6611). In Korea used only for geodetic applications before being replaced by Korean 1985 (code 6162).
103	Tristan Astro 1968	A geodetic datum first defined in 1968 suitable for use in St Helena, Ascension and Tristan da Cunha - Tristan da Cunha island group including Tristan, Inaccessible, Nightingale, Middle and Stoltenhoff Islands. Tristan 1968 references the International 1924 ellipsoid and the Greenwich prime meridian. Tristan 1968 is a geodetic datum for Military and topographic mapping. It was defined by information from DMA / NIMA / NGA TR8350.2 (3rd edition, Amendment 1, 3 January 2000).
104	Viti Levu 1916	A geodetic datum first defined in 1912 suitable for use in Fiji - Viti Levu island. Viti Levu 1912 references the Clarke 1880 (international foot) ellipsoid and the Greenwich prime meridian. Viti Levu 1912 Latitude origin was obtained astronomically at station Monavatu = 1753'28.285"S, longitude origin was obtained astronomically at station Suva = 17825'35.835"E. Viti Levu 1912 is a geodetic datum for Geodetic survey, cadastre, topographic mapping, engineering survey. It was defined by information from Clifford J. Mugnier in Photogrammetric Engineering and Remote Sensing, October 2000, www.asprs.org . For topographic mapping, replaced by Fiji 1956. For other purposes, replaced by Fiji 1986.
105	Wake-Eniwetok 1960	A geodetic datum first defined in 1960 suitable for use in Marshall Islands - onshore. Wake atoll onshore. Marshall Islands 1960 references the Hough 1960 ellipsoid and the Greenwich prime meridian. Marshall Islands 1960 is a geodetic datum for Military and topographic mapping. It was defined by information from DMA / NIMA / NGA TR8350.2 (3rd edition, Amendment 1, 3 January 2000).

106	Wake Island Astro 1952	A geodetic datum first defined in 1952 suitable for use in Wake atoll - onshore. Wake Island 1952 references the International 1924 ellipsoid and the Greenwich prime meridian. Wake Island 1952 is a geodetic datum for Military and topographic mapping. It was defined by information from DMA / NIMA / NGA TR8350.2 (3rd edition, Amendment 1, 3 January 2000).
107	Yacare	A geodetic datum suitable for use in Uruguay - onshore. Yacare references the International 1924 ellipsoid and the Greenwich prime meridian. Yacare origin is Fundamental point: Yacare. Latitude: 3035'53.68"S, longitude: 5725'01.30"W (of Greenwich). Yacare is a geodetic datum for Topographic mapping. It was defined by information from NIMA http://earth-info.nima.mil/
108	Zanderij	A geodetic datum suitable for use in Suriname - onshore and offshore. Zanderij references the International 1924 ellipsoid and the Greenwich prime meridian. Zanderij is a geodetic datum for Topographic mapping.
109	American Samoa 1962	A geodetic datum first defined in 1962 suitable for use in American Samoa - Tutuila, Aunu'u, Ofu, Olesaga and Ta'u islands. American Samoa 1962 references the Clarke 1866 ellipsoid and the Greenwich prime meridian. American Samoa 1962 origin is Fundamental point: Betty 13 eccentric. Latitude: 1420'08.34"S, longitude: 17042'52.25"W (of Greenwich). American Samoa 1962 is a geodetic datum for Topographic mapping. It was defined by information from NIMA TR8350.2 revision of January 2000. Oil industry sources for origin description details.
110	Deception Island	A geodetic datum suitable for use in Antarctica - South Shetland Islands - Deception Island. Deception Island references the Clarke 1880 (RGS) ellipsoid and the Greenwich prime meridian. Deception Island is a geodetic datum for Military and scientific mapping. It was defined by information from DMA / NIMA / NGA TR8350.2 (3rd edition, Amendment 1, 3 January 2000).
111	Indian 1960	A geodetic datum suitable for use in Cambodia - onshore; Vietnam - onshore and offshore Cuu Long basin. Indian 1960 references the Everest 1830 (1937 Adjustment) ellipsoid and the Greenwich prime meridian. Indian 1960 origin is DMA extension over IndoChina of the Indian 1954 network adjusted to better fit local geoid. Indian 1960 is a geodetic datum for Topographic mapping. Also known as Indian (DMA Reduced).
112	Indonesian 1974	A geodetic datum first defined in 1974 suitable for use in Indonesia - onshore. Indonesian Datum 1974 references the Indonesian National Spheroid ellipsoid and the Greenwich prime meridian. Indonesian Datum 1974 origin is Fundamental point: Padang. Latitude: 056'38.414"S, longitude: 10022' 8.804"E (of Greenwich). Ellipsoidal height 3.190m, gravity-related height 14.0m above mean sea level. Indonesian Datum 1974 is a geodetic datum for Topographic mapping. It was defined by information from Bakosurtanal 1979 paper by Jacob Rais. Replaced by DGN95.
113	North Sahara 1959	A geodetic datum first defined in 1959 suitable for use in Algeria - onshore and offshore. Nord Sahara 1959 references the Clarke 1880 (RGS) ellipsoid and the Greenwich prime meridian. Nord Sahara 1959 origin is Coordinates of primary network readjusted on ED50 datum and then transformed conformally to Clarke 1880 (RGS) ellipsoid. Nord Sahara 1959 is a geodetic datum for Topographic mapping. It was defined by information from "Le System Geodesique Nord-Sahara"; IGN Paris Adjustment includes Morocco and Tunisia but use only in Algeria. Within Algeria the adjustment is north of 32N but use has been extended southwards in many disconnected projects, some based on independent astro stations rather than the

		geodetic network.
114	Pulkovo 1942	A geodetic datum first defined in 1942 suitable for use in Armenia; Azerbaijan; Belarus; Estonia - onshore; Georgia - onshore; Kazakhstan; Kyrgyzstan; Latvia - onshore; Lithuania - onshore; Moldova; Russian Federation - onshore; Tajikistan; Turkmenistan; Ukraine - onshore; Uzbekistan. Pulkovo 1942 references the Krassowsky 1940 ellipsoid and the Greenwich prime meridian. Pulkovo 1942 origin is Fundamental point: Pulkovo observatory. Latitude: 5946'18.550"N, longitude: 3019'42.090"E (of Greenwich). Pulkovo 1942 is a geodetic datum for Topographic mapping.
116	S-JTSK	A geodetic datum suitable for use in Czech Republic; Slovakia. System Jednotne Trigonometrické Site Katastrální references the Bessel 1841 ellipsoid and the Greenwich prime meridian. System Jednotne Trigonometrické Site Katastrální origin is Modification of Austrian MGI datum, code 6312. System Jednotne Trigonometrické Site Katastrální is a geodetic datum for Geodetic survey, cadastre, topographic mapping, engineering survey. It was defined by information from Research Institute for Geodesy Topography and Cartography (VUGTK); Prague. S-JTSK = System of the Unified Trigonometrical Cadastral Network.
117	Voirol 1950	Voirol 1950 datum.
118	Average Terrestrial System 1977	A geodetic datum first defined in 1977 suitable for use in Canada - New Brunswick; Nova Scotia; Prince Edward Island. Average Terrestrial System 1977 references the Average Terrestrial System 1977 ellipsoid and the Greenwich prime meridian. Average Terrestrial System 1977 is a geodetic datum for Topographic mapping. It was defined by information from New Brunswick Geographic Information Corporation land and water information standards manual. In use from 1979.
119	Compensation Geodesique du Quebec 1977	Compensation Geodesique du Quebec 1977.
120	Finnish (KKJ)	A geodetic datum first defined in 1966 suitable for use in Finland - onshore. Kartastokoordinaattijärjestelmä (1966) references the International 1924 ellipsoid and the Greenwich prime meridian. Kartastokoordinaattijärjestelmä (1966) origin is Adjustment with fundamental point SF31 based on ED50 transformed to best fit the older VVJ adjustment. Kartastokoordinaattijärjestelmä (1966) is a geodetic datum for Geodetic survey, cadastre, topographic mapping, engineering survey. It was defined by information from National Land Survey of Finland; http://www.maanmittauslaitos.fi . Adopted in 1970.
121	Ordnance Survey of Ireland	A geodetic datum first defined in 1952 suitable for use in United Kingdom (UK) - Northern Ireland (Ulster) - onshore. OSNI 1952 references the Airy 1830 ellipsoid and the Greenwich prime meridian. OSNI 1952 origin is Position fixed to the coordinates from the 19th century Principle Triangulation of station Divis. Scale and orientation controlled by position of Principle Triangulation stations Knocklyd and Trostan. OSNI 1952 is a geodetic datum for Geodetic survey and topographic mapping. It was defined by information from Ordnance Survey of Northern Ireland. Replaced by Geodetic Datum of 1965 alias 1975 Mapping Adjustment or TM75 (datum code 6300).
122	Revised Kertau	A geodetic datum first defined in 1969 suitable for use in Malaysia - West Malaysia; Singapore. Kertau (RSO) references the Everest 1830 (RSO 1969) ellipsoid and the Greenwich prime meridian. Kertau (RSO) is a geodetic datum for Metrication of RSO grid. It was defined by information from Defence Geographic Centre. Adopts metric conversion of 0.914398 metres per yard exactly. This is a truncation of the Sears 1922 ratio.

123	Revised Nahrwan	A geodetic datum first defined in 1967 suitable for use in Arabian Gulf; Qatar - offshore; United Arab Emirates (UAE) - Abu Dhabi; Dubai; Sharjah; Ajman; Fujairah; Ras Al Kaimah; Umm Al Qaiwain - onshore and offshore. Nahrwan 1967 references the Clarke 1880 (RGS) ellipsoid and the Greenwich prime meridian. Nahrwan 1967 origin is Fundamental point: Nahrwan south base. Latitude: 3319'10.87"N, longitude: 4443'25.54"E (of Greenwich). Nahrwan 1967 is a geodetic datum for Topographic mapping. In Iraq, replaces Nahrwan 1934.
124	GGRS 76 (Greece)	A geodetic datum first defined in 1987 suitable for use in Greece - onshore. Greek Geodetic Reference System 1987 references the GRS 1980 ellipsoid and the Greenwich prime meridian. Greek Geodetic Reference System 1987 origin is Fundamental point: Dionysos. Latitude 3804'33.8"N, longitude 2355'51.0"E of Greenwich; geoid height 7.0 m. Greek Geodetic Reference System 1987 is a geodetic datum for Topographic mapping. It was defined by information from L. Portokalakis; Public Petroleum Corporation of Greece. Replaced (old) Greek datum. Oil industry work based on ED50.
125	Nouvelle Triangulation de France	A geodetic datum first defined in 1895 suitable for use in France - onshore - mainland and Corsica. Nouvelle Triangulation Francaise references the Clarke 1880 (IGN) ellipsoid and the Greenwich prime meridian. Nouvelle Triangulation Francaise origin is Fundamental point: Pantheon. Latitude: 4850'46.522"N, longitude: 220'48.667"E (of Greenwich). Nouvelle Triangulation Francaise is a geodetic datum for Topographic mapping.
126	RT 90 (Sweden)	A geodetic datum first defined in 1982 suitable for use in Sweden - onshore and offshore. Rikets koordinatsystem 1990 references the Bessel 1841 ellipsoid and the Greenwich prime meridian. Rikets koordinatsystem 1990 is a geodetic datum for Geodetic survey, cadastre, topographic mapping, engineering survey. It was defined by information from National Land Survey of Sweden Replaces RT38 adjustment (datum code 6308).
127	Geocentric Datum of Australia	A geodetic datum first defined in 1994 suitable for use in Australia including Lord Howe Island, Macquarie Islands, Ashmore and Cartier Islands, Christmas Island, Cocos (Keeling) Islands, Norfolk Island. All onshore and offshore. Geocentric Datum of Australia 1994 references the GRS 1980 ellipsoid and the Greenwich prime meridian. Geocentric Datum of Australia 1994 origin is ITRF92 at epoch 1994.0. Geocentric Datum of Australia 1994 is a geodetic datum for Topographic mapping, geodetic survey. It was defined by information from Australian Surveying and Land Information Group Internet WWW page. http://www.auslig.gov.au/geodesy/datums/gda.htm#specs Coincident with WGS84 to within 1 metre.
128	BJZ54 (A954 Beijing Coordinates)	A geodetic datum first defined in 1954 suitable for use in China - onshore. Beijing 1954 references the Krassowsky 1940 ellipsoid and the Greenwich prime meridian. Beijing 1954 origin is Pulkovo, transferred through Russian triangulation. Beijing 1954 is a geodetic datum for Topographic mapping. It was defined by information from Chinese Science Bulletin, 2009, 54:2714-2721 Scale determined through three baselines in northeast China. Discontinuities at boundaries of adjustment blocks. From 1982 replaced by Xian 1980 and New Beijing.
129	Modified BJZ54	Modified BJZ54 datum.
130	GDZ80	GDZ80 datum.

131	Local Datum	An arbitrary datum defined by a local harbour authority, from which levels and tidal heights are measured by this authority.
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5.24. Reference Point

Definition: -

CamelCase: referencePoint

Alias:

Value type: text

Remarks: -

5.25. Signal Status

Definition: The indication of an element of a signal sequence being a period of light/sound or eclipse/silence.

CamelCase: signalStatus

Alias:

Value type: enumeration

Remarks: No remarks.

Listed Values:

Code	Label	Definition
1	Lit/Sound	The indication of an element of a signal sequence being a period of light or sound.
2	Eclipsed/Silent	The indication of an element of a signal sequence being a period of eclipse or silence.

5.26. Signal Duration

Definition: The time occupied by a single instance of light/sound or eclipse/silence in a signal sequence.

CamelCase: signalDuration

Alias:

Value type: real

Remarks: No remarks.

5.27. Sector Line Length

Definition: A sector is the part of a circle between two straight lines drawn from the centre to the circumference. Sector line length specifies the displayed length of the line, in ground units, defining the limit of the sector.

CamelCase: sectorLineLength

Alias:

Value type: integer

Remarks: No remarks.

5.28. Sector Bearing

Definition: A sector is the part of a circle between two straight lines drawn from the centre to the circumference. Sector bearing specifies the limit of the sector.

CamelCase: sectorBearing

Alias: SECTR1 SECTR2

Value type: real

Remarks: The values given to the common limits of adjacent sectors should be identical. The orientation of bearing is from seaward to the central object. This conforms with the method used in 'List of Lights' publications. A generic term such as 'to shore' cannot be used; a specific bearing must be encoded. Where a light sector limit is defined as 'to the shore', it should be encoded using a value that ensures that, when the limit is drawn, it will fall entirely on land.

5.29. Wave Length Value

Definition: The distance between two successive peaks (or other points of identical phase) on an electromagnetic wave.

CamelCase: waveLengthValue

Alias: Radar Wave Length

Value type: real

Remarks: No remarks.

5.30. Radar Band

Definition: The band code character of the electromagnetic spectrum within which radar wave lengths lie.

CamelCase: radarBand

Alias:

Value type: text

Remarks: No remarks.

5.31. Number of Features

Definition: The number of features of identical character that exist as a co-located group.

CamelCase: numberOfFeatures

Alias:

Value type: integer

Remarks: No remarks.

5.32. Multiplicity Known

Definition: The number of features of identical character that exist as a co-located group is or is not known.

CamelCase: multiplicityKnown

Alias:

Value type: boolean

Remarks: No remarks.

5.33. Sector Arc Extension

Definition: Distance in screen millimetres (mm) by which a sector arc is extended beyond the default.

CamelCase: sectorArcExtension

Alias:

Value type: boolean

Remarks: The sector extension is calculated by ENC production software systems.

5.34. Moire Effect

Definition: A short range (up to 2km) type of directional light. Sodium lighting gives a yellow background to a screen on which a vertical black line will be seen by an observer on the centre line.

CamelCase: moireEffect

Alias:

Value type: boolean

Remarks: No remarks.

5.35. Category of Cable

Definition: Classification of the cable based on the services provided.

CamelCase: categoryOfCable

Alias: CATCBL

Value type: enumeration

Remarks: No remarks.

Listed Values:

Code	Label	Definition
1	Power Line	A cable that transmits or distributes electrical power.
3	Transmission Line	Multiple un-insulated cables usually supported by steel lattice towers. Such features are generally more prominent than normal power lines.
4	Telephone	A cable that transmits telephone signals.
5	Telegraph	An apparatus, system or process for communication at a distance by electric transmission over wire.
6	Mooring Cable	A chain or very strong fibre or wire rope used to anchor or moor vessels or buoys.
7	Ferry	A vessel for transporting passengers, vehicles, and/or goods across a stretch of water, especially as a regular service.
8	Fibre Optic Cable	A cable made of glass or plastic fiber designed to guide light along its length, fibre optic cables are widely used in fiber-optic communication, which permits transmission over longer distances and at higher data rates than other forms of communication.

5.36. Category of Installation Buoy

Definition: Classification of fixed installation buoy.

CamelCase: categoryOfInstallationBuoy

Alias:

Value type: enumeration

Remarks: No remarks.

Listed Values:

Code	Label	Definition
1	Catenary Anchor Leg Mooring	incorporates a large buoy which remains on the surface at all times and is moored by 4 or more anchors. Mooring hawsers and cargo hoses lead from a turntable on top of the buoy, so that the buoy does not turn as the ship swings to wind and stream.
2	Single Buoy Mooring	a mooring structure used by tankers to load and unload in port approaches or in offshore oil and gas fields. The size of the structure can vary between a large mooring buoy and a manned floating structure. Also known as single point mooring (SPM)

5.37. Shackle Type

Definition: Types of shackle.

CamelCase: ShackleType

Alias:

Value type: enumeration

Remarks: -

Listed Values:

Code	Label	Definition
1	forelock shackles	-
2	clenching shackles	-
3	bolt shackles	-
4	screw pin shackles	-
5	kenter shackle	-
6	quick release link	-

5.38. Category of Pile

Definition: Classification of pile, driven into the earth as a foundation or support for a structure.

CamelCase: categoryOfPile

Alias: CATPLE

Value type: enumeration

Remarks: No remarks.

Listed Values:

Code	Label	Definition
1	Stake	An elongated wood or metal pole embedded in the seabed to serve as a marker or support.
3	Post	A vertical piece of timber, metal or concrete forced into the earth or sea bed.
4	Tripodal	A single structure comprising 3 or more piles held together (sections of heavy timber, steel or concrete), and forced into the earth or sea bed.
5	Piling	A number of piles, usually in a straight line, and usually connected or bolted together.
6	Area of Piles	A number of piles, usually in a straight line, but not connected by structural members.
7	Pipe	A vertical hollow cylinder of metal, wood, or other material forced into the earth or seabed.

5.39. Category of Silo/Tank

Definition: Classification based on the product for which a silo or tank is used.

CamelCase: categoryOfSiloTank

Alias: CATSIL

Value type: enumeration

Remarks: No remarks.

Listed Values:

Code	Label	Definition
1	Silo in General	A large storage structure used for storing loose materials.
2	Tank in General	A fixed structure for storing liquids.
3	Grain Elevator	A storage building for grain. Usually a tall frame, metal or concrete structure with an especially compartmented interior.
4	Water Tower	A tower supporting an elevated storage tank of water.

5.40. Building Shape

Definition: The specific shape of the building.

CamelCase: buildingShape

Alias: BUIHP

Value type: enumeration

Remarks: No remarks.

Listed Values:

Code	Label	Definition
5	High-Rise Building	A building having many storeys.
6	Pyramid	A polyhedron of which one face is a polygon of any number of sides, and the other faces are triangles with a common vertex.
7	Cylindrical	Shaped like a cylinder, which is a solid geometrical figure generated by straight lines fixed in direction and describing with one of its points a closed curve, especially a circle.
8	Spherical	Shaped like a sphere, which is a body the surface of which is at all points equidistant from the centre.
9	Cubic	A shape the sides of which are six equal squares; a regular hexahedron.

5.41. Product

Definition: The various substances which are transported, stored or exploited.

CamelCase: product

Alias: PRODDCT

Value type: enumeration

Remarks: No remarks.

Listed Values:

Code	Label	Definition
1	Oil	A thick, slippery liquid that will not dissolve in water, usually petroleum based in the context of storage tanks.
2	Gas	A substance with particles that can move freely, usually a fuel substance in the context of storage tanks.
3	Water	A colourless, odourless, tasteless liquid that is a compound of hydrogen and oxygen.
4	Stone	A general term for rock and rock fragments ranging in size from pebbles and gravel to boulders or large rock masses.
5	Coal	A hard black mineral that is burned as fuel.
6	Ore	A solid rock or mineral from which metal is obtained.
7	Chemicals	Any substance obtained by or used in a chemical process.
8	Drinking Water	Water that is suitable for human consumption.
9	Milk	A white fluid secreted by female mammals as food for their young.
10	Bauxite	A mineral from which aluminum is obtained.
11	Coke	A solid substance obtained after gas and tar have been extracted from coal, used as a fuel.
12	Iron Ingots	An oblong lump of cast iron metal.
13	Salt	Sodium chloride obtained from mines or by the evaporation of sea water.
14	Sand	Loose material consisting of small but easily distinguishable, separate grains, between 0.0625 and 2.000 millimetres in diameter.
15	Timber	Wood prepared for use in building or carpentry.
16	Sawdust/Wood Chips	Powdery fragments of wood made in sawing timber or coarse chips produced for use in manufacturing pressed board.
17	Scrap Metal	Discarded metal suitable for being reprocessed.

18	Liquefied Natural Gas	Natural gas that has been liquefied for ease of transport by cooling the gas to -162 Celsius.
19	Liquefied Petroleum Gas	A compressed gas consisting of flammable light hydrocarbons and derived from petroleum.
20	Wine	The fermented juice of grapes.
21	Cement	A substance made of powdered lime and clay, mixed with water.
22	Grain	A small hard seed, especially that of any cereal plant such as wheat, rice, corn, rye etc.
23	Electricity	Electric charge or current.
24	Ice	The solid form of water.
25	Clay	(Particles of less than 0.002mm); stiff, sticky earth that becomes hard when baked.

5.42. Category of Offshore Platform

Definition: Classification of an offshore raised structure.

CamelCase: categoryOfOffshorePlatform

Alias: CATOFP

Value type: enumeration

Remarks: No remarks.

Listed Values:

Code	Label	Definition
1	Oil Rig	A temporary mobile structure, either fixed or floating, used in the exploration stages of oil and gas fields.
2	Production Platform	A term used to indicate a permanent offshore structure equipped to control the flow of oil or gas. It does not include entirely submarine structures.
3	Observation/Research Platform	A platform from which one's surroundings or events can be observed, noted or recorded such as for scientific study.
4	Articulated Loading Platform	A metal lattice tower, buoyant at one end and attached at the other by a universal joint to a concrete filled base on the sea bed. The platform may be fitted with a helicopter platform, emergency accommodation and hawser/hose retrieval.
5	Single Anchor Leg Mooring	A rigid frame or tube with a buoyancy device at its upper end , secured at its lower end to a universal joint on a large steel or concrete base resting on the sea bed, and at its upper end to a mooring buoy by a chain or wire.
6	Mooring Tower	A platform secured to the sea bed and surmounted by a turntable to which ships moor.
7	Artificial Island	A man-made structure usually built for the exploration or exploitation of marine resources, marine scientific research, tidal observations, etc.
8	Floating Production, Storage and Off-Loading Vessel	An offshore oil/gas facility consisting of a moored tanker/barge by which the product is extracted, stored and exported.
9	Accommodation Platform	A platform used primarily for eating, sleeping and recreation purposes.
10	Navigation, Communication and Control Buoy	A floating structure with control room, power and storage facilities, attached to the sea bed by a flexible pipeline and cables.
11	Floating Oil Tank	A floating structure, anchored to the seabed, for storing oil.

5.43. Horizontal Width

Definition: A measurement of the shorter of two linear axis.

CamelCase: horizontalWidth

Alias: HORWID

Value type: real

Remarks: No remarks.

5.44. Horizontal Length

Definition: A measurement of the longer of two linear axis.

CamelCase: horizontalLength

Alias: HORLEN

Value type: real

Remarks: No remarks.

5.45. Category of Cardinal Mark

Definition: The four quadrants (north, east, south and west) are bounded by the true bearings NW-NE, NE-SE, SE-SW and SW-NW taken from the point of interest. A cardinal mark is named after the quadrant in which it is placed. The name of the cardinal mark indicates that it should be passed to the named side of the mark.

CamelCase: categoryOfCardinalMark

Alias: CATCAM

Value type: enumeration

Remarks: Cardinal marks are used in conjunction with the compass to indicate where a mariner will find safe navigable water. Cardinal marks do not have a distinctive shape but are normally pillar or spar. They are always painted in yellow and black horizontal bands and their distinctive double cone top-marks are always black. (Note that such top-marks are encoded as separate TOPMAR objects). Cardinal marks may also have a special system of flashing white lights and if such lights are fitted they are encoded as separate LIGHTS objects.

Listed Values:

Code	Label	Definition
1	North Cardinal Mark	Quadrant bounded by the true bearing NW-NE taken from the point of interest; it should be passed to the north side of the mark.
2	East Cardinal Mark	Quadrant bounded by the true bearing NE-SE taken from the point of interest. It should be passed to the east side of the mark.
3	South Cardinal Mark	Quadrant bounded by the true bearing SE-SW taken from the point of interest; it should be passed to the south side of the mark.
4	West Cardinal Mark	Quadrant bounded by the true bearing SW-NW taken from the point of interest; it should be passed to the west side of the mark.

5.46. Type of Light

Definition: Type of light equipment.

CamelCase: typeOfLight

Alias:

Value type: text

Remarks: For example bulb type (250mm, 300mm, 400mm), LED type (integral type, 200, 200HI, 250, 300, 350).

5.47. NMEAString

Definition: NMEA string captured from the positioning device.(proposed by CCG)

CamelCase: NMEAString

Alias:

Value type: text

Remarks: .

5.48. Light Characteristic

Definition: The distinct character, such as fixed, flashing, or occulting, which is given to each light to avoid confusion with neighbouring ones.

CamelCase: lightCharacteristic

Alias: LITCHR Character of Light

Value type: enumeration

Remarks: No remarks.

Listed Values:

Code	Label	Definition
1	Fixed	A signal light that shows continuously, in any given direction, with constant luminous intensity and colour.
2	Flashing	A rhythmic light in which the total duration of light in a period is clearly shorter than the total duration of darkness and all the appearances of light are of equal duration.
3	Long-Flashing	A single-flashing light in which a single flash of not less than two seconds duration is regularly repeated.
4	Quick-Flashing	A rhythmic light in which flashes are repeated at a rate of not less than 50 flashes per minutes but less than 80 flashes per minutes. It may be: - Continuous quick-flashing: A quick-flashing light in which a flash is regularly repeated. - Group quick-flashing: A quick-flashing light in which a group of two or more flashes, which are specified in number, is regularly repeated.
5	Very Quick-Flashing	A rhythmic light in which flashes are repeated at a rate of not less than 80 flashes per minute but less than 160 flashes per minute. It may be:- Continuous very quick-flashing: A very quick-flashing light in which a flash is regularly repeated.- Group very quick-flashing: A very quick-flashing light in which a group of two or more flashes, which are specified in number, is regularly repeated.
6	Continuous Ultra Quick-Flashing	A rhythmic light in which flashes are regularly repeated at a rate of not less than 160 flashes per minute.
7	Isophased	A light with all durations of light and darkness equal.
8	Occulting	A rhythmic light in which the total duration of light in a period is clearly longer than the total duration of darkness and all the eclipses are of equal duration. It may be: - Single-occulting: An occulting light in which an eclipse is regularly repeated. - Group-occulting: An occulting light in which a group of two or more eclipses, which are specified in number, is regularly repeated. - Composite group-occulting: An occulting light in which a sequence of groups of one or more eclipses, which are specified in number, is regularly repeated, and the groups comprise different numbers of eclipses.
12	Morse	A rhythmic light in which appearances of light of two clearly different durations are grouped to represent a character or characters in the Morse code.

13	Fixed and Flash	A rhythmic light in which a fixed light is combined with a flashing light of higher luminous intensity.
14	Flash and Long-Flash	A rhythmic light in which a flashing light is combined with a long-flashing light of higher luminous intensity.
15	Occulting and Flash	A rhythmic light in which an occulting light is combined with a flashing light of higher luminous intensity.
16	Fixed and Long-Flash	A rhythmic light in which a fixed light is combined with a long-flashing light of higher luminous intensity.
17	Occulting Alternating	An alternating light in which the total duration of light in each period is clearly longer than the total duration of darkness and in which the intervals of darkness (occultations) are all of equal duration.
18	Long-Flash Alternating	An alternating single-flashing light in which an appearance of light of not less than two seconds duration is regularly repeated.
19	Flash Alternating	An alternating rhythmic light in which the total duration of light in a period is clearly shorter than the total duration of darkness and all the appearances of light are of equal duration.
20	Group Alternating	Occulting light in which the occultations are combined in groups, each group including the same number of occultations, and in which the groups are repeated at regular intervals.
25	Quick-Flash Plus Long-Flash	A rhythmic light in which a group of quick flashes is followed by one or more long flashes in a regularly repeated sequence with a regular periodicity.
26	Very Quick-Flash Plus Long-Flash	A rhythmic light in which a group of very quick flashes is followed by one or more long flashes in a regularly repeated sequence with a regular periodicity.
27	Ultra Quick-Flash Plus Long-Flash	A rhythmic light in which a group of ultra quick flashes is followed by one or more long flashes in a regularly repeated sequence with a regular periodicity.
28	Alternating	A signal light that shows, in any given direction, two or more colours in a regularly repeated sequence with a regular periodicity.
29	Fixed and Alternating Flashing	
30	Group-occulting light	An occulting light in which a group of two or more eclipses, which are specified in number, is regularly repeated.
31	Composite group-occulting light	An occulting light in which a sequence of groups of one or more eclipses, which are specified in number, is regularly repeated, and the groups comprise different numbers of eclipses.
32	Group flashing light	A flashing light in which a group of flashes, specified in number, is regularly repeated.
33	Composite group-flashing light	A light similar to a group-flashing light except that successive groups in a period have different numbers of flashes.
34	Group quick light	A quick-flashing light in which a group of two or more flashes, which are specified in number, is regularly repeated.
35	Group very quick light	A very quick-flashing light in which a group of two or more flashes, which are specified in number, is regularly repeated.

5.49. Category Of Power Source

Definition: -

CamelCase: CategoryOfPowerSource

Alias:

Value type: enumeration

Remarks: -

Listed Values:

Code	Label	Definition
1	battery	-
2	generator	-
3	solar panel	-
4	electrical service	-

5.50. Category Of Synthetic AIS Aid To Navigation

Definition: -

CamelCase: CategoryOfSyntheticAISaidtoNavigation

Alias:

Value type: enumeration

Remarks: -

Listed Values:

Code	Label	Definition
1	predicted	-
2	monitored	-

5.51. Category Of Physical AIS Aid To Navigation

Definition: A classification of AIS AtoNs that correspond to an actual, physical Aid to Navigation at a real-world location.

CamelCase: CategoryOfPhysicalAIS AidToNavigation

Alias:

Value type: enumeration

Remarks: -

Listed Values:

Code	Label	Definition
1	Physical AIS Type 1	Simple transmission of static, pre-programmed information.
2	Physical AIS Type 2	Transmission of dynamic, real-time updated information via connected sensors.
3	Physical AIS Type 3	Full two-way communication: transmission + remote control / configuration.

5.52. Virtual AIS Aid to Navigation Type

Definition: A purpose of a virtual AIS Aid to Navigation.

CamelCase: virtualAIS AidToNavigationType

Alias:

Value type: enumeration

Remarks: No remarks.

Listed Values:

Code	Label	Definition
1	North Cardinal	Indicates that it should be passed to the north side of the aid.
2	East Cardinal	Indicates that it should be passed to the east side of the aid.
3	South Cardinal	Indicates that it should be passed to the south side of the aid.
4	West Cardinal	Indicates that it should be passed to the west side of the aid.
5	Port Lateral	Indicates the port boundary of a navigational channel or suggested route when proceeding in the conventional direction of buoyage.
6	Starboard Lateral	Indicates the starboard boundary of a navigational channel or suggested route when proceeding in the conventional direction of buoyage.
7	Preferred Channel to Port	At a point where a channel divides, when proceeding in the conventional direction of buoyage, the preferred channel (or primary route) is indicated by a modified port-hand lateral mark.
8	Preferred Channel to Starboard	At a point where a channel divides, when proceeding in the conventional direction of buoyage, the preferred channel (or primary route) is indicated by a modified starboard-hand lateral mark.
9	Isolated Danger	A mark used alone to indicate a dangerous reef or shoal. The mark may be passed on either hand.
10	Safe Water	Indicates that there is navigable water around the mark.
11	Special Purpose	A special purpose aid is primarily used to indicate an area or feature, the nature of which is apparent from reference to a chart, Sailing Directions or Notice to Mariners
12	New Danger Marking	A mark used to indicate the existence of a recently identified new danger, such as a wreck.

5.53. MMSI Code

Definition: The Maritime Mobile Service Identity (MMSI) Code is formed of a series of nine digits which are transmitted over the radio path in order to uniquely identify ship stations, ship earth stations, coast stations, coast earth stations, and group calls. These identities are formed in such a way that the identity or part thereof can be used by telephone and telex subscribers connected to the general telecommunications network principally to call ships automatically.

CamelCase: mMSICode

Alias:

Value type: text

Remarks: No remarks.

5.54. Category of Radar Transponder Beacon

Definition: Classification of radar transponder beacon based on functionality.

CamelCase: categoryOfRadarTransponderBeacon

Alias: CATRTB

Value type: enumeration

Remarks: No remarks.

Listed Values:

Code	Label	Definition
1	Ramark, Radar Beacon Transmitting Continuously	A radar marker beacon which continuously transmits a signal appearing as a radial line on a radar screen, the line indicating the direction of the beacon. Ramarks are intended primarily for marine use. The name 'ramark' is derived from the words radar marker.
2	Racon, Radar Transponder Beacon	A radar beacon which returns a coded signal which provides identification of the beacon, as well as range and bearing. The range and bearing are indicated by the location of the first character received on the radar screen. The name 'racon' is derived from the words radar beacon.
3	Leading Racon/Radar Transponder Beacon	A radar beacon that may be used (in conjunction with at least one other radar beacon) to indicate a leading line.

5.55. Topmark/Daymark Shape

Definition: The shape a topmark or daymark exhibits.

CamelCase: topmarkDaymarkShape

Alias: TOPSHP

Value type: enumeration

Remarks: No remarks.

Listed Values:

Code	Label	Definition
1	Cone (Point Up)	Is where the vertex points up. A cone is a solid figure generated by straight lines drawn from a fixed point (the vertex) to a circle in a plane not containing the vertex. Cones are commonly used as International Association of Marine Aids to Navigation and Lighthouse Authorities - IALA topmarks, lateral.
2	Cone (Point Down)	Is where the vertex points down. A cone is a solid figure generated by straight lines drawn from a fixed point (the vertex) to a circle in a plane not containing the vertex. Cones are commonly used as International Association of Marine Aids to Navigation and Lighthouse Authorities - IALA topmarks, lateral.
3	Sphere	A curved surface all points of which are equidistant from a fixed point within, called the centre.
4	2 Spheres	Two spheres, one above the other. Two black spheres are commonly used as an International Association of Marine Aids to Navigation and Lighthouse Authorities - IALA topmark (isolated danger).
5	Cylinder	A solid geometrical figure generated by straight lines fixed in direction and describing with one of point a closed curve, especially a circle (in which case the figure is circular cylinder, it's ends being parallel circles). Cylinders are commonly used as International Association of Marine Aids to Navigation and Lighthouse Authorities - IALA topmarks lateral.
6	Board	Usually of rectangular shape, made from timber or metal and used to provide a contrast with the natural background of a daymark. The actual daymark is often painted on to this board.
7	X-Shaped	Having a shape or a cross-section like the capital letter X. An x-shape as an International Association of Marine Aids to Navigation and Lighthouse Authorities - IALA topmark should be 3 dimensional in shape. It is made of at least three crossed bars.
8	Upright Cross	A cross with one vertical member and one horizontal member; that is, similar in shape to the character '+'.
9	Cube (Point Up)	A cube standing on one of its vertexes. A cube is a solid contained by six equal squares, a regular hexahedron.
10	2 Cones (Point to Point)	2 cones, one above the other, with their vertices together in the centre.

11	2 Cones (Base to Base)	2 cones, one above the other, with their bases together in the centre and their vertices pointing up and down.
12	Rhombus	A plane figure having four equal sides and equal opposite angles (two acute and two obtuse); an oblique equilateral parallelogram.
13	2 Cones (Points Upward)	2 cones, one above the other, with their vertices pointing up.
14	2 Cones (Points Downward)	2 cones, one above the other, with their vertices pointing down.
15	Besom (Point Up)	Besom: A bundle of rods or twigs. Perch: A staff placed on top of a buoy, rock or shoal as a mark for navigation. A besom, point up is where the thicker (untied) end of the besom is at the bottom.
16	Besom (Point Down)	Besom: A bundle of rods or twigs. Perch: A staff placed on top of a buoy, rock or shoal as a mark for navigation. A besom, point down is where the thinner (tied) end of the besom is at the bottom.
17	Flag	A flag mounted on a short pole.
18	Sphere Over a Rhombus	A sphere located above a rhombus.
19	Square	A plane figure with four right angles and four equal straight sides.
20	Rectangle (Horizontal)	Where the two longer opposite sides are standing horizontally. A rectangle is a plane figure with four right angles and four straight sides, opposite sides being parallel and equal in length.
21	Rectangle (Vertical)	Where the two longer opposite sides are standing vertically. A rectangle is a plane figure with four right angles and four straight sides, opposite sides being parallel and equal in length.
22	Trapezium (Up)	A quadrilateral having one pair of opposite sides parallel, and which stands on its longer parallel side.
23	Trapezium (Down)	A quadrilateral having one pair of opposite sides parallel, and which stands on its shorter parallel side.
24	Triangle (Point Up)	A figure having three angles and three sides, and which has a vertex at the top.
25	Triangle (Point Down)	A figure having three angles and three sides, and which has a side at the top.
26	Circle	A perfectly round plane figure whose circumference is everywhere equidistant from its centre.
27	Two Upright Crosses (One Over the Other)	Two upright crosses, generally vertically disposed one above the other.
28	T-Shape	Having a shape like the capital letter T.
29	Triangle Pointing Up Over a Circle	A triangle, vertex uppermost, located above a circle.
30	Upright Cross Over a Circle	An upright cross located above a circle.
31	Rhombus Over a Circle	A rhombus located above a circle.
32	Circle Over a Triangle Pointing Up	A circle located over a triangle, vertex uppermost.

33	Other Shape (See Shape Information)	An uncommon and/or non-standardized shape as textually described using an associated attribute.
34	Tubular	Having the form of or consisting of a tube.

5.56. Orientation Value

Definition: The angular distance measured from true north to the major axis of the feature.

CamelCase: orientationValue

Alias: ORIENT

Value type: real

Remarks: No remarks.

5.57. Category of Special Purpose Mark

Definition: Classification of an aid to navigation which signifies some special purpose.

CamelCase: categoryOfSpecialPurposeMark

Alias: CATSPM

Value type: enumeration

Remarks: A mark may be a beacon, a buoy, a signpost or may take another form.

Listed Values:

Code	Label	Definition
1	Firing Danger Mark	A mark used to indicate a firing danger area, usually at sea.
2	Target Mark	Any object toward which something is directed. The distinctive marking or instrumentation of a ground point to aid its identification on a photograph.
3	Marker Ship Mark	A mark marking the position of a ship which is used as a target during some military exercise.
4	Degaussing Range Mark	A mark used to indicate a degaussing range.
5	Barge Mark	A mark of relevance to barges.
6	Cable Mark	A mark used to indicate the position of submarine cables or the point at which they run on to the land.
7	Spoil Ground Mark	A mark used to indicate the limit of a spoil ground.
8	Outfall Mark	A mark used to indicate the position of an outfall or the point at which it leaves the land.
9	ODAS	Ocean Data Acquisition System.
10	Recording Mark	A mark used to record data for scientific purposes.
11	Seaplane Anchorage Mark	A mark used to indicate a seaplane anchorage.
12	Recreation Zone Mark	A mark used to indicate a recreation zone.
13	Private Mark	A privately maintained mark.
14	Mooring Mark	A mark indicating a mooring or moorings.
15	LANBY	A large buoy designed to take the place of a lightship where construction of an offshore light station is not feasible.
16	Leading Mark	Aids to navigation or other indicators so located as to indicate the path to be followed. Leading marks identify a leading line when they are in transit.
17	Measured Distance Mark	A mark forming part of a transit indicating one end of a measured distance.

18	Notice Mark	A notice board or sign indicating information to the mariner.
19	TSS Mark	A mark indicating a Traffic Separation Scheme.
20	Anchoring Prohibited Mark	A mark indicating an anchoring prohibited area.
21	Berthing Prohibited Mark	A mark indicating that berthing is prohibited.
22	Overtaking Prohibited Mark	A mark indicating that overtaking is prohibited.
23	Two-Way Traffic Prohibited Mark	A mark indicating a one-way route.
24	Reduced Wake Mark	A mark indicating that vessels must not generate excessive wake.
25	Speed Limit Mark	A mark indicating that a speed limit applies.
26	Stop Mark	A mark indicating the place where the bow of a ship must stop when traffic lights show red.
27	General Warning Mark	A mark indicating that special caution must be exercised in the vicinity of the mark.
28	Sound Ship's Siren Mark	A mark indicating that a ship should sound its siren or horn.
29	Restricted Vertical Clearance Mark	A mark indicating the minimum vertical space available for passage.
30	Maximum Vessel's Draught Mark	A mark indicating the maximum draught of vessel permitted.
31	Restricted Horizontal Clearance Mark	A mark indicating the minimum horizontal space available for passage.
32	Strong Current Warning Mark	A mark warning of strong currents.
33	Berthing Permitted Mark	A mark indicating that berthing is allowed.
34	Overhead Power Cable Mark	A mark indicating an overhead power cable.
35	Channel Edge Gradient Mark	A mark indicating the gradient of the slope of a dredge channel edge.
36	Telephone Mark	A mark indicating the presence of a telephone.
37	Ferry Crossing Mark	A mark indicating that a ferry route crosses the ship route; often used with a 'sound ship's siren' mark.
39	Pipeline Mark	A mark used to indicate the position of submarine pipelines or the point at which they run on to the land.
40	Anchorage Mark	A mark indicating an anchorage area.
41	Clearing Mark	A mark used to indicate a clearing line.
42	Control Mark	A mark indicating the location at which a restriction or requirement exists.
43	Diving Mark	A mark indicating that diving may take place in the vicinity.
44	Refuge Beacon	A mark providing or indicating a place of safety.
45	Foul Ground Mark	A mark indicating a foul ground.

46	Yachting Mark	A mark installed for use by yachtsmen.
47	Heliport Mark	A mark indicating an area where helicopters may land.
48	GNSS Mark	A mark indicating a location at which a GNSS position has been accurately determined.
49	Seaplane Landing Mark	A mark indicating an area where sea-planes land.
50	Entry Prohibited Mark	A mark indicating that entry is prohibited.
51	Work in Progress Mark	A mark indicating that work (generally construction) is in progress.
52	Mark With Unknown Purpose	A mark whose detailed characteristics are unknown.
53	Wellhead Mark	A mark indicating a borehole that produces or is capable of producing oil or natural gas.
54	Channel Separation Mark	A mark indicating the point at which a channel divides separately into two channels.
55	Marine Farm Mark	A mark indicating the existence of a fish, mussel, oyster or pearl farm/culture.
56	Artificial Reef Mark	A mark indicating the existence or the extent of an artificial reef.
57	Ice Mark	A mark, used year round, that may be submerged when ice passes through the area.
58	Nature Reserve Mark	A mark used to define the boundary of a nature reserve.
59	Fish Aggregating Device	A fish aggregating (or aggregation) device (FAD) is a man-made object used to attract ocean going pelagic fish such as marlin, tuna and mahi-mahi (dolphin fish). They usually consist of buoys or floats tethered to the ocean floor with concrete blocks.
60	Wreck Mark	A mark used to indicate the existence of a wreck.
61	Customs Mark	A mark used to indicate the existence of a customs checkpoint.
62	Causeway Mark	A mark used to indicate the existence of a causeway.
63	Wave Recorder	A surface following buoy used to measure wave activity.
64	Jetski Prohibited	A mark indicating a jetski prohibited area.
65	Facility Protection Mark	-
66	Oil Pipeline Protection Mark	-
67	Marine Cable Protection Mark	-

5.58. Estimated Range of Transmission

Definition: The estimated range of a non-optical electromagnetic transmission.

CamelCase: estimatedRangeOfTransmission

Alias: ESTRNG

Value type: real

Remarks: The estimated range (distance) assumes 'in vacuo' transmission and a standard antenna height of 5 metres. Thus it gives a hint to the mariner whether they are likely to receive transmission at a certain distance from an object.

5.59. Category of Radio Station

Definition: Classification of radio services offered by a radio station.

CamelCase: categoryOfRadioStation

Alias: CATROS

Value type: enumeration

Remarks: A radiobeacon is a radio transmitter which emits a distinctive or characteristic signal on which a bearing may be taken.

Listed Values:

Code	Label	Definition
1	Circular (Non-Directional) Marine or Aero-Marine Radiobeacon	A radio station which need not necessarily be manned, the emissions of which, radiated around the horizon, enable its bearing to be determined by means of the radio direction finder of a ship.
2	Directional Radiobeacon	A special type of radiobeacon station the emissions of which are intended to provide a definite track for guidance.
3	Rotating Pattern Radiobeacon	A special type of radiobeacon station emitting a beam of waves to which a uniform turning movement is given, the bearing of the station being determined by means of an ordinary listening receiver and a stop watch. Also referred to as a rotating loop radiobeacon.
4	Consol Beacon	A type of long range position fixing beacon.
5	Radio Direction-Finding Station	A radio station intended to determine only the direction of other stations by means of transmission from the latter.
6	Coast Radio Station Providing QTG Service	A radio station which is prepared to provide QTG service; that is to say, to transmit upon request from a ship a radio signal, the bearing of which can be taken by that ship.
7	Aeronautical Radiobeacon	A radio beacon designed for aeronautical use.
8	Decca	The Decca Navigator System is a high accuracy, short to medium range radio navigational aid intended for coastal and landfall navigation.
9	Loran C	A low frequency electronic position fixing system using pulsed transmissions at 100 KHz.
10	Differential GNSS	A radiobeacon transmitting DGPS correction signals.
11	Toran	An electronic position fixing system used mainly by aircraft.
12	Omega	A long-range radio navigational aid which operates within the VLF frequency band. The system comprises eight land based stations.
13	Syledis	A ranging position fixing system operating at 420-450 MHz over a range of up to 400 Km.
14	Chaika	Chaika is a low frequency electronic position fixing system using

		pulsed transmissions at 100 Khz.
19	Radio Telephone Station	The equipment needed at one station to carry on two way voice communication by radio waves only.
20	AIS Base Station	An onshore AIS unit that monitors traffic in the waterways.

5.60. Type of Environmental Observation Equipment

Definition: Type of sensor used to observe the environment. For example Anemometer, fog monitor, etc.

CamelCase: typeOfEnvironmentalObservationEquipment

Alias:

Value type: text

Remarks: No remarks.

5.61. Value of Maximum Range

Definition: The extreme distance at which an feature can be seen or a signal detected.

CamelCase: valueOfMaximumRange

Alias: VALMXR

Value type: real

Remarks: Does not apply to lights where the 'value of nominal range' should be used.

5.62. Signal Period

Definition: The time occupied by an entire cycle of intervals of light and eclipse.

CamelCase: signalPeriod

Alias: SIGPER

Value type: real

Remarks: No remarks.

5.63. Signal Output

Definition: Strength of signal output.

CamelCase: signalOutput

Alias:

Value type: real

Remarks: No remarks.

5.64. Signal Group

Definition: The number of signals, the combination of signals or the morse character(s) within one period of full sequence.

CamelCase: signalGroup

Alias: SIGGRP

Value type: text

Remarks: No remarks.

5.65. Signal Frequency

Definition: The frequency of a signal.

CamelCase: signalFrequency

Alias: SIGFRQ

Value type: integer

Remarks: No remarks.

5.66. Flare Bearing

Definition: The bearing about which the light flare symbol is rotated to be displayed in ECDIS.

CamelCase: flareBearing

Alias:

Value type: integer

Remarks: No remarks.

5.67. Category of Fog Signal

Definition: Classification of the various means of generating the fog signal.

CamelCase: categoryOfFogSignal

Alias: CATFOG

Value type: enumeration

Remarks: The classification 'horn' is the generic term for fog signals 'nautophone', 'reed' and 'tyfon'.

Listed Values:

Code	Label	Definition
1	Explosive	A signal produced by the firing of explosive charges.
2	Diaphone	A diaphone uses compressed air and generally emits a powerful low-pitched sound, which often concludes with a brief sound of suddenly lowered pitch, termed the 'grunt'.
3	Siren	A type of fog signal apparatus which produces sound by virtue of the passage of air through slots or holes in a revolving disk.
4	Nautophone	A horn having a diaphragm oscillated by electricity.
5	Reed	[1] A reed uses compressed air and emits a weak, high pitched sound. [2] Any of various water or marsh plants with a firm stem. (Concise Oxford English Dictionary)
6	Tyfon	A diaphragm horn which operates under the influence of compressed air or steam.
7	Bell	A ringing sound with a short range.
8	Whistle	A distinctive sound made by a jet of air passing through an orifice. The apparatus may be operated automatically, by hand or by air being forced up a tube by waves acting on a buoy.
9	Gong	A sound produced by vibration of a disc when struck.
10	Horn	A horn uses compressed air or electricity to vibrate a diaphragm and exists in a variety of types which differ greatly in their sound and power.

5.68. Value of Nominal Range

Definition: The luminous range of a light in a homogenous atmosphere in which the meteorological visibility is 10 sea miles.

CamelCase: valueOfNominalRange

Alias: VALNMR

Value type: real

Remarks: No remarks.

5.69. Major Light

Definition: A statement expressing if a light is considered to be a major light in terms of ECDIS display in a particular area.

CamelCase: majorLight

Alias:

Value type: boolean

Remarks: Major light is only intended to provide an indication to the ECDIS that the light is considered to be an important light in terms of its display. As such this is a cartographic attribute to aid the compiler in determining the most appropriate display for a light; it is not intended to be used as a formal classification method for lights.

5.70. Light Visibility

Definition: The specific visibility of a light, with respect to the light's intensity and ease of recognition.

CamelCase: lightVisibility

Alias: LITVIS

Value type: enumeration

Remarks: No remarks.

Listed Values:

Code	Label	Definition
1	High Intensity	Non-marine lights with a higher power than marine lights and visible from well off shore (often 'Aero' lights).
2	Low Intensity	Non-marine lights with lower power than marine lights.
3	Faint	A decrease in the apparent intensity of a light which may occur in the case of partial obstructions.
4	Intensified	A light in a sector is intensified (that is, has longer range than other sectors).
5	Unintensified	A light in a sector is unintensified (that is, has shorter range than other sectors).
6	Visibility Deliberately Restricted	A light sector is deliberately reduced in intensity, for example to reduce its effect on a built-up area.
7	Obscured	Said of the arc of a light sector designated by its limiting bearings in which the light is not visible from seaward.
8	Partially Obscured	This value specifies that parts of the sector are obscured.
9	Visible in Line of Range	Lights that must in line to be visible.

5.71. Signal Generation

Definition: The mechanism used to generate a fog or light signal.

CamelCase: signalGeneration

Alias: SIGGEN

Value type: enumeration

Remarks: No remarks.

Listed Values:

Code	Label	Definition
1	Automatically	Signal generation is initiated by a self regulating mechanism such as a timer or light sensor.
2	By Wave Action	The signal is generated by the motion of the sea surface such as a bell in a buoy.
3	By Hand	The signal is generated by a manually operated mechanism such as a hand cranked siren.
4	By Wind	The signal is generated by the motion of air such as a wind driven whistle.
5	Radio Activated	Activated by radio signal.
6	Call Activated	Activated by making a call to a manned station.

5.72. Exhibition Condition of Light

Definition: The outward display of the light.

CamelCase: exhibitionConditionOfLight

Alias: EXCLIT

Value type: enumeration

Remarks: No remarks.

Listed Values:

Code	Label	Definition
1	Light Shown Without Change of Character	A light shown throughout the 24 hours without change of character.
2	Daytime Light	A light which is only exhibited by day.
3	Fog Light	A light which is exhibited in fog or conditions of reduced visibility.
4	Night Light	A light which is only exhibited at night.

5.73. Category of Light

Definition: Classification of different light types.

CamelCase: categoryOfLight

Alias: CATLIT

Value type: enumeration

Remarks: All lights are considered to be marine lights unless the category of light indicates otherwise.

Listed Values:

Code	Label	Definition
1	Directional Function	A light illuminating a sector of very narrow angle and intended to mark a direction to follow.
4	Leading Light	A light associated with other lights so as to form a leading line to be followed.
5	Aero Light	An aero light is established for aeronautical navigation and may be of higher power than marine lights and visible from well offshore.
6	Air Obstruction Light	A light marking an obstacle which constitutes a danger to air navigation.
8	Flood Light	A broad beam light used to illuminate a structure or area.
9	Strip Light	A light whose source has a linear form generally horizontal, which can reach a length of several metres.
10	Subsidiary Light	A light placed on or near the support of a main light and having a special use in navigation.
11	Spotlight	A powerful light focused so as to illuminate a small area.
12	Front	Term used with leading lights to describe the position of the light on the lead as viewed from seaward.
13	Rear	Term used with leading lights to describe the position of the light on the lead as viewed from seaward.
14	Lower	Term used with leading lights to describe the position of the light on the lead as viewed from seaward.
15	Upper	Term used with leading lights to describe the position of the light on the lead as viewed from seaward.
17	Emergency	A light available as a backup to a main light which will be illuminated should the main light fail.
18	Bearing Light	A light which enables its approximate bearing to be obtained without the use of a compass.
19	Horizontally Disposed	A group of lights of identical character and almost identical position, that are disposed horizontally.

20	Vertically Disposed	A group of lights of identical character and almost identical position, that are disposed vertically.
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5.74. Traffic Flow

Definition: Direction of vessels passing a reference point.

CamelCase: trafficFlow

Alias: TRAFIC

Value type: enumeration

Remarks: No remarks.

Listed Values:

Code	Label	Definition
1	Inbound	Traffic flow in a general direction toward a port or similar destination.
2	Outbound	Traffic flow in a general direction away from a port or similar point of origin.
3	One-Way	Traffic flow in one general direction only.
4	Two-Way	Traffic flow in two generally opposite directions.

5.75. Technique of Vertical Measurement

Definition: Survey method used to obtain depth information.

CamelCase: techniqueOfVerticalMeasurement

Alias: TECSOU

Value type: enumeration

Remarks: No remarks.

Listed Values:

Code	Label	Definition
1	Found by Echo Sounder	The depth was determined by using an instrument that determines depth of water by measuring the time interval between emission of a sonic or ultrasonic signal and return of its echo from the bottom.
2	Found by Side Scan Sonar	The depth was computed from a record produced by active sonar in which fixed acoustic beams are directed into the water perpendicularly to the direction of travel to scan the seabed and generate a record of the seabed configuration.
3	Found by Multi Beam	The depth was determined by using a wide swath echo sounder that uses multiple beams to measure depths directly below and transverse to the ship's track.
4	Found by Diver	The depth was determined by a person skilled in the practice of diving.
5	Found by Lead Line	The depth was determined by using a line, graduated with attached marks and fastened to a sounding lead.
6	Swept by Wire-Drag	The given area was determined to be free from navigational dangers to a certain depth by towing a buoyed wire at the desired depth by two launches, or a least depth was identified using the same technique.
7	Found by Laser	The depth was determined by using an instrument that measures distance by emitting timed pulses of laser light and measuring the time between emission and reception of the reflected pulses.
8	Swept by Vertical Acoustic System	The given area has been swept using a system comprised of multiple echo sounder transducers attached to booms deployed from the survey vessel.
9	Found by Electromagnetic Sensor	The depth was determined by using an instrument that compares electromagnetic signals.
10	Photogrammetry	The science or art of obtaining reliable measurements from photographs.
11	Satellite Imagery	The depth was determined by using instruments placed aboard an artificial satellite.
12	Found by Levelling	The depth was determined by using levelling techniques to find the elevation of the point relative to a datum.

13	Swept by Side Scan Sonar	The given area was determined to be free from navigational dangers to a certain depth by towing a side scan sonar.
14	Computer Generated	The sounding was determined from a bottom model constructed using a computer.
15	Found by LIDAR	The depth was measured by using an instrument that measures distance by emitting timed pulses of laser light and measuring the time between emission and reception of the reflected pulses.
16	Synthetic Aperture Radar	A radar with a synthetic aperture antenna which is composed of a large number of elementary transducing elements. The signals are electronically combined into a resulting signal equivalent to that of a single antenna of a given aperture in a given direction.
17	Hyperspectral Imagery	Term used to describe the imagery derived from subdividing the electromagnetic spectrum into very narrow bandwidths. These narrow bandwidths may be combined with or subtracted from each other in various ways to form images useful in precise terrain or target analysis.

5.76. Quality of Vertical Measurement

Definition: The reliability of the value of a sounding.

CamelCase: qualityOfVerticalMeasurement

Alias: QUASOU

Value type: enumeration

Remarks: No remarks.

Listed Values:

Code	Label	Definition
1	Depth Known	The depth from the chart datum to the seabed (or to the top of a drying feature) is known.
2	Depth or Least Depth Unknown	The depth from chart datum to the seabed, or the shoalest depth of the feature is unknown.
3	Doubtful Sounding	A depth that may be less than indicated.
4	Unreliable Sounding	A depth that is considered to be an unreliable value.
5	No Bottom Found at Value Shown	Upon investigation the bottom was not found at this depth.
6	Least Depth Known	The shoalest depth over a feature is of known value.
7	Least Depth Unknown, Safe Clearance at Value Shown	The least depth over a feature is unknown, but there is considered to be safe clearance at this depth.
8	Value Reported (Not Surveyed)	Depth value obtained from a report, but not fully surveyed.
9	Value Reported (Not Confirmed)	Depth value obtained from a report, which it has not been possible to confirm.
10	Maintained Depth	The depth at which a channel is kept by human influence, usually by dredging.
11	Not Regularly Maintained	Depths may be altered by human influence, but will not be routinely maintained.

5.77. Maximal Permitted Draught

Definition: The maximal permitted draught of a vessel or convoy according to the particular article/clause of the applicable law/regulation.

CamelCase: maximalPermittedDraught

Alias: lg_drt

Value type: real

Remarks: No remarks.

5.78. Depth Range Minimum Value

Definition: The minimum (shoalest) value of a depth range.

CamelCase: depthRangeMinimumValue

Alias: DRVAL1

Value type: real

Remarks: Where the area dries, the value is negative.

5.79. Based On Fixed Marks

Definition: A straight route (known as a recommended track, range or leading line), which comprises: a. at least two structures (usually beacons or daymarks) and/or natural features, which may carry lights and/or top-marks. The structures/features are positioned so that when observed to be in line, a vessel can follow a known bearing with safety. (Adapted from International Association of Marine Aids to Navigation and Lighthouse Authorities - IALA Aids to Navigation Guide, 1990); or b. a single structure or natural feature, which may carry lights and/or a topmark, and a specified bearing which can be followed with safety. (S-57 Edition 3.1, Appendix A Chapter 2, Page 2.72, November 2000, as amended).

CamelCase: basedOnFixedMarks

Alias: CATTRK = 1

Value type: boolean

Remarks: No remarks.

5.80. Category of Navigation Line

Definition: Classification of route guidance given to vessels.

CamelCase: categoryOfNavigationLine

Alias: CATNAV

Value type: enumeration

Remarks: No remarks.

Listed Values:

Code	Label	Definition
1	Clearing Line	A straight line that marks the boundary between a safe and a dangerous area or that passes clear of a navigational danger.
2	Transit Line	A line passing through one or more fixed marks.
3	Leading Line Bearing a Recommended Track	A line passing through one or more clearly defined objects, along the path of which a vessel can approach safely up to a certain distance off.

5.81. Category of Lateral Mark

Definition: Classification of lateral marks in the IALA Buoyage System.

CamelCase: categoryOfLateralMark

Alias: CATLAM

Value type: enumeration

Remarks: There are two international buoyage regions, A and B, between which lateral marks differ. When top-marks, retro reflectors and/or lights are fitted to these marks, they are encoded as separate features.

Listed Values:

Code	Label	Definition
1	Port-Hand Lateral Mark	Indicates the port boundary of a navigational channel or suggested route when proceeding in the "conventional direction of buoyage".
2	Starboard-Hand Lateral Mark	Indicates the starboard boundary of a navigational channel or suggested route when proceeding in the "conventional direction of buoyage".
3	Preferred Channel to Starboard Lateral Mark	At a point where a channel divides, when proceeding in the "conventional direction of buoyage", the preferred channel (or primary route) is indicated by a modified port-hand lateral mark.
4	Preferred Channel to Port Lateral Mark	At a point where a channel divides, when proceeding in the "conventional direction of buoyage", the preferred channel (or primary route) is indicated by a modified starboard-hand lateral mark.

5.82. Manned Structure

Definition: An expression of the feature being permanently manned or not.

CamelCase: mannedStructure

Alias:

Value type: boolean

Remarks: No remarks.

5.83. Vertical Datum

Definition: The reference level used for expressing the vertical measurements of points on the earth's surface. Also called datum level, reference plane, levelling datum, datum for sounding reduction, datum for heights.

CamelCase: verticalDatum

Alias: VERDAT Datum Level Reference Plane Levelling Datum Datum for Sounding Reduction Datum for Heights

Value type: enumeration

Remarks: No remarks.

Listed Values:

Code	Label	Definition
1	Mean Low Water Springs	The average height of the low waters of spring tides. This level is used as a tidal datum in some areas. Also called spring low water.
2	Mean Lower Low Water Springs	The average height of lower low water springs at a place.
3	Mean Sea Level	The average height of the surface of the sea at a tide station for all stages of the tide over a 19-year period, usually determined from hourly height readings measured from a fixed predetermined reference level.
4	Lowest Low Water	An arbitrary level conforming to the lowest tide observed at a place, or some what lower.
5	Mean Low Water	The average height of all low waters at a place over a 19-year period.
6	Lowest Low Water Springs	An arbitrary level conforming to the lowest water level observed at a place at spring tides during a period of time shorter than 19 years.
7	Approximate Mean Low Water Springs	An arbitrary level, usually within 0.3m from that of Mean Low Water Springs (MLWS).
8	Indian Spring Low Water	An arbitrary tidal datum approximating the level of the mean of the lower low water at spring tides. It was first used in waters surrounding India.
9	Low Water Springs	An arbitrary level, approximating that of mean low water springs (MLWS).
10	Approximate Lowest Astronomical Tide	An arbitrary level, usually within 0.3m from that of Lowest Astronomical Tide (LAT).
11	Nearly Lowest Low Water	An arbitrary level approximating the lowest water level observed at a place, usually equivalent to the Indian Spring Low Water (ISLW).
12	Mean Lower Low Water	The average height of the lower low waters at a place over a 19-year period.
13	Low Water	The lowest level reached at a place by the water surface in one oscillation. Also called low tide.
14	Approximate Mean Low Water	An arbitrary level, usually within 0.3m from that of Mean Low Water

		(MLW).
15	Approximate Mean Lower Low Water	An arbitrary level, usually within 0.3m from that of Mean Lower Low Water (MLLW).
16	Mean High Water	The average height of all high waters at a place over a 19-year period.
17	Mean High Water Springs	The average height of the high waters of spring tides. Also called spring high water.
18	High Water	The highest level reached at a place by the water surface in one oscillation.
19	Approximate Mean Sea Level	An arbitrary level, usually within 0.3m from that of Mean Sea Level (MSL).
20	High Water Springs	An arbitrary level, approximating that of mean high water springs (MHWS).
21	Mean Higher High Water	The average height of higher high waters at a place over a 19-year period.
22	Equinoctial Spring Low Water	The level of low water springs near the time of an equinox.
23	Lowest Astronomical Tide	The lowest tide level which can be predicted to occur under average meteorological conditions and under any combination of astronomical conditions.
24	Local Datum	An arbitrary datum defined by a local harbour authority, from which levels and tidal heights are measured by this authority.
25	International Great Lakes Datum 1985	A vertical reference system with its zero based on the mean water level at Rimouski/Pointe-au-Pere, Quebec, over the period 1970 to 1988.
26	Mean Water Level	The average of all hourly water levels over the available period of record.
27	Lower Low Water Large Tide	The average of the lowest low waters, one from each of 19 years of observations.
28	Higher High Water Large Tide	The average of the highest high waters, one from each of 19 years of observations.
29	Nearly Highest High Water	An arbitrary level approximating the highest water level observed at a place, usually equivalent to the high water springs.
30	Highest Astronomical Tide	The highest tidal level which can be predicted to occur under average meteorological conditions and under any combination of astronomical conditions.
31	Local Low Water Reference Level	Low water reference level of the local area.
32	Local High Water Reference Level	High water reference level of the local area.
33	Local Mean Water Reference Level	Mean water reference level of the local area.
34	Equivalent Height of Water	A low water level which is the result of a defined low water discharge

	(German GIW)	- called "equivalent discharge".
35	Highest Shipping Height of Water (German HSW)	Upper limit of water levels where navigation is allowed.
36	Reference Low Water Level According to Danube Commission	The water level at a discharge, which is exceeded 94 % of the year within a period of 30 years.
37	Highest Shipping Height of Water According to Danube Commission	The water level at a discharge, which is exceeded 1% of the year within a period of 30 years.
38	Dutch River Low Water Reference Level (OLR)	The water level at a discharge, which is exceeded 95% of the year within a period of 20 years.
39	Russian Project Water Level	Conditional low water level with established probability.
40	Russian Normal Backwater Level	Highest water level derived from the upper backwater stream in watercourse or reservoir under the normal operational conditions.
41	Ohio River Datum	The Ohio River datum.
43	Dutch High Water Reference Level	Dutch High Water Reference Level.
44	Baltic Sea Chart Datum 2000	The datum refers to each Baltic country's realization of the European Vertical Reference System (EVRS) with land-uplift epoch 2000, which is connected to the Normaal Amsterdams Peil (NAP).
45	Dutch Estuary Low Water Reference Level (OLW)	Dutch Estuary Low Water Reference Level (OLW)
46	International Great Lakes Datum 2020	-
47	Sea Floor	-
48	Sea Surface	-
49	Hydrographic Zero	-

5.84. Function

Definition: A specific role that describes a feature.

CamelCase: function

Alias: FUNCTN

Value type: enumeration

Remarks: No remarks.

Listed Values:

Code	Label	Definition
2	Harbour-Masters Office	A local official who has charge of mooring and berthing of vessels, collecting harbour fees, etc.
3	Customs Office	Serves as a government office where customs duties are collected, the flow of goods are regulated and restrictions enforced, and shipments or vehicles are cleared for entering or leaving a country.
4	Health Office	The office which is charged with the administration of health laws and sanitary inspections.
5	Hospital	An institution or establishment providing medical or surgical treatment for the ill or wounded.
6	Post Office	The public department, agency or organisation responsible primarily for the collection, transmission and distribution of mail.
7	Hotel	An establishment, especially of a comfortable or luxurious kind, where paying visitors are provided with accommodation, meals and other services.
8	Railway Station	A building with platforms where trains arrive, load, discharge and depart.
9	Police Station	The headquarters of a local police force and that is where those under arrest are first charged.
10	Water-Police Station	The headquarters of a local water-police force.
11	Pilot Office	The office or headquarters of pilots; the place where the services of a pilot may be obtained.
12	Pilot Lookout	A distinctive structure or place on shore from which personnel keep watch upon events at sea or along the coast.
13	Bank Office	An office for custody, deposit, loan, exchange or issue of money.
14	Headquarters for District Control	The quarters of an executive officer (director, manager, etc.) with responsibility for an administrative area.
15	Transit Shed/Warehouse	A building or part of a building for storage of wares or goods.
16	Factory	A building or buildings with equipment for manufacturing; a workshop.

17	Power Station	A stationary plant containing apparatus for large scale conversion of some form of energy (such as hydraulic, steam, chemical or nuclear energy) into electrical energy.
18	Administrative	A building for the management of affairs.
19	Educational Facility	A building concerned with education (for example school, college, university, etc).
20	Church	A building for public Christian worship.
21	Chapel	A place for Christian worship other than a parish, cathedral or church, especially one attached to a private house or institution.
22	Temple	A building for public Jewish worship.
23	Pagoda	A Hindu or Buddhist temple or sacred building.
24	Shinto Shrine	A building for public Shinto worship.
25	Buddhist Temple	A building for public Buddhist worship.
26	Mosque	A Muslim place of worship.
27	Marabout	A shrine marking the burial place of a Muslim holy man.
28	Lookout	Keeping a watch upon events at sea or along the coast.
29	Communication	Transmitting and/or receiving electronic communication signals.
30	Television	A system for reproducing on a screen visual images transmitted (usually with sound) by radio signals.
31	Radio	Transmitting and/or receiving radio-frequency electromagnetic waves as a means of communication.
32	Radar	A method, system or technique of using beamed, reflected, and timed radio waves for detecting, locating, or tracking objects, and for measuring altitudes.
33	Light Support	A structure serving as a support for one or more lights.
34	Microwave	Broadcasting and receiving signals using microwaves.
35	Cooling	Generation of chilled liquid and/or gas for cooling purposes.
36	Observation	A place from which the surroundings can be observed but at which a watch is not habitually maintained.
37	Timeball	A visual time signal in the form of a ball.
38	Clock	Instrument for measuring time and recording hours.
39	Control	Used to control the flow of traffic within a specified range of an installation.
40	Airship Mooring	Equipment or structure to secure an airship.
41	Stadium	An arena for holding and viewing events.
42	Bus Station	A building where buses and coaches regularly stop to take on and/or

		let off passengers, especially for long-distance travel.
43	Passenger Terminal Building	A building within a terminal for the loading and unloading of passengers.
44	Sea Rescue Control	A unit responsible for promoting efficient organization of search and rescue services and for coordinating the conduct of search and rescue operations within a search and rescue region.
45	Observatory	A building designed and equipped for making observations of astronomical, meteorological, or other natural phenomena.
46	Ore Crusher	A building or structure used to crush ore.
47	Boathouse	A building or shed, usually built partly over water, for sheltering a boat or boats.
48	Pumping Station	A facility to move solids, liquids or gases by means of pressure or suction.

5.85. Category of Landmark

Definition: Classification of prominent cultural and natural features in the landscape.

CamelCase: categoryOfLandmark

Alias: CATLMK

Value type: enumeration

Remarks: No remarks.

Listed Values:

Code	Label	Definition
1	Cairn	A mound of stones, usually conical or pyramidal, raised as a landmark or to designate a point of importance in surveying.
2	Cemetery	A site and associated structures devoted to the burial of the dead.
3	Chimney	A vertical structure containing a passage or flue for discharging smoke and gases of combustion.
4	Dish Aerial	A parabolic aerial for the receipt and transmission of high frequency radio signals.
5	Flagstaff	A staff or pole on which flags are raised.
6	Flare Stack	A tall structure used for burning-off waste oil or gas.
7	Mast	A relatively tall structure usually held vertical by guy lines.
8	Windsock	A tapered fabric sleeve mounted so as to catch and swing with the wind, thus indicating the wind direction.
9	Monument	A structure erected and/or maintained as a memorial to a person and/or event.
10	Column/Pillar	A cylindrical or slightly tapering body of considerably greater length than diameter erected vertically.
11	Memorial Plaque	A slab of metal, usually ornamented, erected as a memorial to a person or event.
12	Obelisk	A tapering shaft usually of stone or concrete, square or rectangular in section, with a pyramidal apex.
13	Statue	A representation of a living being, sculptured, moulded, or cast in a variety of materials (for example: marble, metal, or plaster).
14	Cross	A monument, or other structure in form of a cross.
15	Dome	A landmark comprising a hemispherical or spheroidal shaped structure.
16	Radar Scanner	A device used for directing a radar beam through a search pattern.

17	Tower	A relatively tall, narrow structure that may either stand alone or may form part of another structure.
18	Windmill	A system of vanes attached to a tower and driven by wind (excluding wind turbines).
19	Windmotor	A modern structure for the use of wind power.
20	Spire/Minaret	A tall conical or pyramid-shaped structure often built on the roof or tower of a building, especially a church or mosque.
21	Large Rock or Boulder on Land	An isolated rocky formation or a single large stone.
22	Triangulation Mark	A recoverable point on the earth, whose geographic position has been determined by angular methods with geodetic instruments. A triangulation point is a selected point, which has been marked with a station mark, or it is a conspicuous natural or artificial feature.
23	Boundary Mark	A marker identifying the location of a surveyed boundary line.
24	Observation Wheel	Wheels with passenger cars mounted external to the rim and independently rotated by electric motors.
25	Torii	A form of decorative gateway or portal, consisting of two upright wooden posts connected at the top by two horizontal crosspieces, commonly found at the entrance to Shinto temples.
26	Bridge	(1) An elevated structure extending across or over the weather deck of a vessel, or part of such a structure. The term is sometimes modified to indicate the intended use, such as navigating bridge or signal bridge. (2) A structure erected over a depression or an obstacle such as a body of water, railroad, etc., to provide a roadway for vehicles or pedestrians.
27	Dam	A barrier to check or confine anything in motion; particularly one constructed to hold back water and raise its level to form a reservoir, or to prevent flooding.

5.86. Type of Buoy

Definition: Type equipment used as a buoy in a particular installation.

CamelCase: typeOfBuoy

Alias:

Value type: text

Remarks: Types of light buoy; for example LANBY-100, LS-35, LL-30, LL-28, LL-26, LL-26(M), LL-24, LS-24, LSP-24, LT-10Types of buoy e.g. U-17C(P), U-17S(P), U-17C(S), U-17S(S), UR-17C(P), UR-17S(P), UR-17C(S), UR-17.

5.87. Buoy Shape

Definition: The principal shape and/or design of a buoy.

CamelCase: buoyShape

Alias: BOYSHP

Value type: enumeration

Remarks: The principal shapes are those recommended in the International Association of Marine Aids to Navigation and Lighthouse Authorities - IALA System.

Listed Values:

Code	Label	Definition
1	Conical	The upper part of the body above the water-line, or the greater part of the superstructure, has approximately the shape or the appearance of a pointed cone with the point upwards.
2	Can	The upper part of the body above the water-line, or the greater part of the superstructure, has the shape of a cylinder, or a truncated cone that approximates to a cylinder, with a flat end uppermost.
3	Spherical	Shaped like a sphere, which is a body the surface of which is at all points equidistant from the centre.
4	Pillar	The upper part of the body above the water-line, or the greater part of the superstructure is a narrow vertical structure, pillar or lattice tower.
5	Spar	The upper part of the body above the water-line, or the greater part of the superstructure, has the form of a pole, or of a very long cylinder, floating upright.
6	Barrel	The upper part of the body above the water-line, or the greater part of the superstructure, has the form of a barrel or cylinder floating horizontally.
7	Superbuoy	A very large buoy designed to carry a signal light of high luminous intensity at a high elevation.
8	Ice Buoy	A specially constructed shuttle shaped buoy which is used in ice conditions.

5.88. Visual Prominence

Definition: The extent to which a feature, either natural or artificial, is visible from seaward.

CamelCase: visualProminence

Alias: CONVIS

Value type: enumeration

Remarks: No remarks.

Listed Values:

Code	Label	Definition
1	Visually Conspicuous	Term applied to a feature either natural or artificial which is distinctly and notably visible from seaward.
2	Not Visually Conspicuous	An object that may be visible from seaward, but cannot be used as a fixing mark and is not conspicuous.
3	Prominent	Objects which are easily identifiable, but do not justify being classed as conspicuous.

5.89. Vertical Length

Definition: The total vertical length of a feature.

CamelCase: verticalLength

Alias: VERLEN

Value type: real

Remarks: For floating objects: the vertical distance from the surface of water to the highest point of that object. For fixed objects: the vertical distance from seabed or ground to the highest point of that object. For objects on top of other objects: the vertical distance from the lowest to the highest point of that object. Vertical length measurements do not require a datum.

5.90. Status

Definition: The condition of an object at a given instant in time.

CamelCase: status

Alias: STATUS

Value type: enumeration

Remarks: No remarks.

Listed Values:

Code	Label	Definition
1	Permanent	Intended to last or function indefinitely.
2	Occasional	Acting on special occasions; happening irregularly.
3	Recommended	Presented as worthy of confidence, acceptance, use, etc.
4	Not in Use	Use has ceased, but the facility still exists intact; disused.
5	Periodic/Intermittent	Recurring at intervals.
6	Reserved	Set apart for some specific use.
7	Temporary	Meant to last only for a time.
8	Private	Administered by an individual or corporation, rather than a State or a public body.
9	Mandatory	Compulsory; enforced.
11	Extinguished	No longer lit.
12	Illuminated	Lit by floodlights, strip lights, etc.
13	Historic	Famous in history; of historical interest.
14	Public	Belonging to, available to, used or shared by, the community as a whole and not restricted to private use.
15	Synchronized	Occur at a time, coincide in point of time, be contemporary or simultaneous.
16	Watched	Looked at or observed over a period of time especially so as to be aware of any movement or change.
17	Unwatched	Usually automatic in operation, without any permanently-stationed personnel to superintend it.
18	Existence Doubtful	A feature that has been reported but has not been definitely determined to exist.
19	On Request	When you ask for it.

20	Drop Away	To become lower in level.
21	Rising	To become higher in level.
22	Increasing	Becoming larger in magnitude.
23	Decreasing	Becoming smaller in magnitude.
24	Strong	Not easily broken or destroyed.
25	Good	In a satisfactory condition to use.
26	Moderately	Fairly but not very.
27	Poor	Not as good as it could be or should.
28	Buoyed	Marked by buoys.
29	Fully Operational	Entire observation platform is operating in accordance with, or exceeding, manufacturer specifications.
30	Partially Operational	At least one instrument that is part of an observation platform is not operating to manufacturer specification.
31	Drifting	Floating platform at the mercy of environmental elements, whether intentional or not.
32	Broken	Fractured or in pieces.
33	Offline	Observation platform is intentionally not reporting an environmental observation.
34	Discontinued	Observation station, suite of instruments, or an individual instrument, for a particular location, has been removed and is no longer at the particular location.
35	Manual Observation	Observations made by a human observer.
36	Unknown Status	Status of an observation platform, suite of instruments, or individual instrument is not known or unspecified.
37	Confirmed	Made certain as to truth, accuracy, validity, availability, etc.
38	Candidate	Item selected for an action.
39	Under Modification	Item that is in the process of being modified.
40	Experimental	
41	Under Removal / Deletion	Item in the process of being removed or deleted.
42	Removed / Deleted	Item that has been removed or deleted.
43	Candidate for Modification	Item selected for modification.

5.91. Radar Conspicuous

Definition: A feature which returns a strong radar echo.

CamelCase: radarConspicuous

Alias: CONRAD

Value type: boolean

Remarks: No remarks.

5.92. Nature of Construction

Definition: The building's primary construction material.

CamelCase: natureOfConstruction

Alias: NATCON

Value type: enumeration

Remarks: No remarks.

Listed Values:

Code	Label	Definition
1	Masonry	Constructed of stones or bricks, usually quarried, shaped, and mortared.
2	Concreted	Constructed of concrete, a material made of sand and gravel that is united by cement into a hardened mass used for roads, foundations, etc.
3	Loose Boulders	Constructed from large stones or blocks of concrete, often placed loosely for protection against waves or water turbulence.
4	Hard Surfaced	Constructed with a surface of hard material, usually a term applied to roads surfaced with asphalt or concrete.
5	Unsurfaced	Constructed with no extra protection, usually a term applied to roads not surfaced with a hard material.
6	Wooden	Constructed from wood.
7	Metal	Constructed from metal.
8	Glass Reinforced Plastic	Constructed from a plastic material strengthened with fibres of glass.
9	Painted	The application of paint to some other construction or natural feature.
10	Framework	Constructed from a lattice framework of, often diagonal, intersecting struts.
11	Latticed	A structure of crossed wooden or metal strips usually arranged to form a diagonal pattern of open spaces between the strips.
12	Glass	[1] Any artificial or natural substance having similar properties and composition, as fused borax, obsidian, or the like. [2] Something made of such a substance, as a windowpane.
13	Fiberglass	Constructed from fiberglass.
14	Plastic	Constructed from plastic.

5.93. Marks Navigational - System Of

Definition: The system of navigational buoyage a region complies with.

CamelCase: marksNavigationalSystemOf

Alias: MARSYS

Value type: enumeration

Remarks: No remarks.

Listed Values:

Code	Label	Definition
1	IALA A	Navigational aids conform to the International Association of Marine Aids to Navigation and Lighthouse Authorities - IALA A system.
2	IALA B	Navigational aids conform to the International Association of Marine Aids to Navigation and Lighthouse Authorities - IALA B system.
9	No System	Navigational aids do not conform to any defined system.
10	Other System	Navigational aids conform to a defined system other than International Association of Marine Aids to Navigation and Lighthouse Authorities - IALA.
11	CEVNI	CEVNI (European Code for Navigation on Inland Waterways) is the European code for rivers, canals land lakes in most of Europe.
12	Russian Inland Waterway Regulations	Navigational aids conform to the Russian inland waterway regulations.
13	Brazilian National Inland Waterway Regulations - Two Sides	Navigational aids conform to the Brazilian national inland waterway regulations for two sides.
14	Brazilian National Inland Waterway Regulations - Side Independent	Navigational aids conform to the Brazilian national inland waterway regulations - side independent.
15	Paraguay-Parana Waterway - Brazilian Complementary Aids	Navigational aids conform to the Brazilian complementary aids on the Paraguay-Parana waterway.

5.94. Height

Definition: The value of the vertical distance to the highest point of the object, measured from a specified vertical datum.

CamelCase: height

Alias: HEIGHT

Value type: real

Remarks: Height must not be used for floating objects.

5.95. Elevation

Definition: The altitude of the ground level of an object, measured from a specified vertical datum.

CamelCase: elevation

Alias: ELEVAT

Value type: real

Remarks: No remarks.

5.96. Colour Pattern

Definition: A regular repeated design containing more than one colour.

CamelCase: colourPattern

Alias: COLPAT

Value type: enumeration

Remarks: No remarks.

Listed Values:

Code	Label	Definition
1	Horizontal Stripes	Straight bands or stripes of differing colours oriented horizontally.
2	Vertical Stripes	Straight bands or stripes of differing colours oriented vertically.
3	Diagonal Stripes	Straight bands or stripes of differing colours oriented diagonally (that is, not horizontally or vertically).
4	Squared	Often referred to as checker plate, where alternate colours are used to create squares similar to a chess or draught board. The pattern may be straight or diagonal.
5	Stripes (Direction Unknown)	Straight bands or stripes of differing colours oriented in an unknown direction.
6	Border Stripe	A band or stripe of colour which is displayed around the outer edge of the object, which may also form a border to an inner pattern or plain colour.
7	Single Colour	One solid colour of uniform coverage.
8	Rectangle	A four-sided shape that is made up of two pairs of parallel lines and that has four right angles, on a different coloured background.
9	Triangle	A shape that is made up of three lines and three angles, on a different coloured background.

5.97. Colour

Definition: The property possessed by an object of producing different sensations on the eye as a result of the way it reflects or emits light.

CamelCase: colour

Alias: COLOUR

Value type: enumeration

Remarks: No remarks.

Listed Values:

Code	Label	Definition
1	White	The achromatic object colour of greatest lightness characteristically perceived to belong to objects that reflect diffusely nearly all incident energy throughout the visible spectrum.
2	Black	The achromatic color of least lightness characteristically perceived to belong to objects that neither reflect nor transmit light.
3	Red	A color whose hue resembles that of blood or of the ruby or is that of the long-wave extreme of the visible spectrum.
4	Green	Of the color green.
5	Blue	A color whose hue is that of the clear sky or that of the portion of the color spectrum lying between green and violet.
6	Yellow	A color whose hue resembles that of ripe lemons or sunflowers or is that of the portion of the spectrum lying between green and orange.
7	Grey	Of the color grey.
8	Brown	Any of a group of colors between red and yellow in hue, of medium to low lightness, and of moderate to low saturation.
9	Amber	A variable color averaging a dark orange yellow.
10	Violet	Any of a group of colors of reddish-blue hue, low lightness, and medium saturation.
11	Orange	Any of a group of colors that are between red and yellow in hue.
12	Magenta	A deep purplish red.
13	Pink	Any of a group of colors bluish red to red in hue, of medium to high lightness, and of low to moderate saturation.
14	Green A	-
15	Green B	-
16	White Temporary	-

17	Red Temporary	-
18	Yellow Temporary	-
19	Green Preferred	-
20	Green Temporary	-

5.98. Beacon Shape

Definition: Describes the characteristic geometric form of the beacon.

CamelCase: beaconShape

Alias: BCNSHP

Value type: enumeration

Remarks: No remarks.

Listed Values:

Code	Label	Definition
1	Stake, Pole, Perch, Post	An elongated wood or metal pole, driven into the ground or seabed, which serves as a navigational aid or a support for a navigational aid.
2	Withy	A tree without roots stuck or spoiled into the bottom of the sea to serve as a navigational aid.
3	Beacon Tower	A solid structure of the order of 10 metres in height used as a navigational aid.
4	Lattice Beacon	A structure consisting of strips of metal or wood crossed or interlaced to form a structure to serve as an aid to navigation or as a support for an aid to navigation.
5	Pile Beacon	A long heavy timber(s) or section(s) of steel, wood, concrete, etc., forced into the seabed to serve as an aid to navigation or as a support for an aid to navigation.
6	Cairn	A mound of stones, usually conical or pyramidal, raised as a landmark or to designate a point of importance in surveying.
7	Buoyant Beacon	A tall spar-like beacon fitted with a permanently submerged buoyancy chamber, the lower end of the body is secured to seabed sinker either by a flexible joint or by a cable under tension.

5.99. Date Start

Definition: The earliest date on which an object (for example a buoy) will be present.

CamelCase: dateStart

Alias: DATSTA

Value type: S100_TruncatedDate

Remarks: The Date Start should be encoded using 4 digits for the calendar year (YYYY), 2 digits for the month (MM) (for example April = 04) and 2 digits for the day (DD). When no specific month and/or day is required/known, indication of the month and/or day is omitted, and replaced with dashes (-). When no specific year is required (that is, the event or date range ends at the same time each year) the following two cases may be considered:- same day each year: ----MMDD- same month each year: ----MM--This conforms to ISO 8601: 2004. Date Start indicates the earliest date of an event or the start of a date range. It is used to indicate the start of a fixed date range, the start of a periodic date range, or the deployment or implementation of a feature at a specific date in the future.

5.100. Date End

Definition: The latest date on which an object (for example a buoy) will be present.

CamelCase: dateEnd

Alias: DATEND

Value type: S100_TruncatedDate

Remarks: The Date End should be encoded using 4 digits for the calendar year (YYYY), 2 digits for the month (MM) (for example April = 04) and 2 digits for the day (DD). When no specific month and/or day is required/known, indication of the month and/or day is omitted, and replaced with dashes (-). When no specific year is required (that is, the event or date range ends at the same time each year) the following two cases may be considered:- same day each year: ----MMDD- same month each year: ----MM--This conforms to ISO 8601: 2004. Date End indicates the latest date of an event or the end of a date range. It is used to indicate the end of a fixed date range, the end of a periodic date range, or the removal or cancellation of a feature at a specific date in the future.

5.101. Installation Date

Definition: The date when an item was installed.

CamelCase: installationDate

Alias:

Value type: date

Remarks: No remarks.

5.102. AtoN Number

Definition: Identifier from a list of Aids to Navigation publication, such as List of Lights.

CamelCase: AtoNNumber

Alias:

Value type: URI

Remarks: -

5.103. ID Code

Definition: Identification code as specified in predefined system. Also called identification number.

CamelCase: iDCode

Alias: Identification Number Identification Code

Value type: text

Remarks: No remarks.

5.104. Scale Minimum

Definition: The minimum scale at which the feature may be used for example for ECDIS presentation.

CamelCase: scaleMinimum

Alias: SCAMIN

Value type: integer

Remarks: The modulus of the scale is indicated, that is 1:1 250 000 is encoded as 1250000.

5.105. Source Date

Definition: The production date of the source; for example the date of measurement.

CamelCase: sourceDate

Alias: SORDAT

Value type: date

Remarks: No remarks.

5.106. Source

Definition: The publication, document, or reference work from which information comes or is acquired.

CamelCase: source

Alias:

Value type: text

Remarks: May be populated with the corresponding paper chart Notice to Mariners numbers, although other references are permitted.

5.107. Pictorial Representation

Definition: Indicates whether a pictorial representation of the feature is available.

CamelCase: pictorialRepresentation

Alias: PICREP

Value type: text

Remarks: The 'pictorial representation' could be a drawing or a photo. The string encodes the file name of an external graphic file (pixel/vector).

5.108. Inspection Frequency

Definition: A statement of how frequently an item is inspected.

CamelCase: inspectionFrequency

Alias:

Value type: text

Remarks: No remarks.

5.109. Inspection Requirements

Definition: A statement of what requirements are in place for how inspection of an item is carried out.

CamelCase: inspectionRequirements

Alias:

Value type: text

Remarks: No remarks.

5.110. AtoN Maintenance Record

Definition: A reference following the Uniform Resource Identifier (URI) principles to a record of maintenance.

CamelCase: aToNMaintenanceRecord

Alias:

Value type: text

Remarks: No remarks.

5.111. Display Name

Definition: A statement expressing if a feature name is to be displayed in certain system display settings or not.

CamelCase: displayName

Alias:

Value type: boolean

Remarks: Where it is allowable to encode multiple instances of feature name for a single feature instance, only one feature name instance can indicate that the name is to be displayed (display name set to True).

5.112. Name

Definition: The individual name of a feature.

CamelCase: name

Alias: OBJNAM

Value type: text

Remarks: No remarks.

5.113. Aid Availability Category

Definition: A Category denoting the significance of an Aid to Navigation, expressed in terms of the probability that an AtoN or system of AtoN, as defined by the Competent Authority, is performing its specified function at any randomly chosen time. This is expressed as a percentage of total time that an AtoN or system of AtoN should be performing their specified function.

CamelCase: aidAvailabilityCategory

Alias:

Value type: enumeration

Remarks: No remarks.

Listed Values:

Code	Label	Definition
1	Category 1	An AtoN or system of AtoN that is considered by the Competent Authority to be of vital navigational significance.
2	Category 2	An AtoN or system of AtoN that is considered by the Competent Authority to be of important navigational significance.
3	Category 3	An AtoN or system of AtoN that is considered by the Competent Authority to be of necessary navigational significance.

5.114. Condition

Definition: The various conditions of buildings and other constructions.

CamelCase: condition

Alias: CONDTN

Value type: enumeration

Remarks: The default 'condition' should be considered to be completed, undamaged and working normally.

Listed Values:

Code	Label	Definition
1	Under Construction	Being built but not yet capable of function.
2	Ruined	A structure in a decayed or deteriorated condition resulting from neglect or disuse, or a damaged structure in need of repair.
3	Under Reclamation	An area of the sea, a lake or the navigable part of a river that is being reclaimed as land, usually by the dumping of earth and other material.
4	Wingless	A windmill or wind turbine from which the vanes or turbine blades are missing.
5	Planned Construction	Detailed planning has been completed but construction has not been initiated.

5.115. Remote Monitoring System

Definition: Classification or name of system used to remotely monitor a feature.

CamelCase: remoteMonitoringSystem

Alias:

Value type: text

Remarks: No remarks.

5.116. Delivery Point

Definition: Details of where post can be delivered such as the apartment, name and/or number of a street, building or PO Box.

CamelCase: deliveryPoint

Alias: DELPNT

Value type: text

Remarks: No remarks.

5.117. City Name

Definition: The name of a town or city.

CamelCase: cityName

Alias: CITYNM

Value type: text

Remarks: No remarks.

5.118. Administrative Division

Definition: A generic term for an administrative region within a country at a level below that of the sovereign state.

CamelCase: administrativeDivision

Alias:

Value type: text

Remarks: No remarks.

5.119. Country Name

Definition: The name of a nation.

CamelCase: countryName

Alias:

Value type: text

Remarks: No remarks.

5.120. Postal Code

Definition: Known in various countries as a postcode, or ZIP code, the postal code is a series of letters and/or digits that identifies each postal delivery area.

CamelCase: postalCode

Alias: POSCOD Postcode ZIP Code

Value type: text

Remarks: No remarks.

5.121. Category Of Aggregation

Definition: named aggregations between two or more aids to navigation and/or navigationally relevant features

CamelCase: CategoryOfAggregation

Alias:

Value type: S100_CodeList

Remarks: -

Listed Values:

Code	Label	Definition
1	leading line	-
3	measured distance	-
2	range system	-

5.122. Text

Definition: A non-formatted digital text string.

CamelCase: text

Alias: INFORM NINFOM

Value type: text

Remarks: Should be used, for example, to hold the information that is for short cautionary or explanatory notes. Therefore, text populated in text must not exceed 300 characters. Text may be in English, or in a national language. No formatting of text is possible within text. If formatted text, or text strings exceeding 300 characters, is required, then an alternate concept should be used.

5.123. Language

Definition: The method of human communication, either spoken or written, consisting of the use of words in a structured and conventional way.

CamelCase: language

Alias:

Value type: text

Remarks: The language is encoded by a 3 character code following ISO 639-2/T.

5.124. Headline

Definition: Words set at the head of a passage or page to introduce or categorize.

CamelCase: headline

Alias:

Value type: text

Remarks: No remarks.

5.125. File Reference

Definition: The file name of an externally referenced text file.

CamelCase: fileReference

Alias: TXTDSC NTXTDS

Value type: text

Remarks: No remarks.

5.126. File Locator

Definition: The location of a fragment of text or other information in a support file.

CamelCase: fileLocator

Alias:

Value type: text

Remarks: Application schemas must describe how the associated file is identified. The associated file will commonly be named in a file reference co-attribute of the same complex attribute. Each DCEG must specify requirements for the format of the associated file and the semantics of file locator. For example, the value of file locator may be an HTML ID in an HTML file, line number in a text file) or a bookmark in a PDF file.

5.127. Uncertainty Variable Factor

Definition: The factor to be applied to the variable component of an uncertainty equation so as to provide the best estimate of the variable horizontal or vertical accuracy component for positions, depths, heights, vertical distances and vertical clearances.

CamelCase: uncertaintyVariableFactor

Alias:

Value type: real

Remarks: No remarks.

5.128. Uncertainty Fixed

Definition: The best estimate of the fixed horizontal or vertical accuracy component for positions, depths, heights, vertical distances and vertical clearances.

CamelCase: uncertaintyFixed

Alias: POSACC SOUACC VERACC

Value type: real

Remarks: No remarks.

5.129. Information in National Language

Definition: Textual information in national language characters.

CamelCase: informationInNationalLanguage

Alias: NINFOM

Value type: text

Remarks: Encodes any textual information about an object using a specified national language. The textual information could be, for example, a list, a table or a text. This should be used, for example, to hold the information that is shown on paper charts by cautionary and explanatory notes.

5.130. Horizontal Distance Uncertainty

Definition: The best estimate of the horizontal accuracy of horizontal clearances and distances.

CamelCase: horizontalDistanceUncertainty

Alias: HORACC

Value type: real

Remarks: The error is assumed to be positive and negative. The plus/minus character must not be encoded.

5.131. Orientation Uncertainty

Definition: The best estimate of the accuracy of a bearing.

CamelCase: orientationUncertainty

Alias:

Value type: real

Remarks: No remarks.

5.132. Category of Temporal Variation

Definition: An assessment of the likelihood of change over time.

CamelCase: categoryOfTemporalVariation

Alias:

Value type: enumeration

Remarks: No remarks.

Listed Values:

Code	Label	Definition
1	Extreme Event	Indication of the possible impact of a significant event (for example hurricane, earthquake, volcanic eruption, landslide, etc), which is considered likely to have changed the seafloor or landscape significantly.
2	Likely to Change and Significant Shoaling Expected	Continuous or frequent change (for example river siltation, sand waves, seasonal storms, icebergs, etc) that is likely to result in new significant shoaling.
3	Likely to Change But Significant Shoaling Not Expected	Continuous or frequent change (for example sand wave shift, seasonal storms, icebergs, etc) that is not likely to result in new significant shoaling.
4	Likely to Change	Continuous or frequent change to non-bathymetric features (for example river siltation, glacier creep/recession, sand dunes, buoys, marine farms, etc).
5	Unlikely to Change	Significant change to the seafloor is not expected.
6	Unassessed	Not having been assessed.

5.133. Maximum Display Scale

Definition: The value considered by the Data Producer to be the maximum (largest) scale at which the data is to be displayed before it can be considered to be “grossly overscaled”.

CamelCase: maximumDisplayScale

Alias:

Value type: integer

Remarks: No remarks.

5.134. Minimum Display Scale

Definition: The smallest intended viewing scale for the data.

CamelCase: minimumDisplayScale

Alias:

Value type: integer

Remarks: No remarks.

5.135. Aton Commissioning

Definition: .

CamelCase: atonCommissioning

Alias:

Value type: enumeration

Remarks: No remarks.

Listed Values:

Code	Label	Definition
1	Buoy establishment	.
2	Light establishment	.
3	Beacon establishment	.
4	Audible signal establishment	.
5	Fog signal establishment	.
6	AIS transmitter establishment	.
7	V-AIS establishment	.
8	RACON establishment	.
9	DGPS station establishment	.
10	eLORAN station establishment	.
11	DGLONASS station establishment	.
12	e-Chayka station establishment	.
13	EGNOS establishment	.

5.136. Aton Removal

Definition: .

CamelCase: atonRemoval

Alias:

Value type: enumeration

Remarks: No remarks.

Listed Values:

Code	Label	Definition
1	Buoy removal	.
2	Buoy temporary removal	.
3	Light removal	.
4	Light temporary removal	.
5	Beacon removal	.
6	Beacon temporary removal	.
7	Fog signal removal	.
8	Fog signal temporary removal	.
9	Audible signal removal	.
10	Audible signal temporary removal	.
11	V-AIS removal	.
12	V-AIS temporary removal	.
13	RACON signal removal	.
14	RACON temporary removal	.
15	DGPS removal	.
16	DGPS temporary removal	.
17	EGNOS removal	.
18	EGNOS temporary removal	.
19	LORAN C station removal	.
20	LORAN C station temporary removal	.

21	eLORAN removal	.
22	eLORAN temporary removal	.
23	Chayka station removal	.
24	Chayka station temporary removal	.
25	e-Chayka station removal	.
26	e-Chayka station temporary removal	.

5.137. Aton Replacement

Definition: .

CamelCase: atonReplacement

Alias:

Value type: enumeration

Remarks: No remarks.

Listed Values:

Code	Label	Definition
1	Buoy change	.
2	Buoy temporary change	.
3	Light change	.
4	Light temporary change	.
5	Sector light change	.
6	Sector light temporary change	.
7	Beacon change	.
8	Beacon temporary change	.
9	Fog signal change	.
10	Fog signal temporary change	.
11	Audible signal change	.
12	Audible signal temporary change	.
13	V-AIS change	.
14	V-AIS temporary change	.
15	RACON signal change	.
16	RACON temporary change	.

5.138. Fixed Aton Change

Definition: .

CamelCase: fixedAtonChange

Alias:

Value type: enumeration

Remarks: No remarks.

Listed Values:

Code	Label	Definition
1	Beacon missing	.
2	Beacon damaged	.
3	Light beacon Unlit	.
4	Light beacon Unreliable	.
5	Light beacon Not synchronized	.
6	Light beacon damaged	.
7	Beacon topmark missing	.
8	Beacon topmark damaged	.
9	Beacon daymark unreliable	.
10	Floodlit beacon Unlit	.
11	Beacon restored to normal	.

5.139. Floating Aton Change

Definition: .

CamelCase: floatingAtonChange

Alias:

Value type: enumeration

Remarks: No remarks.

Listed Values:

Code	Label	Definition
1	Buoy adrift	.
2	Buoy damaged	.
3	Buoy daymark unreliable	.
4	Buoy destroyed	.
5	Buoy missing	.
6	Buoy move	.
7	Buoy off position	.
8	Buoy re-establishment	.
9	Buoy restored to normal	.
10	Buoy topmark damaged	.
11	Buoy topmark missing	.
12	Buoy will be withdrawn	.
13	Buoy withdrawn	.
14	Decommissioned for winter	.
15	Lifted for Winter	.
16	Light buoy Light damaged	.
17	Light buoy Light not synchronized	.
18	Light buoy Light unlit	.
19	Light buoy Light unreliable	.
20	Marine Aids to Navigation unreliable	.

21	Recommissioned for navigation season	.
22	Replaced by Winter Spar	.
23	Seasonal decommissioning complete	.
24	Seasonal decommissioning in progress	.
25	Seasonal recommissioning complete	.
26	Seasonal recommissioning in progress	.

5.140. Audible Signal Aton Change

Definition: .

CamelCase: audibleSignalAtonChange

Alias:

Value type: enumeration

Remarks: No remarks.

Listed Values:

Code	Label	Definition
1	Audible signal out of service	.
2	Fog signal out of service	.
3	Audible signal operating properly	.
4	Fog signal operating properly	.

5.141. Lighted Aton Change

Definition: .

CamelCase: lightedAtonChange

Alias:

Value type: enumeration

Remarks: No remarks.

Listed Values:

Code	Label	Definition
1	Light unlit	.
2	Light unreliable	.
3	Light re-establishment	.
4	Light range reduced	.
5	Light without rhythm	.
6	Light out of synchronization	.
7	Light daymark unreliable	.
8	Light operating properly	.
9	Sector light Sector obscured	.
10	Front leading/range light Unlit	.
11	Rear leading/range light Unlit	.
12	Front leading/range light Unreliable	.
13	Rear leading/range light Unreliable	.
14	Front leading/range light Light range reduced	.
15	Rear leading/range light Light range reduced	.
16	Front leading/range light without rhythm	.
17	Rear leading/range light without rhythm	.
18	Leading/range lights out of	.

	synchronization	
19	Front leading/range beacon Unreliable	.
20	Rear leading/range beacon Unreliable	.
21	Front leading/range light is operating properly	.
22	Rear leading/range light is operating properly	.
23	Front leading/range beacon restored to normal	.
24	Rear leading/range beacon restored to normal	.

5.142. Electronic Aton Change

Definition: .

CamelCase: electronicAtonChange

Alias:

Value type: enumeration

Remarks: No remarks.

Listed Values:

Code	Label	Definition
1	AIS transmitter out of service	.
2	AIS transmitter unreliable	.
3	AIS transmitter operating properly	.
4	V-AIS out of service	.
5	V-AIS unreliable	.
6	V-AIS operating properly	.
7	RACON out of service	.
8	RACON unreliable	.
9	RACON operating properly	.
10	DGPS out of service	.
11	DGPS operating properly	.
12	DGPS unreliable	.
13	LORAN C operating properly	.
14	LORAN C unreliable	.
15	LORAN C out of service	.
16	eLORAN operating properly	.
17	eLORAN unreliable	.
18	eLORAN out of service	.
19	DGLOANSS operating properly	.
20	DGLOANSS unreliable	.

21	DGLOANSS out of service	.
22	Chayka operating properly	.
23	Chayka unreliable	.
24	Chayka out of service	.
25	e-Chayka operating properly	.
26	e-Chayka unreliable	.
27	e-Chayka out of service	.
28	EGNOS operating properly	.
29	EGNOS unreliable	.
30	EGNOS out of service	.

5.143. Positioning Equipment

Definition: .

CamelCase: positioningEquipment

Alias:

Value type: enumeration

Remarks: No remarks.

Listed Values:

Code	Label	Definition
1	DGPS Receiver	.
2	GLONASS Receiver	.
3	GPS Receiver	.
4	GPS/WAAS Receiver	.

6. Complex Attributes

6.1. Contact Address

Definition: Direction or superscription of a letter, package, etc., specifying the name of the place to which it is directed, and optionally a contact person or organisation who should receive it.

CamelCase: contactAddress

Alias:

Remarks: No remarks.

SubAttribute Bindings:

S-10x Attribute	S-57 Acronym	Allowable Encoding Value	Type	Multiplicity
Delivery Point	(DELPNT)		TE	0, 1
City Name	(CITYNM)		TE	0, 1
Administrative Division			TE	0, 1
Country Name			TE	0, 1
Postal Code	(POSCOD) (Postcode) (ZIP Code)		TE	0, 1

6.2. Directional Character

Definition: A directional light is a light illuminating a sector of very narrow angle and intended to mark a direction to follow.

CamelCase: directionalCharacter

Alias:

Remarks: No remarks.

SubAttribute Bindings:

S-10x Attribute	S-57 Acronym	Allowable Encoding Value	Type	Multiplicity
Moire Effect			BO	0, 1
Orientation Value	(ORIENT)		(S) RE	1, 1

6.3. Feature Name

Definition: Provides the name of an entity, defines the national language of the name, and provides the option to display the name at various system display settings.

CamelCase: featureName

Alias:

Remarks: No remarks.

SubAttribute Bindings:

S-10x Attribute	S-57 Acronym	Allowable Encoding Value	Type	Multiplicity
Display Name			BO	0, 1
Language			TE	0, 1
Name	(OBJNAM)		TE	1, 1

6.4. Fixed Date Range

Definition: An active period of a single fixed event or occurrence, as the date range between discrete start and end dates.

CamelCase: fixedDateRange

Alias:

Remarks: Dates must be encoded in the format YYYYMMDD; using 4 digits for the calendar year (YYYY) and, optionally, 2 digits for the month (MM) (for example April = 04) and 2 digits for the day (DD). When no specific month and/or day is required/known, the values are replaced with dashes (-). The date range of a recurring event or occurrence must be encoded using periodicDateRange.

SubAttribute Bindings:

S-10x Attribute	S-57 Acronym	Allowable Encoding Value	Type	Multiplicity
Date End	(DATEND)		TD	0, 1
Date Start	(DATSTA)		TD	0, 1

6.5. Light Sector

Definition: A sector is the part of a circle between two straight lines drawn from the centre to the circumference.

CamelCase: lightSector

Alias:

Remarks: No remarks.

SubAttribute Bindings:

S-10x Attribute	S-57 Acronym	Allowable Encoding Value	Type	Multiplicity
Colour	(COLOUR)	1 : White 2 : Black 3 : Red 4 : Green 5 : Blue 6 : Yellow 7 : Grey 8 : Brown 9 : Amber 10 : Violet 11 : Orange 12 : Magenta 13 : Pink	EN	1, * (ordered)
Directional Character			C	0, 1
Moire Effect			(S) BO	0, 1
Orientation			(S) C	1, 1
Orientation Uncertainty			(S) RE	0, 1
Orientation Value	(ORIENT)		(S) RE	1, 1
Sector Arc Extension			BO	0, 1

6.6. Multiplicity of Features

Definition: The number of features of identical character that exist as a colocated group.

CamelCase: multiplicityOfFeatures

Alias:

Remarks: No remarks.

SubAttribute Bindings:

S-10x Attribute	S-57 Acronym	Allowable Encoding Value	Type	Multiplicity
Multiplicity Known			BO	1, 1
Number of Features			IN	0, 1

6.7. Orientation

Definition: (1) The angular distance measured from true north to the major axis of the feature. (2) In ECDIS, the mode in which information on the ECDIS is being presented. Typical modes include: north-up - as shown on a nautical chart, north is at the top of the display; Ships head-up - based on the actual heading of the ship, (e.g. Ships gyrocompass); course-up display - based on the course or route being taken.

CamelCase: orientation

Alias:

Remarks: No remarks.

SubAttribute Bindings:

S-10x Attribute	S-57 Acronym	Allowable Encoding Value	Type	Multiplicity
Orientation Uncertainty			RE	0, 1
Orientation Value	(ORIENT)		RE	1, 1

6.8. Periodic Date Range

Definition: The active period of a recurring event or occurrence.

CamelCase: periodicDateRange

Alias:

Remarks: No remarks.

SubAttribute Bindings:

S-10x Attribute	S-57 Acronym	Allowable Encoding Value	Type	Multiplicity
Date End	(DATEND)		TD	1, 1
Date Start	(DATSTA)		TD	1, 1

6.9. Radar Wave Length

Definition: The distance between two successive peaks (or other points of identical phase) on an electromagnetic wave in the radar band of the electromagnetic spectrum.

CamelCase: radarWaveLength

Alias:

Remarks: No remarks.

SubAttribute Bindings:

S-10x Attribute	S-57 Acronym	Allowable Encoding Value	Type	Multiplicity
Radar Band			TE	1, 1
Wave Length Value	(RadarWaveLength)		RE	1, 1

6.10. Rhythm of Light

Definition: The sequence of times occupied by intervals of light/sound and eclipse/silence for all light characteristics or sound signals.

CamelCase: rhythmOfLight

Alias:

Remarks: No remarks.

SubAttribute Bindings:

S-10x Attribute	S-57 Acronym	Allowable Encoding Value	Type	Multiplicity
Light Characteristic	(LITCHR) (Character of Light)	1 : Fixed 2 : Flashing 3 : Long-Flashing 4 : Quick-Flashing 5 : Very Quick-Flashing 6 : Continuous Ultra Quick-Flashing 7 : Isophased 8 : Occulting 12 : Morse 13 : Fixed and Flash 14 : Flash and Long-Flash 15 : Occulting and Flash 16 : Fixed and Long-Flash 17 : Occulting Alternating 18 : Long-Flash Alternating 19 : Flash Alternating 20 : Group Alternating 25 : Quick-Flash Plus Long-Flash 26 : Very Quick-Flash Plus Long-Flash 27 : Ultra Quick-Flash Plus Long-Flash 28 : Alternating 29 : Fixed and Alternating Flashing 30 : Group-occulting light 31 : Composite group-occulting light 32 : Group flashing light	EN	1, 1

		33 : Composite group-flashing light 34 : Group quick light 35 : Group very quick light		
Signal Group	(SIGGRP)		TE	0, 10 (ordered)
Signal Period	(SIGPER)		RE	0, 1
Signal Status		1 : Lit/Sound 2 : Eclipsed/Silent	(S) EN	1, 1

6.11. Sector Characteristics

Definition: Describes the characteristics of a light sector.

CamelCase: sectorCharacteristics

Alias:

Remarks: No remarks.

SubAttribute Bindings:

S-10x Attribute	S-57 Acronym	Allowable Encoding Value	Type	Multiplicity
Light Characteristic	(LITCHR) (Character of Light)	1 : Fixed 2 : Flashing 3 : Long-Flashing 4 : Quick-Flashing 5 : Very Quick-Flashing 6 : Continuous Ultra Quick-Flashing 7 : Isophased 8 : Occulting 12 : Morse 13 : Fixed and Flash 14 : Flash and Long-Flash 15 : Occulting and Flash 16 : Fixed and Long-Flash 17 : Occulting Alternating 18 : Long-Flash Alternating 19 : Flash Alternating 20 : Group Alternating 25 : Quick-Flash Plus Long-Flash 26 : Very Quick-Flash Plus Long-Flash 27 : Ultra Quick-Flash Plus Long-Flash 28 : Alternating 29 : Fixed and Alternating Flashing 30 : Group-occulting light 31 : Composite group-occulting light 32 : Group flashing light 33 : Composite group-	EN	1, 1

		flashing light 34 : Group quick light 35 : Group very quick light		
Light Sector			C	1, 10
Colour	(COLOUR)	1 : White 2 : Black 3 : Red 4 : Green 5 : Blue 6 : Yellow 7 : Grey 8 : Brown 9 : Amber 10 : Violet 11 : Orange 12 : Magenta 13 : Pink	(S) EN	1, * (ordered)
Directional Character			(S) C	0, 1
Signal Status		1 : Lit/Sound 2 : Eclipsed/Silent	(S) EN	1, 1

6.12. Sector Information

Definition: Additional textual information about a light sector.

CamelCase: sectorInformation

Alias:

Remarks: No remarks.

SubAttribute Bindings:

S-10x Attribute	S-57 Acronym	Allowable Encoding Value	Type	Multiplicity
Language			TE	0, 1
Text	(INFORM) (NINFOM)		TE	1, 1

6.13. Sector Limit

Definition: A sector is the part of a circle between two straight lines drawn from the centre to the circumference. The sector limit specifies the limits of the sector In a clockwise direction around the central feature (for example a light).

CamelCase: sectorLimit

Alias:

Remarks: No remarks.

SubAttribute Bindings:

S-10x Attribute	S-57 Acronym	Allowable Encoding Value	Type	Multiplicity
Sector Limit One	(SECTR1)		C	1, 1
Sector Line Length			(S) IN	0, 1

6.14. Sector Limit One

Definition: A sector is the part of a circle between two straight lines drawn from the centre to the circumference. Sector limit one specifies the first limit of the sector. The order of sector limit one and sector limit two is clockwise around the central feature (for example a light).

CamelCase: sectorLimitOne

Alias: SECTR1

Remarks: No remarks.

SubAttribute Bindings:

S-10x Attribute	S-57 Acronym	Allowable Encoding Value	Type	Multiplicity
Sector Bearing	(SECTR1) (SECTR2)		RE	1, 1
Sector Line Length			IN	0, 1

6.15. Sector Limit Two

Definition: A sector is the part of a circle between two straight lines drawn from the centre to the circumference. Sector limit two specifies the second limit of the sector. The order of sector limit one and sector limit two is clockwise around the central feature (for example a light).

CamelCase: sectorLimitTwo

Alias: SECTR2

Remarks: No remarks.

SubAttribute Bindings:

S-10x Attribute	S-57 Acronym	Allowable Encoding Value	Type	Multiplicity
Sector Bearing	(SECTR1) (SECTR2)		RE	1, 1
Sector Line Length			IN	0, 1

6.16. Shape Information

Definition: Textual information about the shape of a non-standard topmark.

CamelCase: shapeInformation

Alias:

Remarks: No formatting of text is possible within shape information. If formatted text is required, then an associated text file referenced by the complex attribute textual description must be used.

SubAttribute Bindings:

S-10x Attribute	S-57 Acronym	Allowable Encoding Value	Type	Multiplicity
Language			TE	0, 1
Text	(INFORM) (NINFOM)		TE	1, 1

6.17. Signal Sequence

Definition: The sequence of times occupied by intervals of light and eclipse for all light characteristics.

CamelCase: signalSequence

Alias: SIGSEQ

Remarks: No remarks.

SubAttribute Bindings:

S-10x Attribute	S-57 Acronym	Allowable Encoding Value	Type	Multiplicity
Signal Duration			RE	1, 1
Signal Status		1 : Lit/Sound 2 : Eclipsed/Silent	EN	1, 1

6.18. Spatial Accuracy

Definition: Provides an indication of the vertical and horizontal positional uncertainty of bathymetric data, optionally within a specified date range.

CamelCase: spatialAccuracy

Alias:

Remarks: No remarks.

SubAttribute Bindings:

S-10x Attribute	S-57 Acronym	Allowable Encoding Value	Type	Multiplicity
Fixed Date Range			C	0, 1
Date End	(DATEND)		(S) TD	0, 1
Uncertainty Variable Factor			(S) RE	0, 1

6.19. Cable Dimensions

Definition: The dimensions of a cable to give its length and diameter.

CamelCase: CableDimensions

Alias:

Remarks: -

SubAttribute Bindings:

S-10x Attribute	S-57 Acronym	Allowable Encoding Value	Type	Multiplicity
Cable Length			RE	1, 1
Height Length Units		1 : Metres 2 : Feet 3 : Kilometres 4 : Hectometres 5 : Statute Miles 6 : Nautical Miles	EN	1, 1
Diameter			RE	1, 1

6.20. Change Details

Definition: -

CamelCase: ChangeDetails

Alias:

Remarks: -

SubAttribute Bindings:

S-10x Attribute	S-57 Acronym	Allowable Encoding Value	Type	Multiplicity
Aton Commissioning		1 : Buoy establishment 2 : Light establishment 3 : Beacon establishment 4 : Audible signal establishment 5 : Fog signal establishment 6 : AIS transmitter establishment 7 : V-AIS establishment 8 : RACON establishment 9 : DGPS station establishment 10 : eLORAN station establishment 11 : DGLONASS station establishment 12 : e-Chayka station establishment 13 : EGNOS establishment	EN	0, 1
Aton Removal		1 : Buoy removal 2 : Buoy temporary removal 3 : Light removal 4 : Light temporary removal 5 : Beacon removal 6 : Beacon temporary removal 7 : Fog signal removal 8 : Fog signal temporary removal 9 : Audible signal removal 10 : Audible signal temporary removal	EN	0, 1

		11 : V-AIS removal 12 : V-AIS temporary removal 13 : RACON signal removal 14 : RACON temporary removal 15 : DGPS removal 16 : DGPS temporary removal 17 : EGNOS removal 18 : EGNOS temporary removal 19 : LORAN C station removal 20 : LORAN C station temporary removal 21 : eLORAN removal 22 : eLORAN temporary removal 23 : Chayka station removal 24 : Chayka station temporary removal 25 : e-Chayka station removal 26 : e-Chayka station temporary removal 27 :		
Aton Replacement		1 : Buoy change 2 : Buoy temporary change 3 : Light change 4 : Light temporary change 5 : Sector light change 6 : Sector light temporary change 7 : Beacon change 8 : Beacon temporary change 9 : Fog signal change 10 : Fog signal temporary change 11 : Audible signal change 12 : Audible signal temporary change 13 : V-AIS change 14 : V-AIS temporary change 15 : RACON signal change 16 : RACON temporary	EN	0, 1

		change		
Fixed Aton Change		1 : Beacon missing 2 : Beacon damaged 3 : Light beacon Unlit 4 : Light beacon Unreliable 5 : Light beacon Not synchronized 6 : Light beacon damaged 7 : Beacon topmark missing 8 : Beacon topmark damaged 9 : Beacon daymark unreliable 10 : Floodlit beacon Unlit 11 : Beacon restored to normal	EN	0, 1
Floating Aton Change		1 : Buoy adrift 2 : Buoy damaged 3 : Buoy daymark unreliable 4 : Buoy destroyed 5 : Buoy missing 6 : Buoy move 7 : Buoy off position 8 : Buoy re-establishment 9 : Buoy restored to normal 10 : Buoy topmark damaged 11 : Buoy topmark missing 12 : Buoy will be withdrawn 13 : Buoy withdrawn 14 : Decommissioned for winter 15 : Lifted for Winter 16 : Light buoy Light damaged 17 : Light buoy Light not synchronized 18 : Light buoy Light unlit 19 : Light buoy Light unreliable 20 : Marine Aids to Navigation unreliable 21 : Recommissioned for navigation season 22 : Replaced by Winter Spar 23 : Seasonal	EN	0, 1

		decommissioning complete 24 : Seasonal decommissioning in progress 25 : Seasonal recommissioning complete 26 : Seasonal recommissioning in progress		
Audible Signal Aton Change		1 : Audible signal out of service 2 : Fog signal out of service 3 : Audible signal operating properly 4 : Fog signal operating properly	EN	0, 1
Lighted Aton Change		1 : Light unlit 2 : Light unreliable 3 : Light re-establishment 4 : Light range reduced 5 : Light without rhythm 6 : Light out of synchronization 7 : Light daymark unreliable 8 : Light operating properly 9 : Sector light Sector obscured 10 : Front leading/range light Unlit 11 : Rear leading/range light Unlit 12 : Front leading/range light Unreliable 13 : Rear leading/range light Unreliable 14 : Front leading/range light Light range reduced 15 : Rear leading/range light Light range reduced 16 : Front leading/range light without rhythm 17 : Rear leading/range light without rhythm 18 : Leading/range lights out of synchronization 19 : Front leading/range beacon Unreliable 20 : Rear leading/range beacon Unreliable 21 : Front leading/range light	EN	0, 1

		is operating properly 22 : Rear leading/range light is operating properly 23 : Front leading/range beacon restored to normal 24 : Rear leading/range beacon restored to normal		
Electronic Aton Change		1 : AIS transmitter out of service 2 : AIS transmitter unreliable 3 : AIS transmitter operating properly 4 : V-AIS out of service 5 : V-AIS unreliable 6 : V-AIS operating properly 7 : RACON out of service 8 : RACON unreliable 9 : RACON operating properly 10 : DGPS out of service 11 : DGPS operating properly 12 : DGPS unreliable 13 : LORAN C operating properly 14 : LORAN C unreliable 15 : LORAN C out of service 16 : eLORAN operating properly 17 : eLORAN unreliable 18 : eLORAN out of service 19 : DGLOANSS operating properly 20 : DGLOANSS unreliable 21 : DGLOANSS out of service 22 : Chayka operating properly 23 : Chayka unreliable 24 : Chayka out of service 25 : e-Chayka operating properly 26 : e-Chayka unreliable 27 : e-Chayka out of service 28 : EGNOS operating properly 29 : EGNOS unreliable	EN	0, 1

		30 : EGNOS out of service		
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6.21. Obscured Sector

Definition: -

CamelCase: ObscuredSector

Alias:

Remarks: -

SubAttribute Bindings:

S-10x Attribute	S-57 Acronym	Allowable Encoding Value	Type	Multiplicity
Sector Limit			C	1, 1
Text	(INFORM) (NINFOM)		(S) TE	1, 1

6.22. Sinker Dimensions

Definition: The dimensions of a sinker/anchor to give its three dimensional shape measurements.

CamelCase: sinkerDimensions

Alias:

Remarks: -

SubAttribute Bindings:

S-10x Attribute	S-57 Acronym	Allowable Encoding Value	Type	Multiplicity
Height Length Units		1 : Metres 2 : Feet 3 : Kilometres 4 : Hectometres 5 : Statute Miles 6 : Nautical Miles	EN	1, 1
Horizontal Length	(HORLEN)		RE	0, 1
Horizontal Width	(HORWID)		RE	0, 1
Vertical Length	(VERLEN)		RE	0, 1

6.23. Positioning Method

Definition: A description of the method used to obtain a position.(proposed by CCG)

CamelCase: positioningMethod

Alias:

Remarks: -

SubAttribute Bindings:

S-10x Attribute	S-57 Acronym	Allowable Encoding Value	Type	Multiplicity
Positioning Equipment		1 : DGPS Receiver 2 : GLONASS Receiver 3 : GPS Receiver 4 : GPS/WAAS Receiver	EN	1, 1
NMEAString			TE	1, 1

6.24. Horizontal Position Uncertainty

Definition: The best estimate of the accuracy of a position.

CamelCase: horizontalPositionUncertainty

Alias: POSACC

Remarks: The expected input is the maximum of the two-dimensional error. The error is assumed to be positive and negative.

SubAttribute Bindings:

S-10x Attribute	S-57 Acronym	Allowable Encoding Value	Type	Multiplicity
Uncertainty Fixed	(POSACC) (SOUACC) (VERACC)		RE	1, 1
Uncertainty Variable Factor			RE	0, 1

6.25. Information

Definition: Textual information about the feature. The information may be provided as a string of text or as a file name of a single external text file that contains the text.

CamelCase: information

Alias: INFORM

Remarks: At least one of the sub-attributes file reference or text must be populated. The sub-attribute file reference is generally used for long text strings or those that require formatting, however, there is no restriction on the type of text (except for lexical level) that can be held in files referenced by sub-attribute file reference.

SubAttribute Bindings:

S-10x Attribute	S-57 Acronym	Allowable Encoding Value	Type	Multiplicity
File Locator			TE	0, 1
File Reference	(TXTDSC) (NTXTDS)		TE	0, 1
Headline			TE	0, 1
Language			TE	1, 1
Text	(INFORM) (NINFOM)		TE	0, 1

6.26. Textual Description

Definition: Encodes the file name of a single external text file that contains the text in a defined language, which provides additional textual information that cannot be provided using other allowable attributes for the feature.

CamelCase: textualDescription

Alias: TXTDSC

Remarks: No remarks.

SubAttribute Bindings:

S-10x Attribute	S-57 Acronym	Allowable Encoding Value	Type	Multiplicity
File Reference	(TXTDSC) (NTXTDS)		TE	1, 1 (ordered)
Language			TE	0, 1 (ordered)

6.27. Vertical Uncertainty

Definition: The best estimate of the vertical accuracy of depths, heights, vertical distances and vertical clearances.

CamelCase: verticalUncertainty

Alias: VERACC SOUACC

Remarks: Encodes the vertical uncertainty associated with any vertical measurement.

SubAttribute Bindings:

S-10x Attribute	S-57 Acronym	Allowable Encoding Value	Type	Multiplicity
Uncertainty Fixed	(POSACC) (SOUACC) (VERACC)		RE	1, 1
Uncertainty Variable Factor			RE	0, 1

7. Roles

7.1. Status Part

Definition: -

CamelCase: Statuspart

Alias:

Remarks: No remarks.

7.2. Buoy Part

Definition: TBD

CamelCase: buoyPart

Alias:

Remarks: No remarks.

7.3. Topmark part

Definition: TBD

CamelCase: topmarkPart

Alias:

Remarks: No remarks.

7.4. Parent

Definition: -

CamelCase: parent

Alias:

Remarks: No remarks.

7.5. Child

Definition: -

CamelCase: child

Alias:

Remarks: No remarks.

7.6. Physical AIS Broadcast By

Definition: -

CamelCase: physicalAISbroadcastBy

Alias:

Remarks: No remarks.

7.7. Physical AIS Broadcasts

Definition: -

CamelCase: physicalAISbroadcasts

Alias:

Remarks: No remarks.

7.8. Synthetic AIS Broadcast By

Definition: -

CamelCase: syntheticAISbroadcastBy

Alias:

Remarks: No remarks.

7.9. Synthetic AIS Broadcasts

Definition: -

CamelCase: syntheticAISbroadcasts

Alias:

Remarks: No remarks.

7.10. Virtual AIS Broadcast By

Definition: -

CamelCase: virtualAISbroadcastBy

Alias:

Remarks: No remarks.

7.11. Virtual AIS Broadcasts

Definition: -

CamelCase: virtualAISbroadcasts

Alias:

Remarks: No remarks.

7.12. Buoy Attached

Definition: -

CamelCase: buoyattached

Alias:

Remarks: No remarks.

7.13. Counter Weightholds

Definition: -

CamelCase: counterWeightholds

Alias:

Remarks: No remarks.

7.14. Buoy Hangs

Definition: -

CamelCase: buoyhangs

Alias:

Remarks: No remarks.

7.15. Bridle Holds

Definition: -

CamelCase: bridleholds

Alias:

Remarks: No remarks.

7.16. Shackle To Cableconnected To

Definition: -

CamelCase: shackleToCableconnectedTo

Alias:

Remarks: No remarks.

7.17. Shackle To Cableconnected

Definition: -

CamelCase: shackleToCableconnected

Alias:

Remarks: No remarks.

7.18. Swivel Attached

Definition: -

CamelCase: swivelattached

Alias:

Remarks: No remarks.

7.19. Bridle Attached

Definition: -

CamelCase: bridleattached

Alias:

Remarks: No remarks.

7.20. Cable Holds

Definition: -

CamelCase: cableholds

Alias:

Remarks: No remarks.

7.21. Shackle To Bridle Connecteda

Definition: -

CamelCase: shackleToBridleconnected

Alias:

Remarks: No remarks.

7.22. Shackle To Bridle Connected To

Definition: -

CamelCase: shackleToBridleconnectedTo

Alias:

Remarks: No remarks.

7.23. Shackle To Buoy Connected

Definition: -

CamelCase: shackleToBuoyconnected

Alias:

Remarks: No remarks.

7.24. Shackle To Buoy Connected To

Definition: -

CamelCase: shackleToBuoyconnectedTo

Alias:

Remarks: No remarks.

7.25. Shackle To Swivel Connected

Definition: -

CamelCase: shackleToSwivelconnected

Alias:

Remarks: No remarks.

7.26. Shackle To Swivel Connected To

Definition: -

CamelCase: shackleToSwivelconnectedTo

Alias:

Remarks: No remarks.

7.27. Shackle To Anchor Connected To

Definition: -

CamelCase: shackleToAnchorconnectedTo

Alias:

Remarks: No remarks.

7.28. Shackle To Anchor Connected

Definition: -

CamelCase: shackleToAnchorconnected

Alias:

Remarks: No remarks.

7.29. Bridle Hangs

Definition: -

CamelCase: bridlehangs

Alias:

Remarks: No remarks.

7.30. Swivel Holds

Definition: -

CamelCase: swivelholds

Alias:

Remarks: No remarks.

7.31. Peer Aton Aggregation

Definition: TBD

CamelCase: peerAtonAggregation

Alias:

Remarks: No remarks.

7.32. Aton Aggregation By

Definition: TBD

CamelCase: atonAggregationBy

Alias:

Remarks: No remarks.

7.33. Peer Aton Association

Definition: TBD

CamelCase: peerAtonAssociation

Alias:

Remarks: No remarks.

7.34. Aton Association By

Definition: TBD

CamelCase: atonAssociationBy

Alias:

Remarks: No remarks.

7.35. Navigable Track

Definition: The role given to the navigable part of the navigation line.

CamelCase: navigableTrack

Alias:

Remarks: No remarks.

7.36. Navigation Line

Definition: The role given to the navigation line(s) that is generally formed between two or more objects, or by one object and a bearing.

CamelCase: navigationLine

Alias:

Remarks: No remarks.

7.37. Fixing Method

Definition: -

CamelCase: fixingMethod

Alias:

Remarks: No remarks.

7.38. Positioning Method

Definition: -

CamelCase: positioningMethod

Alias:

Remarks: No remarks.

7.39. Danger

Definition: -

CamelCase: danger

Alias:

Remarks: No remarks.

7.40. Marking Aton

Definition: -

CamelCase: markingAton

Alias:

Remarks: No remarks.

8. Information Associations

8.1. Aton Status

Definition:

CamelCase: Atonstatus

Alias:

Remarks: -

Role: Status Part

Role:

8.2. Aton Fixing Method Association

Definition:

CamelCase: AtonFixingMethodAssociation

Alias:

Remarks: -

Role: Fixing Method

Role:

8.3. Aton Positioning Information Association

Definition:

CamelCase: AtonPositioningInformationAssociation

Alias:

Remarks: -

Role: Positioning Method

Role:

9. Feature Associations

9.1. Buoy Topmark

Definition:

CamelCase: BuoyTopmark

Alias:

Remarks: -

Role: Topmark part

Role: Buoy Part

9.2. Structure Equipment

Definition:

CamelCase: StructureEquipment

Alias:

Remarks: -

Role: Parent

Role: Child

9.3. Physical AIS

Definition:

CamelCase: PhysicalAIS

Alias:

Remarks: -

Role: Physical AIS Broadcast By

Role: Physical AIS Broadcasts

9.4. Synthetic AIS

Definition:

CamelCase: SyntheticAIS

Alias:

Remarks: -

Role: Synthetic AIS Broadcast By

Role: Synthetic AIS Broadcasts

9.5. Virtual AIS

Definition:

CamelCase: VirtualAIS

Alias:

Remarks: -

Role: Virtual AIS Broadcast By

Role: Virtual AIS Broadcasts

9.6. Buoy Counter Weight

Definition:

CamelCase: BuoyCounterWeight

Alias:

Remarks: -

Role: Buoy Attached

Role: Counter Weightholds

9.7. Bridle Connection

Definition:

CamelCase: BridleConnection

Alias:

Remarks: -

Role: Buoy Hangs

Role: Bridle Holds

9.8. Shackle Connection

Definition:**CamelCase:** ShackleConnection**Alias:****Remarks:** -**Role:** Shackle To Buoy Connected**Role:** Shackle To Buoy Connected To**9.9. Shackle Connection From Cable****Definition:****CamelCase:** ShackleConnectionFromCable**Alias:****Remarks:** -**Role:** Shackle To Cableconnected**Role:** Shackle To Cableconnected To**9.10. Swivel Cable Connection****Definition:****CamelCase:** SwivelCableConnection**Alias:****Remarks:** -**Role:** Swivel Attached**Role:** Cable Holds**9.11. Bridle Cable Connection****Definition:****CamelCase:** BridleCableConnection**Alias:****Remarks:** -**Role:** Bridle Attached

Role: Cable Holds

9.12. Shackle To Bridle Connection

Definition:

CamelCase: ShackleToBridleConnection

Alias:

Remarks: -

Role: Shackle To Bridle Connecteda

Role: Shackle To Bridle Connected To

9.13. Shackle To Swivel Connection

Definition:

CamelCase: ShackleToSwivelConnection

Alias:

Remarks: -

Role: Shackle To Swivel Connected

Role: Shackle To Swivel Connected To

9.14. ShackleToAnchorConnection

Definition:

CamelCase: ShackleToAnchorConnection

Alias:

Remarks: -

Role: Shackle To Anchor Connected To

Role: Shackle To Anchor Connected

9.15. Swivel Connection

Definition:

CamelCase: SwivelConnection

Alias:

Remarks: -

Role: Bridle Hangs

Role: Swivel Holds

9.16. Aton Aggregations

Definition:

CamelCase: AtonAggregations

Alias:

Remarks: -

Role: Peer Aton Aggregation

Role: Aton Aggregation By

9.17. Aton Associations

Definition:

CamelCase: AtonAssociations

Alias:

Remarks: -

Role: Peer Aton Association

Role: Aton Association By

9.18. Range System

Definition:

CamelCase: RangeSystem

Alias:

Remarks: -

Role: Navigable Track

Role: Navigation Line

9.19. Dangerous Feature Association

Definition:

CamelCase: DangerousFeatureAssociation

Alias:

Remarks: -

Role: Danger

Role: Marking Aton

